Dispersion Calculator^3 User's Manual





Deutsches Zentrum
für Luft- und Raumfahrt
German Aerospace Center

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Dr. Armin Huber

Center for Lightweight Production Technology (ZLP)
Institute of Structures and Design
German Aerospace Center (DLR)
86159 Augsburg
Germany

armin.huber@dlr.de




MADE IN GERMANY
DC
DC
MADE FOR THE WORLD

Dispersion Calculator is dedicated to

Sir Patrick Stewart

Preface

Many applications of ultrasonic guided waves exist, but, as the reader may confirm, getting the dispersion curves of guided waves is difficult. Although there are many researchers who are able to calculate dispersion diagrams with their own code, most people who deal with guided waves do not have such code, either because they don't know where to get it, or they don't have the money to buy DISPERSE. Clearly, writing such code is difficult, in particular in case of multilayered anisotropic laminates, and those who can do it keep the code like a treasure. When I started conducting air-coupled ultrasonic inspection of carbon composites using Lamb waves at DLR in Augsburg, Germany, in 2013, I had no code for calculating dispersion diagrams, too. Fortunately I got the budget in 2016 for buying DISPERSE. With this software, I am able to write and validate DC to this day. I found soon that it would be worth to make DC open to the public because of the high demand and lack of free software, but also because I considered it a waste of resources spending years on the code, and then only I could use it. Therefore, I released DC in 2018, and many user feedback from many countries as well as publications citing DC prove that DC is widely used today. I may say that it is a great pleasure for me to see how my "baby" performs, and I draw much satisfaction and motivation from it to continue the work on it. It is my sincere hope that the scientific community will benefit from DC, and that it might help create connections between humans belonging to different cultures or countries, which tend to compete or even fight each other.

In that sense, I want to express my gratitude to the researchers who have supported me in my work:

- Prof. Dr. Michael Lowe, Imperial College London, London, UK
- Prof. Dr. Michel Castaings, University of Bordeaux, Bordeaux, France
- Prof. Dr. Stanislav Rokhlin, Ohio State University, Columbus, OH, USA
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Armin Huber

Augsburg, Germany
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1 Introduction

The nondestructive inspection (NDI) of components by means of guided waves is an emerging technology in fields like aerospace or pipeline construction an maintenance. The guided waves' capability of propagating many meters in a structure is utilized for swift inspection tasks as well as structural health monitoring and acoustical emission. The German Aerospace Center (DLR) uses Lamb waves excited by air-coupled ultrasound for the quality assurance of large-scale aerospace vehicle components made from carbon composites. However, the use of guided waves in NDI requires knowledge of the dispersion curves. DISPERSE [1, 2] (Imperial College London, London, UK) is the most renown software for the calculation of dispersion curves. It has been developed since the early 1990s by Lowe and Pavlakovic, and it is used for the validation of Dispersion Calculator (DC). But there are also free tools other than DC. GUIGUW [3, 4] is a software package developed by Marzani and Bocchini and released in 2011. Since it is based on the *semi-analytical finite-element method* (SAFE), it has the advantage of being able to model waveguides of arbitrary cross sections. Another promising contender, the open source MATLAB-toolbox ElasticMatrix [5, 6], was released by Ramasawmy et al. in 2020. It is able to compute dispersion curves for multilayered media made of isotropic and transversely isotropic layers where the wave propagation occurs in a material plane of symmetry. However, this covers decoupled cases only, so far excluding the majority of composites where we have arbitrary fiber orientations and wave propagation directions. Nonetheless, it is a promising project meant to be extended in the future. A purely Python-based code package for the simulation of ultrasonic wave propagation in multilayered plates and pipes is "TraFiC", developed by Ducasse since 2020 [7]. Then, the MATLAB-based "The Dispersion Box" was released by Orta et al. in 2022 [8, 9]. This tool is able to compute dispersion and attenuation curves in multilayered, viscoelastic, orthotropic plates. Its special feature is that up to six different methods can be used simultaneously, namely the *global matrix method* (GMM), the *stiffness matrix method* (SMM), the *hybrid compliance-stiffness matrix method* (HCSMM), the SAFE method, the *Legendre polynomial method* (LPM), and the *fifth order shear deformation theory* (5-SDT). The latter was developed also by Orta et al. [10] as an improvement to the previous lower order plate theories. A not to be underestimated advantage is also that The Dispersion Box has an easy-to-use graphical user interface. In 2023, Liu et al. released SAFEDC [11, 12]. This software has a similar layout as DC, but uses SAFE for the modeling of waveguides with arbitrary cross sections and general anisotropy. Also in 2023, Kiefer released the open source "GEWtool" [13] for MATLAB. It uses the *spectral element method* (SEM), enabling extremely fast calculations. In 2024, Gravenkamp released the open source MATLAB-code package "SAMWISE" [14]. SAMWISE uses SAFE and the *scaled boundary finite element method* (SBFEM) to compute dispersion curves and mode shapes of guided waves propagating along plates, cylinders, and more general three-dimensional structures of arbitrary cross-section.

The task that triggered the development of DC was the calculation of incidence angles for the excitation of Lamb waves for the air-coupled ultrasonic inspection of rocket booster pressure vessels of the future launcher Ariane 6. These vessels are made of car-

bon fiber reinforced plastics (CFRP) in order to make them lighter, and therefore enable the launcher to carry more payload. In some areas, such pressure vessels can consist of up to four hundred layers, which is a challenging task to calculate them. To facilitate the calculation, it is a common practice to group layers, but this is not possible here because the layups are very complicated and irregular in terms of the winding angle (fiber orientation) and layer thicknesses. Therefore, every single layer has to be calculated, and SMM, which has been developed by Rokhlin and Wang [15–17] in 2002, has been implemented in MATLAB to perform these calculations. A very useful paper for the coding of SMM is that by Kamal and Giurgiutiu [18]. Tan in 2005 introduced an improvement in efficiency [19], and shortly thereafter he presented HCSMM [20], which solved the instability of SMM for very small frequency-thickness products. Other researchers who use SMM are Barksi et al. [21–23] and Muc et al. [24]. By implementing Guo et al.'s hybrid matrix strategy [25] in DC v3.3, random numerical instability prevalent in hybrid layups, especially in combinations of isotropic and anisotropic layers, was removed. However, initially, the excellent and highly instructive book by Nayfeh [26] featuring the *transfer matrix method* (TMM) served as the backbone for the programming of DC. TMM was developed by Thomson [27] in 1950, and a small correction was conducted by Haskell [28] in 1953. Nayfeh extended TMM to guided SH waves in multilayered anisotropic plates in 1989 [29] and to general guided wave propagation in 1991 [30]. However, SMM solved the well-known numerical instability which TMM suffers from. Moreover, SMM retains the concise form and efficiency of TMM, unlike other stable methods such as GMM, which is used by DISPERSE and ElasticMatrix. GMM was proposed by Knopoff [31] in 1964, and it is discussed in the famous paper by Lowe [32]. The SAFE method, introduced by Gavric [33] in 1995, and used by GUIGUW, The Dispersion Box, and SAFEDC is an alternative to root-finding methods. The SAFE method discretizes the waveguide's cross section into finite elements, allowing the modeling of guided waves in complex geometries. At the same time, the wave propagation direction is solved analytically, which makes this approach more efficient than full FEM modeling. It should be noted, however, that the SAFE method (similar as the *spectral collocation method* (SCM)) is an approximate method, i.e., the accuracy of the results depends on the number of discrete elements set through the thickness of the waveguide. The above-mentioned SCM was introduced by Adamou and Craster for guided wave modeling in elastic media in 2004 [34]. SCM is similar to SAFE in that it is using a one-dimensional mesh over the system's cross section, but SCM possesses a higher accuracy and speed of computation (whereas SEM seems to be even faster than SCM). Instead of solving a differential equation directly, SCM uses a spectral approximation, which satisfies the differential equation and boundary conditions. Hernando-Quintanilla et al. have provided comprehensive studies of guided wave modeling in generally anisotropic media by means of SCM [35–38]. According to the authors, SCM is easier to code than root-finding methods, it is faster, and most importantly, it does not miss modal solutions. More recently, the method has been used to model also leaky waves [39–41]. SCM is currently implemented in DISPERSE. Probably, SCM/SEM are the best discretizing methods and SMM is the best root-finding method.

The theoretical work on guided wave modeling and the implementation into DC is

reported in detail in Refs. [42–44]. Starting in DC v3.0 and completed in DC v3.1, TMM can be used in situations where it is stable. Dispersion curves for isotropic plates are obtained most efficiently by applying the Rayleigh-Lamb equations [45, 46], and the excellent book by Rose [47] served well for the coding. The advanced model for the group/energy velocity of guided waves in anisotropic specimen, introduced in DC v1.11, has been developed with help of the highly recommended book by Giurgiutiu [48]. The coding of the characteristic equations used for fluid-loaded cases, introduced in DC v2.0, was done with help of the excellent book by Rokhlin *et al.* [17]. The modeling of axial guided waves in isotropic cylinders, introduced in DC v3.0, was programmed with the help of several publications. For isotropic rods, the dispersion equations for flexural, longitudinal, and torsional modes were published in 1964 by Meeker and Meitzler [49]. A more efficient equation for flexural modes with circumferential order $n = 1$ is given in the 1960 paper by Pao and Mindlin [50]. The dispersion equation for flexural waves in isotropic hollow cylinders (pipes) was introduced by Gazis in 1959 [51]. However, the equations in his paper contain an error (see the discussion and correction in the DISPERSE manual), and also the way he derived the equations is harder to follow and more unwieldy than the equations presented by Kumar in 1971 [52] (who made no error). Furthermore, Kumar's paper added the fluid-filling (the outer fluid-loading can be added by another equation with a Hankel function for the leaky waves, see Ref. [47]). In fact, Kumar's 1971 paper is everything needed, as the equations for longitudinal and torsional modes in pipes as well as flexural, longitudinal, and torsional modes in rods are naturally contained in his equations (for the rod, remove the equations at the inner radius and also remove the outgoing bulk wave terms; for longitudinal and torsional waves, set $n = 0$ and separate the L- and SV bulk waves terms from the SH terms). Nonetheless, the 1972 paper by Kumar [53], presenting the equations for longitudinal modes, should be mentioned. The dispersion equations for Lamb-type circumferential waves in a pipe are taken from Refs. [54, 55] and those for shear horizontal circumferential waves from Ref. [56].

2 The Dispersion Calculator

2.1 Overview

DC is an interactive and fully validated stand-alone software for the computation of dispersion curves (phase velocity, energy velocity, attenuation, etc. vs frequency) and mode shapes (displacement, stress, strain, etc.) of guided waves in multilayered isotropic plates, rods and pipes (axial and circumferential waves) as well as in multilayered anisotropic plates. The material(s) can be perfectly elastic or viscoelastic, and the waveguide can be surrounded by and, in case of a pipe, be filled with inviscid fluids. DC can handle laminates consisting of several hundreds of layers, and it is able to distinguish the different mode families, like symmetric, anti-symmetric, and non-symmetric (leaky) Lamb and shear horizontal waves, depending on the symmetry and coupling properties of a given layup. DC features a highly efficient and robust dispersion curve tracing algorithm. The initial release of DC took place in November 2018.

DC offers an easy-to-use graphical user interface (GUI) (see for example the **Anisotropic** tab in Fig. 1). Basically, it provides only the parameter setting interfaces like editable fields and buttons as well as some output windows displaying data and other information. All plots like dispersion diagrams or mode shapes are plotted in extra win-

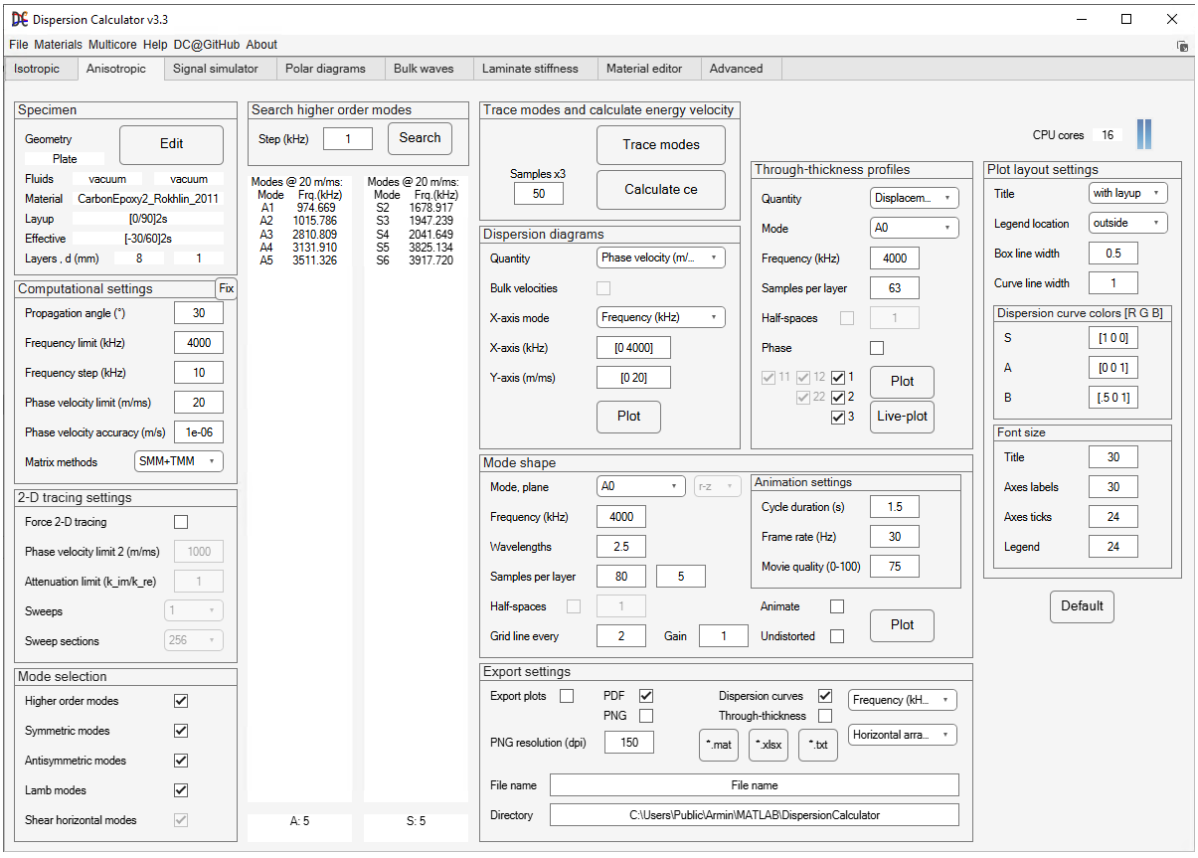


FIG. 1. The Anisotropic tab.

dows, which open when the user presses the corresponding button. This basic arrangement aims to improve user-friendliness by avoiding options in multi-leveled submenus where they are out of sight initially. Instead, DC provides all options that are related to one task in just one tab. For example, in the "Anisotropic" tab, the user can set up the specimen, define the computational parameters, calculate the dispersion curves, through-thickness profiles, and mode shapes, adjust the plot layout, and manage the export of data and plots. DC has eight tabs, of which six cover the main tasks, one allows to enter material parameters, and the last one contains advanced options. The tabs are listed below:

1. **Isotropic**: Calculation of dispersion diagrams and mode shapes for single layer isotropic plates, rods, and pipes (axial and circumferential waves)(fast computation by using the Rayleigh-Lamb equations, respectively, those specific to isotropic cylinders).

2. **Anisotropic** Calculation of dispersion diagrams and mode shapes for multilayered isotropic plates, rods, and pipes and multilayered anisotropic plates (using SMM and TMM).

3. **Signal simulator**: Simulation of the signal of single and multiple guided wave modes (for plates only).

4. **Polar diagrams**: Calculation of polar dispersion diagrams for multilayered anisotropic plates.

5. **Bulk waves**: Calculation of the bulk waves' phase and group velocities, slownesses, and polarization, propagating in anisotropic media as well as their scattering on solid-fluid interfaces.

6. **Laminate stiffness**: Calculation of the homogenized stiffness tensor of laminates.

7. **Material editor**: To enter engineering constants or stiffnesses of orthotropic, transversely isotropic, cubic, and isotropic materials as well as fluids.

8. **Advanced**: Contains the advanced options for the dispersion curve tracing algorithm.

2.2 Limitations

- DC computes only modal solutions with purely real or complex wavenumbers. Purely imaginary wavenumbers are not covered.

- Piezoelectric effects are not supported.

- The half-spaces surrounding a waveguide can only be filled with fluids, not with solids.

- Viscosity in fluids is not supported.

- If the waveguide is immersed in a fluid, the power normalization of the mode shapes is no strict because the components in the half-spaces are ignored in the integration.
- Wave propagation in rods and pipes (axial and circumferential waves) is limited to isotropic layers. Anisotropic layers are not possible.
- In case of circumferential waves in a pipe, no fluid-loading can be applied and viscoelastic damping is ignored.

2.3 Download and installation

DC is available on GitHub [57]. One can download the stand-alone installer, for which no MATLAB is required, and one can download the MATLAB code (notice that Curve Fitting Toolbox is required also). When running the standalone installer, the required MATLAB runtime will be downloaded and installed automatically. In case the MATLAB runtime is not downloaded and installed automatically, download it from

https://www.mathworks.com/products/compiler/matlab-runtime.html

and install it manually. The required runtime version number can be found in the change log file. For more information, see the README on GitHub [57].

2.4 Warnings

The numerical model for obtaining dispersion diagrams in DC has been fully validated against DISPERSE. In some cases, however, DC disagrees with DISPERSE. In some of these cases, validation was made against published data to ensure that DC computes correctly. However, this does not mean that DC is without errors. No software is without them. There are two main sources for erroneous calculations in DC. The first source is the possibility that a dispersion equation is incorrect. I consider this problem rather unlikely in DC. The second source is the dispersion curve tracing algorithm. Dispersion curve tracing is notoriously difficult, in particular when damping is involved in the system. It is impossible to write a tracing routine which works perfectly robust for an infinite number of thinkable combinations of wave guide geometries, material parameters, layups, fluid-loading, etc.. Hence, dispersion curves can be incomplete, sudden jumps can appear, and some modes may be missing completely. Please contact me in case you encounter a bug or if you need help.

3 Getting started quickly

1. Go to the **Material editor**.
2. In the **Anisotropic materials** panel, enter the engineering constants or stiffness components of your material.
3. Enter a **New material's name** and press **Save material**.
4. Go to the **Anisotropic** tab.
5. Press **Edit**, select your waveguide geometry, material, and enter your layup.
6. Press **Trace modes** to calculate the dispersion diagram.
7. Press **Plot** in the **Dispersion diagrams** panel.
8. Click on a dispersion curve to display values.
9. Press **Plot** in the **Through-thickness profiles** panel to display the displacement components of the A_0/B_0 Lamb wave at the highest frequency covered by the dispersion diagram.
10. Activate the **Animate** check box in the **Mode shape panel** panel and press **Plot** below to animate the displacement pattern of the A_0/B_0 Lamb wave at the above frequency.

4 The graphical user interface

4.1 Menu bar

In the **File** menu, a new project can be generated by pressing **New project**. This closes the current DC instance and opens a new one. An existing project file can be loaded via the **Open project** button. By pressing **Save project**, the current project can be saved. Pressing **Quit** closes DC. The **Materials** menu allows to export material lists from DC or to import them from local. Upon pressing **Export**, you are asked to select a directory to which the material lists are exported. Five separate txt-files containing orthotropic, transversely isotropic, cubic, and isotropic materials as well as fluids will be exported to the selected directory. You can rename the material lists. Export material lists when you want to install a new DC version and transfer your custom materials to it. To import material lists, press **Import** and select one or multiple material lists. The names of the imported lists do not matter. DC automatically assigns the material lists to the respective material symmetry class. Once you have imported all lists, you have to restart DC (**File** → **New project**) to make the materials available. You also can open the directory holding the material lists by pressing **Open directory**.

Important note:
To add, edit, and delete materials, use the Material editor (see Sec. 4.8). Do not edit the material lists directly.

In the **Multicore** menu, you can **Enable** parallel computing. The two bars in the upper right corner of the GUI turn blue. The number shown in the **CPU cores** field is the number of CPU cores available for parallel computing. If multi-core computing is enabled, DC uses all CPU cores on your machine for dispersion curve tracing in the **Isotropic** and **Anisotropic** tabs. Each mode family is calculated by one core. If more mode families are scheduled than cores are available, the first idle core takes over the next waiting mode family. This is particularly useful when many mode families are scheduled like many higher circumferential mode orders of axial waves in pipes and rods, thereby reducing computation time drastically compared to sequential computing (multi-core computing is off). In the **Polar diagrams** tab, the dispersion curve tracing is performed on two cores. During parallel computation, the two bars turn green, and turn back to blue when the computation is completed. By pressing **Disable**, multi-core computing can be turned off. The two blue bars turn gray. By default, multi-core computing is off. In this case, the mode families are calculated sequentially, and polar dispersion curves are calculated on a single core. Press **Help** to read DC User's manual. Press **DC@GitHub** to visit the DC repository on GitHub [57], which is the official download platform. Whenever a new DC release is available, this button is highlighted in blue color.

4.2 Isotropic

The **Isotropic** tab consists of several panels. Start by defining the **Specimen** (A1) (see Fig. 2). At first, choose the waveguide **Geometry**. You can select **Plate**, **Rod**, **Pipe**, and **Circumferential**. Table 1 lists the guided wave mode families propagating in the various waveguide geometries. If you select **Plate**, symmetric and anti-symmetric Lamb waves S_m, A_m and shear horizontal waves S_m^{SH}, A_m^{SH} can be distinguished (m is the normal mode order with $m = 0, 1, 2, ..., \infty$). If you add fluid-loading (see further below), Scholte modes appear in addition. If the same fluid is on both sides of the plate, there are symmetric and anti-symmetric leaky Lamb and Scholte modes. However, starting in DC v3.2, both wave types are denoted S_m, A_m without the former distinction of Scholte modes as $S_m^{Scholte}$, $A_m^{Scholte}$ because it turns out that there exists no generally valid rule of how to distinguish them during the computation process.

Important note:
The normal mode order m is allocated in the order the modes are traced and therefore does not necessarily comply with notation established in the literature.

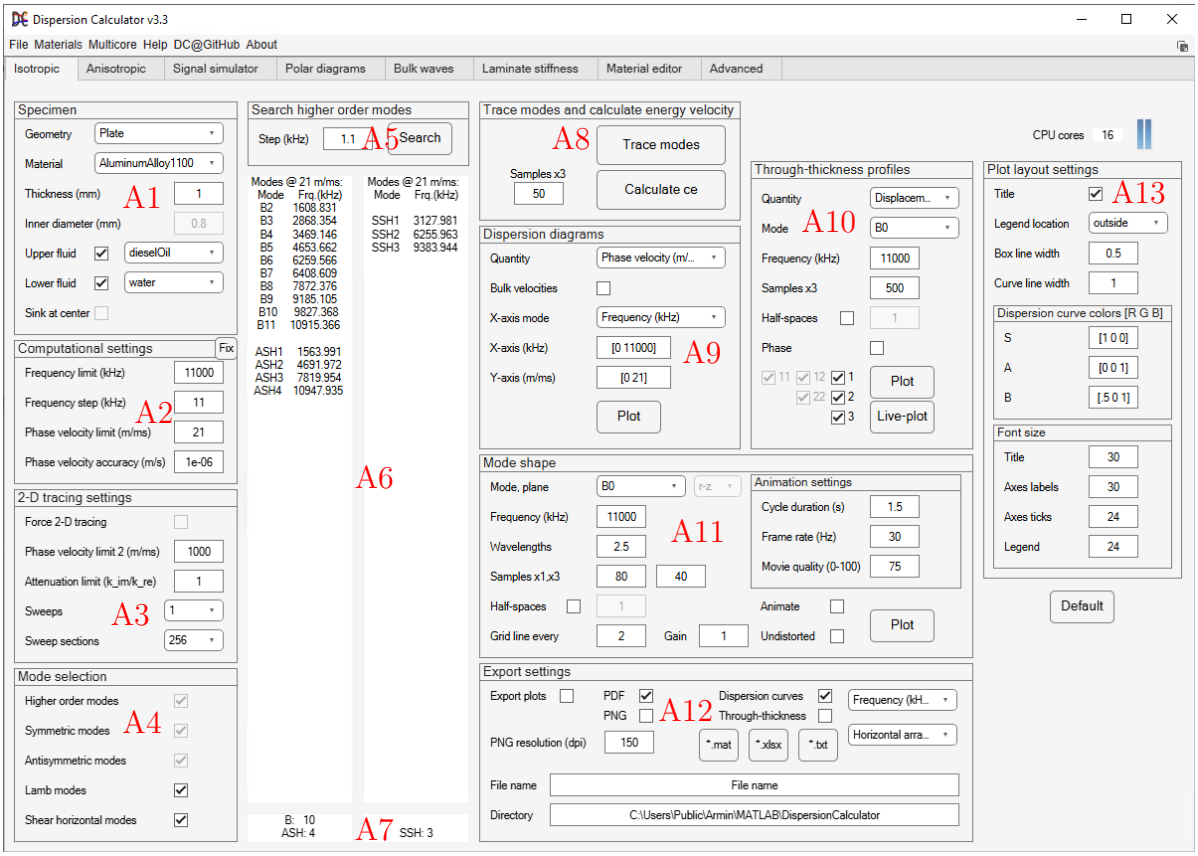


FIG. 2. The tab for the calculation of isotropic specimens.

TABLE 1. Guided wave mode families propagating in the various waveguide geometries (m: normal mode order; n: circumferential mode order).

Geometry	Mode families
Plate	S_m, A_m, B_m [a], S_m^{SH}, A_m^{SH}
Rod, Pipe	T(0,m), L(0,m), F(n,m)
Circumferential	C_m, C_m^{SH}

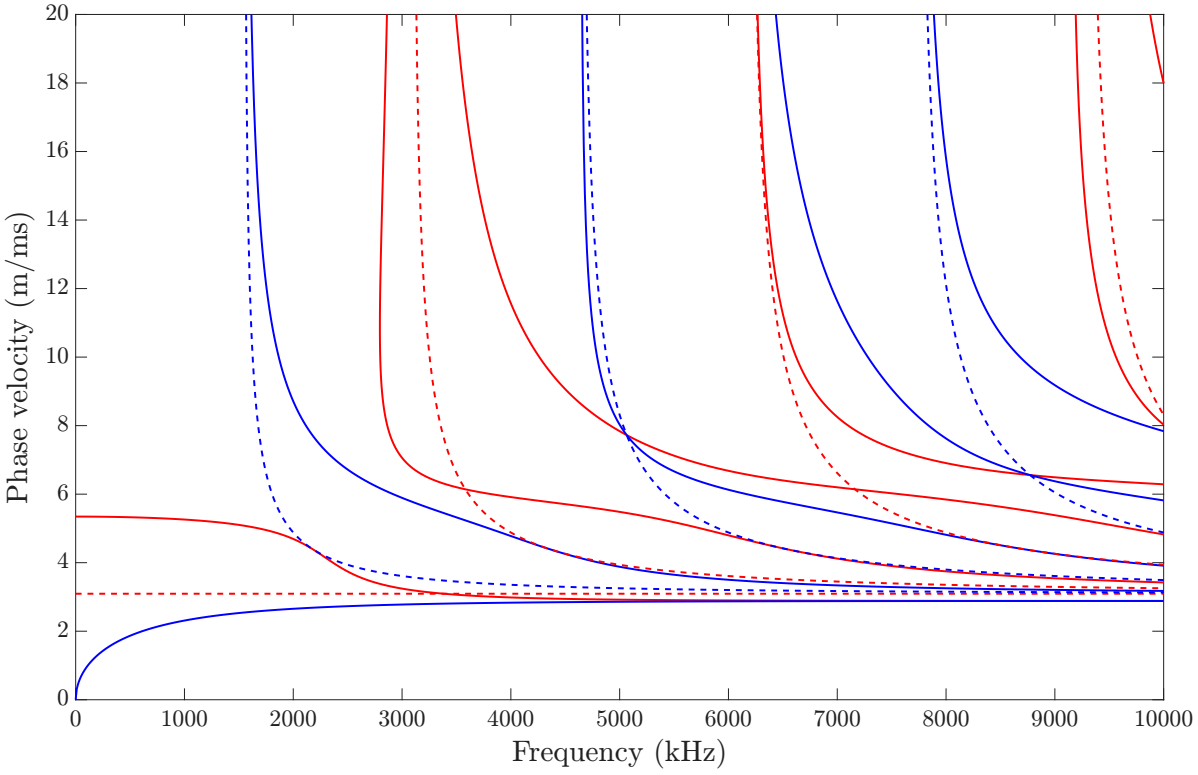
[a] In case of non-symmetric fluid-loading

Use the **Through-thickness profiles** (A10) panel to determine the mode type. Leaky and Scholte modes grow and decay in the fluid-filled half-spaces, respectively, see Fig. 6. If there are different fluids on each side or if only one side is covered with a fluid, non-symmetric leaky Lamb and Scholte modes B_m occur. Note that the shear horizontal waves S_m^{SH}, A_m^{SH} retain their symmetric because they do not couple to inviscid fluids (DC does not support viscosity in fluids). If you select **Rod** or **Pipe**, torsional, longitudinal, and flexural modes T(0,m), L(0,m), F(n,m) are modeled. If you add fluid-loading, L(0,m), F(n,m) contain both leaky and Scholte modes.

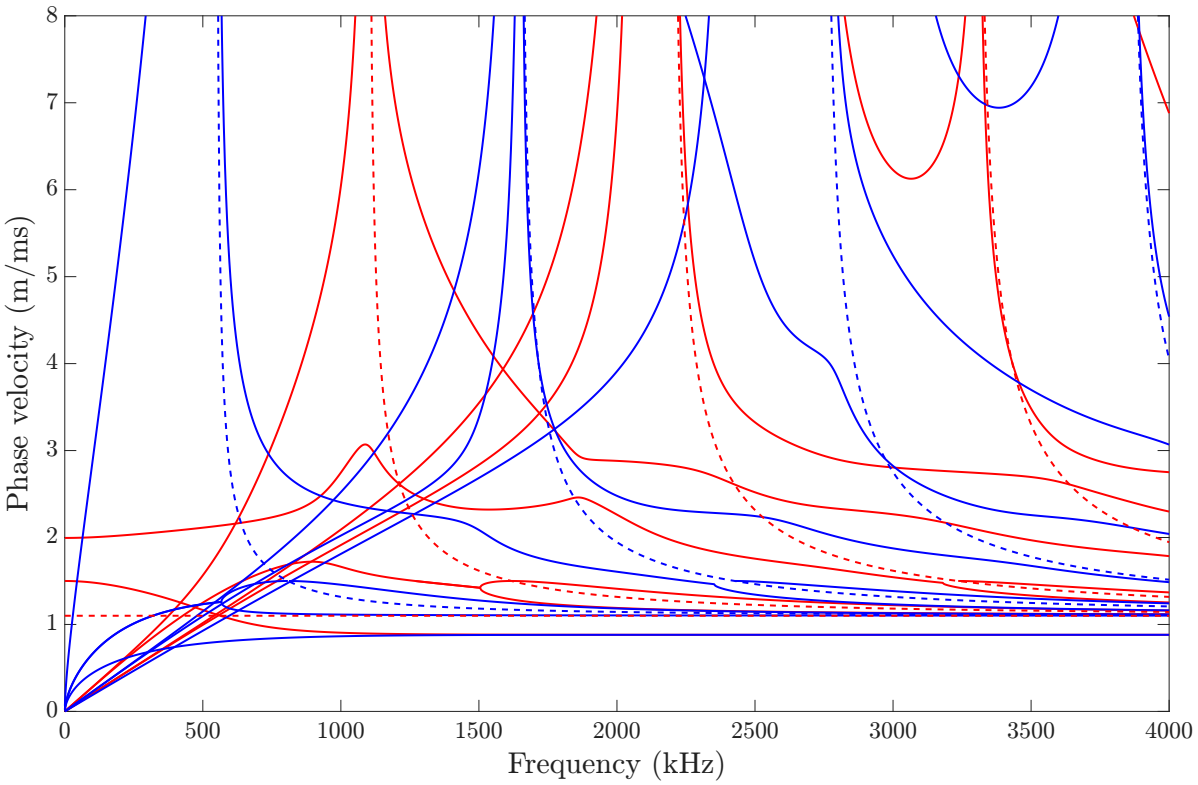
All of these modes propagate in the axial direction of the waveguide. In the nomenclature, n is the circumferential order. For T and L modes, n is zero, but for the flexural modes, n can take any integer number $n = 1, 2, 3, ..., \infty$, and each mode family n has its own set of normal mode orders $m = 1, 2, 3, ..., \infty$. If you select **Circumferential**, the guided waves propagating in the circumferential direction of a pipe are calculated. In analogy to the modes in a plate, there exist Lamb-type modes and SH-type modes C_m, C_m^{SH}. In fact, when the inner radius to outer radius ratio approaches unity, the dispersion curves closely resemble those of a flat plate (DC issues a recommendation to calculate the plate in such cases). DC supports no fluid-loading in the circumferential case, and material damping is ignored.

Next, select a **Material**. Then set the dimension(s) of the waveguide. If you have selected **Plate** as a **Geometry**, you can set its **Thickness**. For a **Rod**, this option is renamed to **Outer diameter**. If you have selected **Pipe** or **Circumferential**, the **Inner diameter** field becomes active. DC does not allow you to set **Inner diameter** larger than **Outer diameter**.

Then come the fluid-loading settings. In case of a **Plate**, you can fill the upper and lower half-spaces with fluids by checking the **Upper fluid** and **Lower fluid** check boxes, and then select the respective fluids from the drop-down menus. If you leave a check box empty, the corresponding half-space is a vacuum. In case of a **Rod**, only the **Outer fluid** option is available, but for a **Pipe**, there is also the **Inner fluid** option. If the pipe is filled with a fluid, the **Sink at center** check box can be used. If it is on, the bulk waves traveling from one side of the inner pipe wall through the fluid-filling to the other side are absorbed such that the (numerous) fluid modes are suppressed, and only the pipe modes are left (see example in Sec. 5.6). No fluid-loading options are available

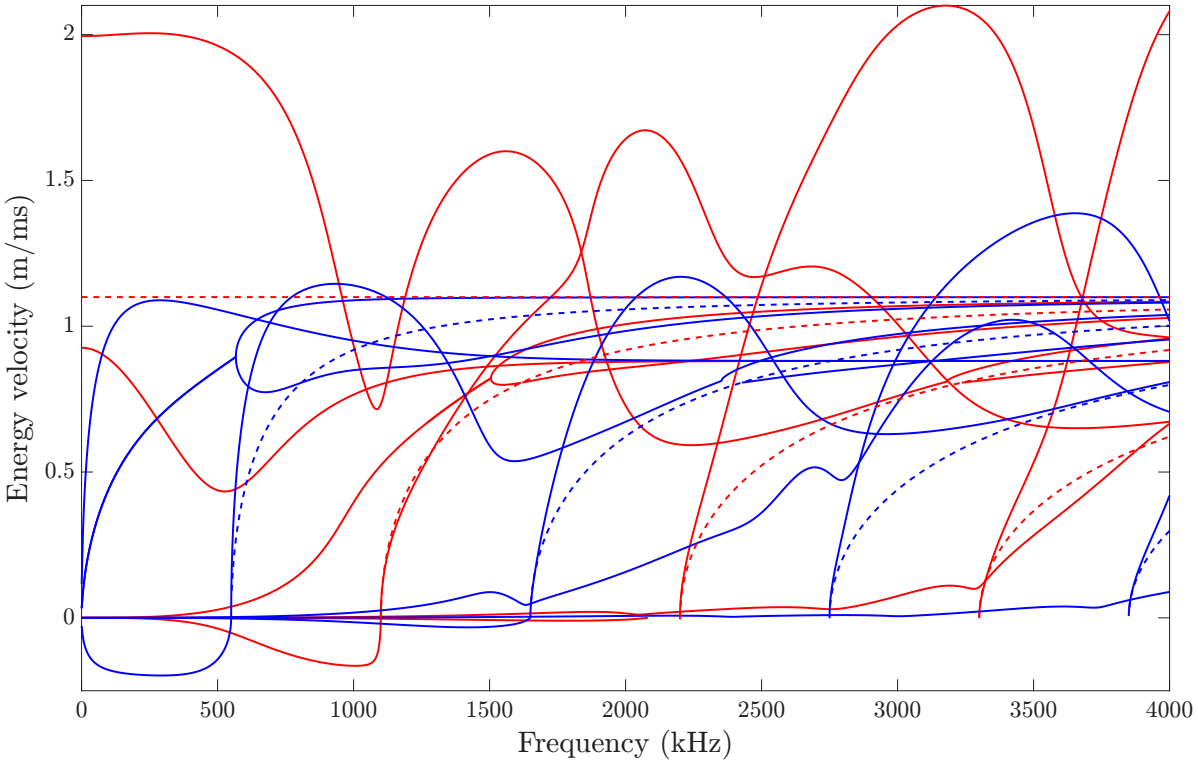


(a)

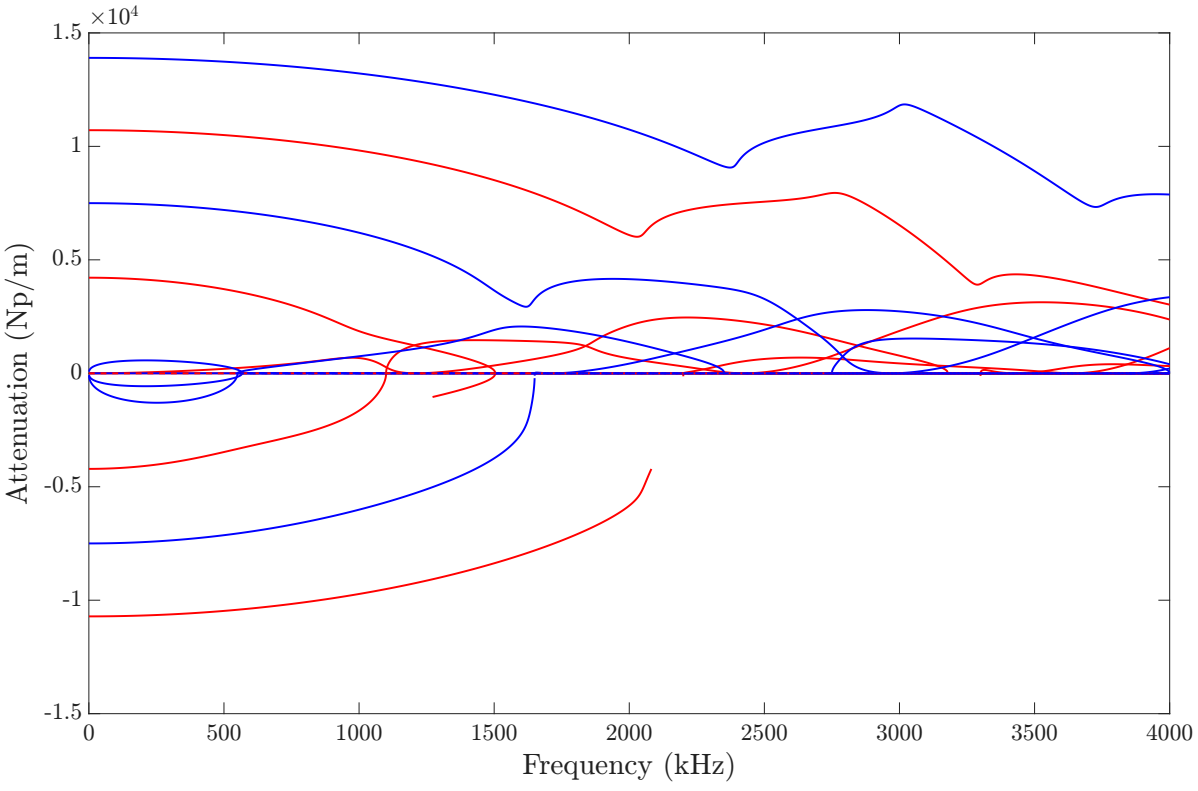


(b)

FIG. 3. Phase velocity dispersion diagrams for 1 mm thick plates: (a) free aluminum alloy 1100 plate and (b) epoxy plate immersed in water. Solid lines indicate (leaky) Lamb modes, dashed lines are shear horizontal modes. Red curves represent symmetric and blue ones anti-symmetric modes.



(a)



(b)

FIG. 4. (a) Energy velocity and (b) attenuation dispersion diagrams for a 1 mm thick epoxy plate immersed in water. Shear horizontal and Scholte modes are not damped.

in case of **Circumferential** wave propagation.

When a half-space is filled with a fluid, energy leaks from the waveguide into the fluid such that the guided waves suffer attenuation as they propagate. Hence, the wavenumber is complex and the dispersion curve tracing is much more demanding compared to the case where the waveguide is surrounded by a vacuum. Without damping, the wavenumber is real, and DC performs only a one-dimensional search on the real wavenumber axis instead of a two-dimensional search in the complex wavenumber plane. Notice that a fluid-filling inside a pipe alone does not cause attenuation because the energy remains in the system. The dispersion curve tracing algorithms in the cases with and without fluid-loading are discussed in Sec. 4.9. Therefore, apply fluid-loading only if you are interested in the damping explicitly and if the fluid has a notable effect on the propagation of the guided waves. The effect of the fluid becomes the more significant the "heavier" the fluid-loading is, i.e., the greater the mass density of the fluid is with respect to that of the plate. For instance, guided wave dispersion curves for a metal plate immersed in air will be hardly different from those for a free metal plate. Skip the fluid-loading in this case because the dispersion curve tracing is much faster and more robust. Figure 3(a) shows the dispersion diagram for a free (in vacuum) aluminum alloy 1100 plate. The opposite situation is present when you have water as a surrounding medium and a low-mass density material such as a (carbon fiber-reinforced) polymer. The guided waves will be strongly damped, and the dispersion curves will look completely different from the free case. Figure 3(b) shows such a heavily damped case phase velocity dispersion diagram, namely that for an epoxy plate immersed in water. The corresponding energy velocity and attenuation dispersion diagrams are plotted in Figs. 4(a) and 4(b), respectively. Interestingly enough, there are modes with negative energy velocity. These are backward propagating modes featuring opposite signs in their phase and energy velocities. Backward propagating modes have negative attenuation since the propagation direction is defined with respect to the phase velocity, not the energy velocity. Another remarkable mode type are the nonpropagating modes. These represent localized disturbances which do not transport energy along the waveguide [58]. Their imaginary wavenumber part does not describe attenuation of energy, but simply the spatial shape of the field. Looking in Fig. 3(b), the nonpropagating modes start in the origin, and then rise more or less steeply until they connect to the propagating modes at the quasi-cut-off frequency. Their localized nature expresses itself by the extreme values of quasi-attenuation, as illustrated in Fig. 4(b). There exists an infinite number of nonpropagating modes. By increasing **Attenuation limit** (see explanation further below), more and more of them can be found and traced. Notice also that Scholte and shear horizontal modes are not damped in this perfectly elastic case. For shear horizontal modes, the reason is that they do not couple to the inviscid water surrounding the plate. Scholte modes can never propagate faster than the speed of sound in the corresponding fluid.

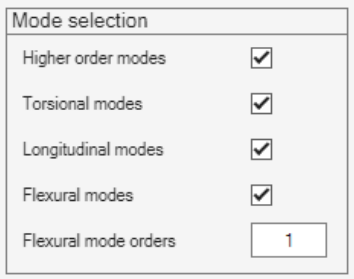
Viscoelasticity is the second physical phenomenon causing attenuation of elastic waves. Viscoelasticity means that vibrational energy is dissipated as heat, thereby reducing the amplitude of the vibration with propagated distance. Its effect on attenuation is generally smaller than that caused by fluid-loading with water (but not necessarily with air). Technically, viscoelasticity is expressed by complex stiffness components, where the

imaginary parts account for the damping. DC uses the hysteretic damping model where the damping loss increases linearly with frequency, i.e., the loss per wavelength is constant. Many viscoelastic default materials are available in DC, listed in Appendix C. To distinguish them from the perfectly elastic materials, they carry the suffix "_viscoelastic". Of course, you can model viscoelastic waveguides which are also fluid-loaded.

You now can change the **Computational settings** (A2). The **Frequency limit** determines up to which frequency the dispersion curves are calculated. The **Frequency step** sets the frequency increments in the dispersion curves. The **Phase velocity limit** determines up to which phase velocity the dispersion curves are calculated in case the standard 1-D tracing algorithm is used. The **Phase velocity accuracy** gives the accuracy of the modal solutions that has to be obtained. Bisections are performed until this value is reached (see Sec. 4.9). Notice that the through-thickness profiles and modes shapes, as well as the simulated signals will be the more accurate the smaller the value of the **Phase velocity accuracy** is. If the **Fix** button is switched off, **Frequency limit**, **Frequency step**, and **Phase velocity limit** are set automatically whenever you make changes in the **Specimen** (A1) settings. Otherwise, if the **Fix** button is switched on, the three settings are not changed automatically.

Whenever damping is present in the system, DC uses the 2-D tracing algorithm, and the **2-D tracing settings** (A3) become active. In cases with no damping, the 2-D tracing algorithm can be forced by checking the **Force 2-D tracing** checkbox (for example to trace the nonpropagating modes). The **Phase velocity limit 2** determines up to which phase velocity the dispersion curves are calculated with 2-D tracing. Before tracing dispersion curves, DC performs sweeps at certain frequencies to find modal solutions; these serve as starting points for dispersion curve tracing. **Attenuation limit** defines the maximum attenuation covered by these initial sweeps as the ratio of the imaginary part of the wavenumber to the real part of the wavenumber. Hence, increasing this ratio increases the attenuation search range. However, note that once DC has found a solution and starts tracing a dispersion curve, DC tries to continue tracing until either **Frequency limit** or **Phase velocity limit 2** are reached. DC does not stop tracing when a dispersion curve exceeds **Attenuation limit**. **Sweeps** defines the number of *extra* initial sweeps for finding modal solutions (DC first performs sweeps at the cut-off frequencies found by the frequency sweep, see **Search higher order modes** (A5)). Performing more extra sweeps increases the likelihood to find all modes. By default, DC performs two extra sweeps close to zero frequency, one sweep close to the **Frequency limit**, and two sweeps evenly distributed along the frequency range. The number of those evenly distributed sweeps is two multiplied by **Sweeps**. Beside the three fixed sweeps, you can distribute a maximum number of $2 \cdot 8 = 16$ sweeps. **Sweep sections** defines the number of samples in the real and imaginary wavenumber parts at which the dispersion equation is evaluated. For example, setting **Sweep sections** to 256 means that the dispersion equation is calculated for a quadratic grid of 256×256 wavenumbers on both the positive and negative attenuation sides (ignoring those where k''/k' is larger than **Attenuation limit**), so the total grid size is 256×512. Increasing **Sweep sections** improves the chance to find weakly pronounced minima (modal solutions) in the dispersion equation amplitude, at the cost of processing

time. The maximum sweep grid size is 2048×4096. For more details on the 2-D tracing algorithm go to Sec. 4.9.



In the **Mode selection** (A4) panel, you can select which mode families shall be calculated and whether the **Higher order modes** shall be calculated in addition to the fundamental modes. The mode family names shown in Fig. 2 appear when the **Geometry** is selected as **Plate** or **Circumferential**. When you select **Pipe** or **Rod**, the panel appearance changes according to Table 1, as shown on the left. A specific option here is the **Flexural mode orders** field. In flexural modes F(n,m), an infinite number $n = 1, 2, 3, ..., \infty$ of circumferential wavenumbers can fit into the circumference of a pipe or rod (n is zero for torsional and longitudinal modes). Enter the maximum number of mode families n to be calculated. Some check boxes become disabled under certain conditions, e.g., **Higher order modes** when 2-D tracing is used, **Symmetric modes** and **Antisymmetric modes** in case of non-symmetric fluid-loading on a **Plate** or if **Geometry** is **Circumferential**.

If you want to calculate the **Higher order modes**, a frequency sweep at the **Phase velocity limit** is performed to find their cut-off frequencies. This can be done in the **Search higher order modes** (A5) panel by pressing **Search**. Otherwise, it will be performed automatically upon pressing **Trace Modes** in the **Trace modes and calculate energy velocity** (A8) panel. The **Step** determines the frequency increments in the sweep. The value is set automatically, but you can change it if you are not sure that you have detected all modes. Try a smaller **Step** in this case. For the standard 1-D tracing algorithm, it is essential that you have detected *all* higher order modes. Instead, when using 2-D tracing, this is not critical. The 2-D tracing algorithm first performs sweeps at the found cut-off frequencies, and in contrast to the 1-D tracing algorithm, it can find and start tracing multiple dispersion curves from each sweep. Then, after performing the sweeps at the cut-off frequencies, it performs the abovementioned *extra* **Sweeps**, to find even more modes, which are not present in nondamped cases, or those which have been missed by the previous sweeps. It is import to note that, even if you are considering a damped case, the frequency sweep is always performed at zero attenuation.

The upper output windows (A6) list the detected modes with their cut-off frequencies, and the lower output windows (A7) summarize how many modes of each family have been found.

Now go to the **Trace modes and calculate energy velocity** (A8) panel and press **Trace modes** to compute the dispersion diagram. A window opens, showing the dispersion curve tracing. Meanwhile the **Trace modes** button turns into a **Stop trace** button. Hit it to cancel the dispersion curve tracing. When the dispersion curve tracing is complete, the window closes and a message informs you about the time elapsed. Directly after that, the energy velocity is calculated automatically. During this, the **Calculate ce** button turns into a **Stop calculate** button. Press it to cancel the energy velocity calculation. You can repeat the energy velocity calculation by pressing **Calculate ce**, for example if you want to increase the accuracy by using a higher number of

through-thickness **Samples x3**. As explained in Appendix A, the energy velocity calculation involves an integration of the mode shape through the thickness of the waveguide. Technically, DC uses trapezoidal numerical integration with respect to the number of through-thickness points defined in the **Samples x3** field. Notice that the accuracy of the energy velocity calculation decreases with an increasing frequency-thickness product because the mode shapes become more complex. Therefore, choose a sufficiently high number of samples. There are some exceptions where DC does not use the numerical integration method, but uses analytic equations (derived using Eq. (10)) for the energy velocity calculation instead. In the exceptions listed below, the prerequisite is that no damping is in the system:

- Lamb and SH waves in plates
- Torsional and longitudinal waves in rods
- Torsional waves in pipes

In case only these mode families are calculated, the **Samples x3** field becomes disabled. The advantage of using the analytic equations is that they compute very fast and that the result is accurate in the limit of the accuracy of the phase velocity dispersion curves.

Now you can go to the **Dispersion diagrams** (A9) panel. Use the **Quantity** drop-down menu to choose between the **Phase velocity**, **Energy velocity**, **Propagation time**, **Coincidence angle**, **Wavelength**, **Wavenumber**, and **Attenuation**. The option below **Quantity** switches depending on what kind of **Quantity** you have selected. If you have selected **Phase velocity** or **Energy velocity**, you have a check box to choose whether the **Bulk velocities** should be displayed in the dispersion diagram or not. If you have selected **Propagation time**, you can enter the **Distance** for which the **Propagation time** shall be calculated. Finally, if you have chosen **Coincidence angle**, you have a drop-down menu that allows to choose the **Couplant** for which the coincidence angle shall be calculated. The **Coincidence angle** angle is given with respect to the normal to the plate. Use the **X-axis mode** drop-down menu to draw the dispersion curves as a function of the frequency in kHz, MHz, or frequency-thickness product in MHz mm. Set the **X-axis**, e.g., [0 1000], and the **Y-axis** limits, e.g., [0 20]. Upon pressing **Plot**, the dispersion diagram is plotted in a new window. In the menu bar of the dispersion diagram, you can export the plot by going to **File** → **Save**, or **Save As**, or **Export Setup**. The **Show modes** menu allows you to show/hide dispersion curves easily. Use the **Analyze** menu to read values at constant frequency, phase velocity, wavelength, etc. Select the quantity which shall be kept constant from the drop-down menu and enter the value into the field on the right. A line will be drawn and all intersections with the dispersion curves are indicated with the corresponding coordinates.



Click on the three dots in the upper right corner of the axes to expand the axes toolbar. It offers up to eight tools, depending on the plot type (see on the left). The first symbol opens a drop-down menu containing three more symbols. The floppy disk symbol allows to export tightly cropped png and pdf-images, a

very useful tool if you want to insert the plots into publications. The other two symbols below copy the image to the clipboard as a pixel or as a vector graphic, respectively. If you want to remove data or replace them by NaN, toggle the **Brush data** tool and left-click on a data point or drag an rectangle over the data while pressing the left mouse button. Then right-click on the selected data and choose an action from the pop-up menu. Note that data are removed only from the plot, but not from the storage, i.e., your changes will be lost once you close the plot window. Activate the **Data tips** and click on a curve to display actual values. You can drag the data tip along a dispersion curve. Move the curves by using **Pan** and **Zoom**. Use **Restore view** to reset the plot.

Now go to the **Through-thickness profiles** (A10) panel. This is a useful tool, which allows you to plot the components of **Displacement**, **Stress**, **Strain**, **Energy density**, and **Power flow density** for a selected **Mode** at a selected **Frequency** or in a live-representation where the profiles are updated as you slide along the dispersion curves. All these quantities change through the thickness of the waveguide and with the frequency-thickness product. For instance, Figs. 5(a) and 5(b) show the through-thickness **Displacement** and **Stress** field components of the A_0 Lamb wave at 10 MHz propagating in a 1 mm thick free aluminum alloy 1100 plate, and in Figs. 5(c) and 5(d) the same is plotted when the plate is immersed in water. The latter plots show the behavior also in the half-spaces above and below the plate to a depth of 1 mm on both sides. For the immersed case, the corresponding **Phase** information is plotted in Figs. 5(e) and 5(f). In all these plots, $x_3 = 0$ mm on the ordinate corresponds to the top surface of the plate, and $x_3 = -d$ (plate thickness d) corresponds to the bottom surface of the plate. The amplitudes, respectively, phase angles are given to the left and right. For the calculation of these amplitudes and phases, the following three conventions are adopted from DISPERSE:

1. **Phase normalization of amplitudes**. The field components of **Displacement**, **Stress**, and **Strain** vary harmonically both in time and space. Therefore, they have amplitudes and phases. For each component of **Displacement** u_i, **Stress** σ_{ij}, and **Strain** ε_{ij}, $i, j = 1, 2, 3$, the **Phase** at the second point from the top surface is taken and the amplitudes at rest of the points are plotted with this **Phase** (the second point is taken because the amplitude at the surface can be zero sometimes). For instance, suppose that $\varphi_{u_{1(2)}}$ is the **Phase** of $u_{1(2)}$ at the second point from the top surface, then all **Phase** normalized points $\hat{u}_1$ are calculated by $\hat{u}_1 = u_1 \mathrm{e}^{-\mathrm{i}\varphi_{u_{1(2)}}}$. This is done for each component separately, e.g., $\hat{\sigma}_{22} = \sigma_{22}\mathrm{e}^{-\mathrm{i}\varphi_{\sigma_{22(2)}}}$. If fluid-filled half-spaces are involved, then the second point from the top of the top half-space is used as a reference, except for components which are zero in the fluid, which, for example, is the case in plates for u_2, σ_{23}, σ_{13}, σ_{12}, ε_{23}, ε_{13}, ε_{12}. In this case, the second point from the top surface of the immersed plate is used, as we had without fluid-loaded half-spaces. The reference point now has zero **Phase**, i.e., it lies on the positive real axis of the complex plane. The other points, however, are still complex in general. DC plots only their real parts, i.e., their projections onto the real axis in a consistent manner. The **Energy density** and **Power flow density**

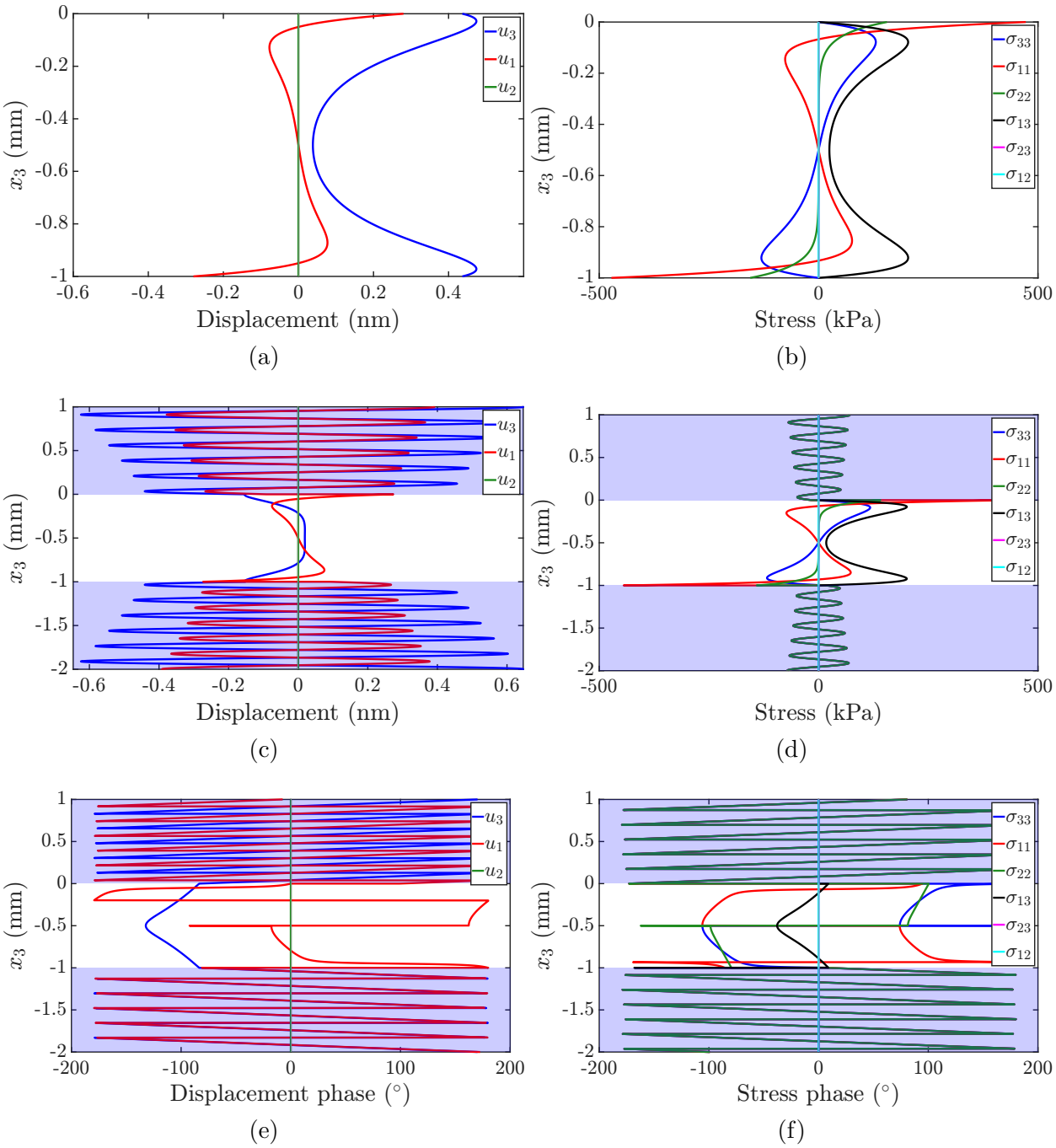


FIG. 5. Displacement (left-hand) and stress (right-hand) field components of the A_0 Lamb wave at 10 MHz in a 1 mm thick aluminum alloy 1100 plate with stress free surfaces (vacuum) (a), (b) and immersed in water (c)-(f). (e) and (f) show the phase information. Leaky modes grow exponentially in the fluid-filled half-spaces. This becomes more obvious when plotting more deeply into the half-spaces.

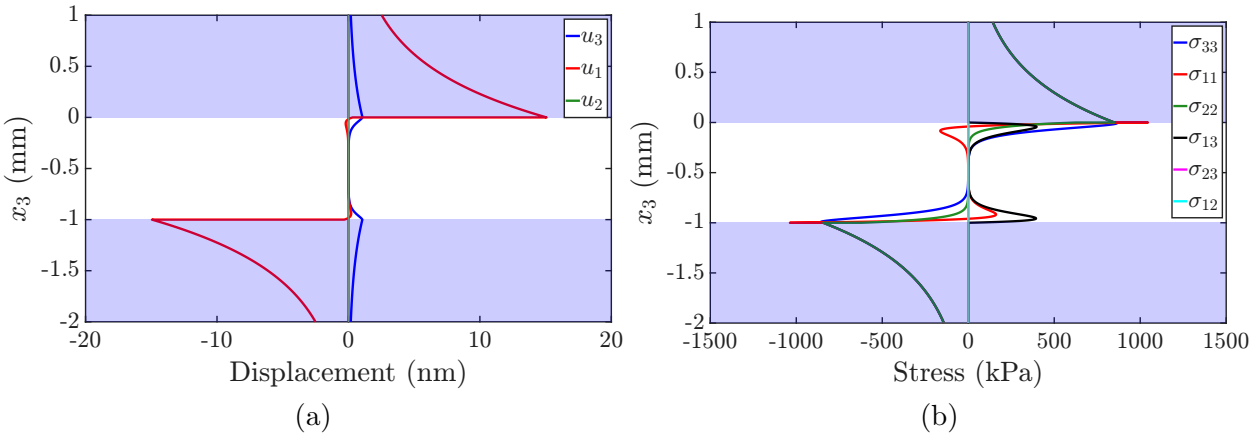


FIG. 6. (a) Displacement and (b) stress field components of a Scholte mode in a 1 mm thick aluminum alloy 1100 plate immersed in water. Scholte modes decay exponentially in the fluid-filled half-spaces.

components are already real-valued because they are calculated using integrals over a complete cycle, according to Eqs. (12), (13), and (14) in Appendix A, so there is no **Phase** information to consider.

2. **Normalization of phases**. The phases of the **Displacement**, **Stress**, and **Strain** components are arbitrary, but the **Phase** differences between them are useful information. As a reference point, the top point of the in-plane **Displacement** component u_1 is taken, no matter if half-spaces are involved or not. The **Phase** at this point is set to zero, and the **Phase** differences of all points of all components are plotted in degrees with respect to it.

3. **Power normalization**. The amplitudes of the field components are calculated by summing up the contributions of the superposing bulk waves, and since their amplitudes are scaled arbitrarily, power normalization is done so that the field components can be compared between different points on the dispersion curves. Therefore, all amplitudes are calculated for guided waves carrying a power flow of one Watt in the propagation direction (x_1 in plates, z in rods and pipes, θ for circumferential waves). The power flow in a plate is per meter normal to the wave propagation direction (x_2). For modes in a rod or pipe, it is for the whole circumference, and for circumferential waves, it is per meter normal to the propagation direction (z). The calculation of the power flow is done using Eq. (14). For instance, to get the power normalized displacements, the displacements are divided through the square root of the power flow in the propagation direction P_1 (in a plate). Similar as in DISPERSE, the power normalization is not strictly right in case of fluid-loading because the power flow in the fluid is neglected. The developer of DISPERSE, Prof. Michael Lowe, describes the benefit of power normalization as follows:

> "[...] once we normalise the mode shapes to power flow then we can do some meaningful comparison from one point on a curve to another, provided our understanding of the comparison is in that context only - e.g., if the mode has unit power flow, then it has a displacement at frequency xxx that is yyy times bigger than it would be at frequency zzz."

Asked about the dimensions of the through-thickness energy density and power flow density, Prof. Lowe explains:

> "The basis for unit power normalisation is that the steady state guided wave carries 1 Watt of energy past the observation point. Clearly the amount of energy depends on the width of the wavefront, so we choose to say that the 1 Watt is per metre width of wavefront (thus units are W/m). This is how this approach was set up by Auld when he was working on guided wave scattering and how his theory of reciprocity contributes to that. Normalisation of the power flow in this manner simplifies the equations for reciprocity and scattering. More detail of that in his book [59] if you want to follow it up.
>
> The strain energy density is a local quantity that is evaluated at any chosen location in the field of the cross section. So this has to be J/m^2. If you integrate this through the thickness of the plate and then across 1 m of wavefront, then you get J/m. This is the energy of the wave per metre width. The rate, per second, of this energy passing the observation point is then W/m, i.e. as we had for the paragraph above."

Displacement is given in nm, **Stress** in kPa, **Strain** is dimensionless, **Energy density** in J/m^2, and **Power flow density** in W/m.

A note on the accuracy of the **Through-thickness profiles** (A10) should be made. Notice that the calculation of the field components depends on the accuracy of the solutions of the dispersion curves. Errors occur if the dispersion curves have not been obtained with high accuracy. These errors become more pronounced at higher frequency-thicknesses. If you need high accuracy profiles, set a smaller value for **Phase velocity accuracy**. Dispersion curve tracing can be slower because it takes more bisections to reach the desired accuracy.

Use the **Quantity** drop-down menu to choose between **Displacement**, **Stress**, **Strain**, **Energy density**, and **Power flow density**. Select the **Mode** you want to analyze from the drop-down list. Set the **Frequency** at which to analyze the selected mode. **Samples x3** determines the number of sample points over the thickness of the waveguide at which the selected quantity is calculated. If you are considering fluid-loading, the **Half-spaces** check box is enabled. With the check box, you can switch the visualization of the **Half-spaces** on and off. In the edit field, you can set how far the **Half-spaces** above

and below the waveguide shall be shown in multiples of the waveguide thickness. For rods and pipes, the bottom of the plot represents the center of the waveguide. Therefore, only the outer half-space is affected by this option. The fluid-filling in a pipe is not a half-space. The **Phase** check box is only enabled if you have selected **Displacement**, **Stress**, or **Strain** in the **Quantity** drop-down menu. Press **Plot** to plot the **Through-thickness profiles** in a new window. If you have checked the **Phase** check box, an extra plot with the phase angles will be shown in addition to the amplitude plot. To the left of the **Plot** button are six check boxes from which you can select which components of the selected **Quantity** shall be plotted. This facilitates analyzing a single component that is otherwise hidden in a plot crowded with up to six curves. In case you have chosen **Displacement**, **Energy density**, and **Power flow density**, only three check boxes are enabled. The labels of the check boxes change according to the selected **Quantity**.

Let us discuss the **Displacement** in a plate. u_1 is the in-plane component along the propagation direction (x_1) of the guided wave. u_2 is in-plane but perpendicular (x_2) to the propagation direction, usually referred to as *shear horizontal*, and u_3 is the out-of-plane or normal **Displacement**. In isotropic materials, there exist only pure Lamb and shear horizontal waves. Pure Lamb waves have only nonzero u_1 and u_3 components (see Fig. 5(a)), and pure shear horizontal modes have only a nonzero u_2 component. A symmetric Lamb wave has the same **Displacement** amplitude u_1 (sign and absolute value) at the top and bottom of the plate, while an anti-symmetric Lamb wave has the same amplitude u_1 at the top and bottom, but with opposite signs, as depicted in Fig. 5(a). The character of a shear horizontal wave is determined by the same rules applied to u_2. Figure 5(b) displays the four nonzero and the two zero stress components present in a pure Lamb wave. The components σ_{i3} vanish at the surfaces, i.e., the surfaces are traction-free, and no energy can leak into the surrounding vacuum. The **Displacement** component u_1, the **Stress** component σ_{11}, and the **Strain** component ε_{11} at the top are always taken positive (to the right) as a convention. Let us now consider the fluid-loaded case shown in Figs. 5(c) and 5(d). The **Half-spaces** are shown up to a depth of one plate thickness above and below the plate. The **Stress** component σ_{33} does not vanish anymore at the plate's surfaces, so energy is transmitted into the water. However, σ_{13} and σ_{23} still vanish because non-viscous fluids such as water do not support shear stresses. For the same reason, only σ_{11}, σ_{22}, and σ_{33} are nonzero in the fluid, having all the same amplitude. Although counter-intuitive, the amplitude of leaky waves grows exponentially with the distance normal to the waveguide. Plotting deeper into the **Half-spaces** than in Figs. 5(c) and 5(d) (and in Figs. 6 and 8) shows this more clearly. The reason for this is that, the further we go away from the plate, the earlier have the rays been emitted from the plate, when the amplitude of the guided wave propagating in the waveguide was higher [40]. Figure 6 displays a Scholte mode for comparison. The difference is that the Scholte mode decays exponentially outside the waveguide. A shear horizontal wave does not couple to a non-viscous fluid surrounding the plate since its only non-zero stress components are the shear stresses σ_{23} and σ_{12}.

A particularly useful facility is the **Live-through-thickness profiles** window displayed in Fig. 7. It opens when you press the **Live-plot** button. The window contains all five through-thickness profile quantities together with two dispersion diagrams from

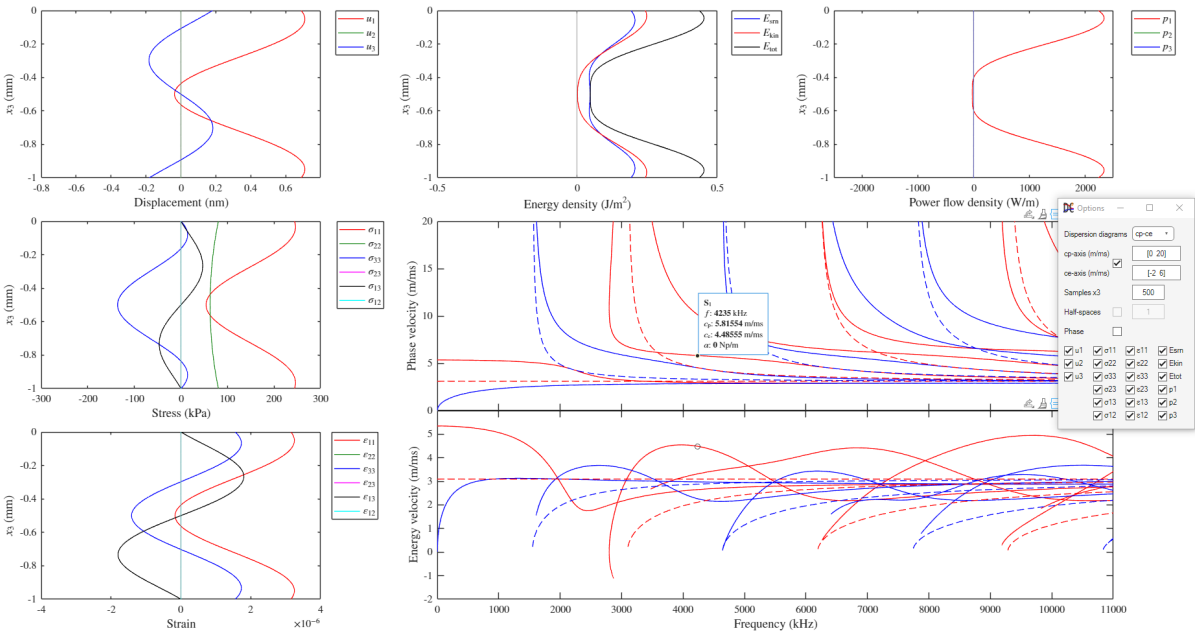


FIG. 7. The Live-through-thickness profiles window.

which you can select the data. When selecting a point in a dispersion diagram, the corresponding point in the other dispersion diagram is indicated by a marker. As you move the cursor along a dispersion curve, the through-thickness profiles are updated automatically. The **Options** panel appears on the right-hand side, but it can also be opened from the **Options** button in the window menu bar (not visible in Fig. 7). From the **Dispersion diagrams** drop-down menu, you can select from three combinations, namely phase and energy velocities, **cp-ce**, phase velocity and attenuation, **cp-α**, and energy velocity and attenuation, **ce-α**. The Y-axis limits in the dispersion diagrams can be adjusted automatically and manually. To adjust them manually, check the check box on the left-hand side between the two editable fields. **Samples x3**, **Half-spaces**, and **Phase** work the same way as explained above. The check boxes below allow to select which field components shall be plotted.

In the **Mode shape** (A11) panel, you can visualize the **Displacement** pattern (mode shape) in a waveguide and in the adjacent **Half-spaces** as they are deformed by continuously propagating guided waves. This is particularly helpful to get an idea of how a certain mode looks like at a certain frequency. For example, Fig. 8 shows the mode shape of ten wavelengths of the A_4 Lamb wave at 10 MHz mm in an aluminum alloy 1100 plate immersed in water. Notice the wave fronts in the water and how the Lamb wave is damped through the loss of energy to the water. Or check out Figs. 9(a) and 9(b), illustrating the mode shape of the L(0,1) mode at 450 kHz in a 10 mm diameter, 1 mm wall thickness aluminum alloy 1100 pipe filled with diesel oil and immersed in water. As in the **Through-thickness profiles** (A10) panel, the bottom in Fig. 9(a) represents the center of the pipe. Again, the guided wave suffers attenuation due to energy leakage into the water (the diesel oil inside the pipe does not cause attenuation). Figure 9(b) shows a

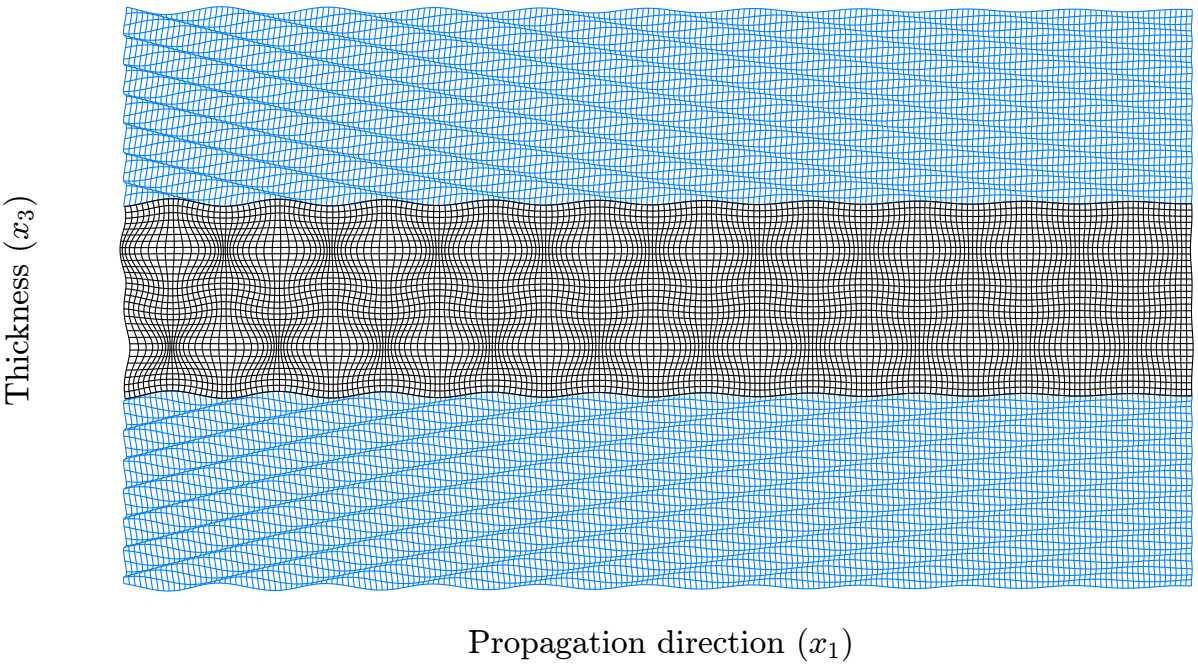
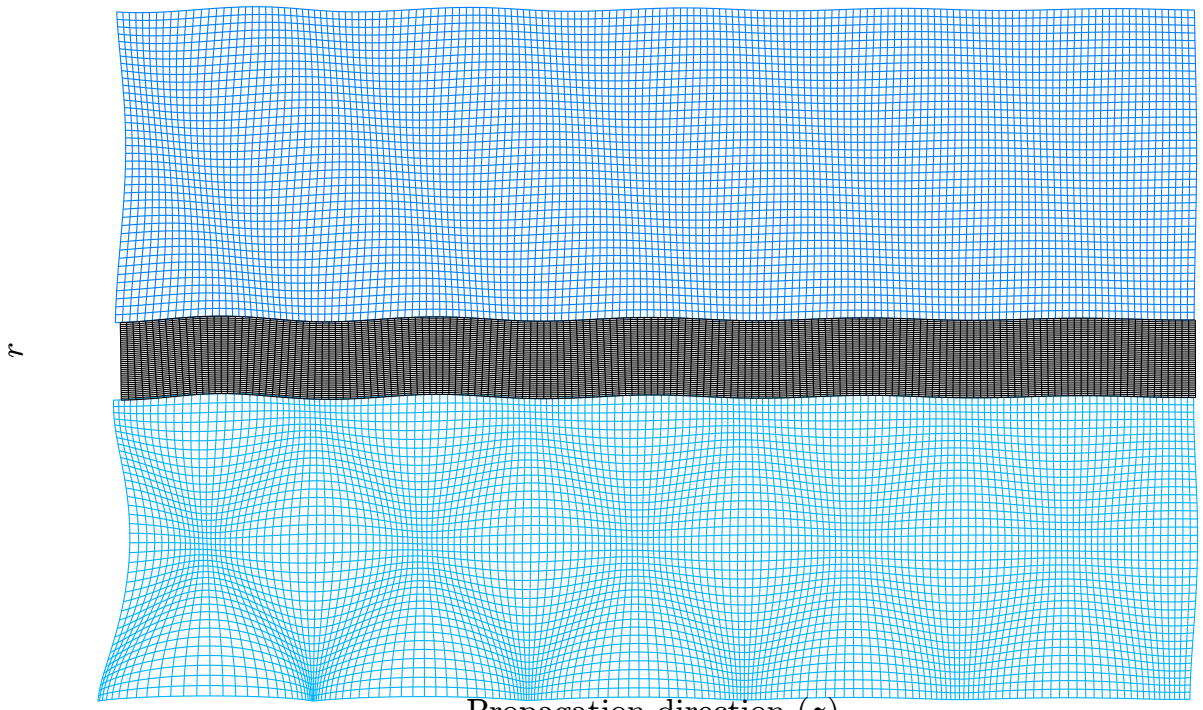
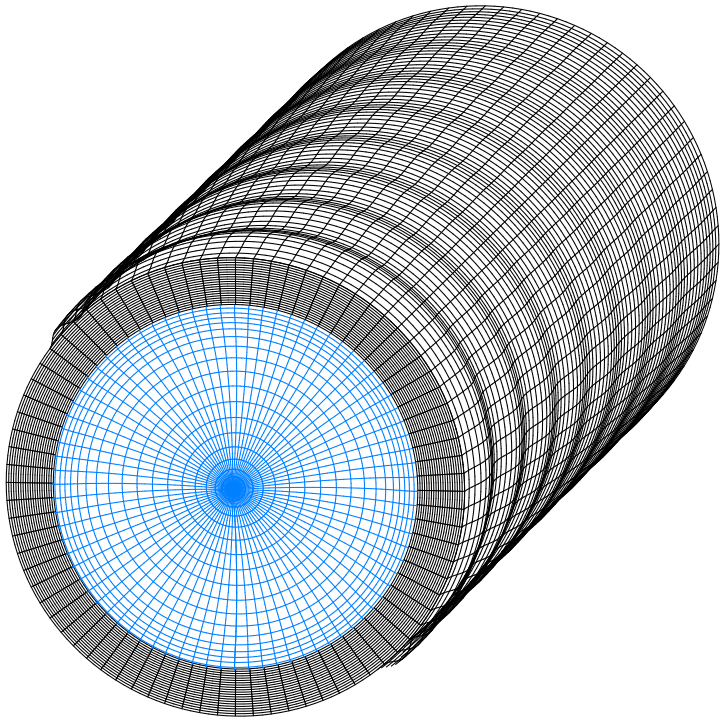


FIG. 8. Mode shape of the A_4 Lamb wave at 10 MHz mm in an aluminum alloy 1100 plate immersed in water.

three-dimensional model of the pipe. The outer half-space is not plotted here (although it can be plotted) to see the pipe from outside. Of course, the **Displacement** is highly exaggerated in these plots. First, select the **Mode** you want to analyze from the drop-down list. In case you have selected **Rod** or **Pipe** as a **Geometry**, you can select the view type from the **plane** drop-down menu. Select **r-z** to plot the normal cross-sectional plot through the thickness of the waveguide. Choose **r-θ** for a three-dimensional plot, which is the most realistic representation of the waveguide. Set the **Frequency** at which to analyze the selected mode. **Wavelengths** determines how many wavelengths along the propagation direction are displayed. **Samples x1,x3** determines at how many sample points along the propagation direction, respectively, through the thickness, the mode shape is calculated. If you are considering fluid-loading, the **Half-spaces** check box is enabled. With the check box, you can switch the visualization of the **Half-spaces** on and off. In the edit field, you can set how far the **Half-spaces** above and below the waveguide shall be shown in multiples of the waveguide thickness. For rods and pipes, only the outer half-space is affected by this option. In the **Grid line every** field, you can specify how many grid lines shall be drawn in the mode shape. This number is updated automatically when you change the number in **Samples x1**. Use **Gain** to magnify the **Displacement** for clarity. Check **Undistorted** to toggle the undistorted grid. Go to the **Animation settings** sub-panel to set up an animation of the grid. The **Cycle duration** determines how long an animated cycle takes. Adjust the **Frame rate** of the movie. The **Movie quality** affects only the quality of the exported movie. Set a value between zero and one hundred. If you check **Animate** and press **Plot**, the animation starts running in an extra window until you close that window. If you leave



(a)



(b)

FIG. 9. Mode shape of the L(0,1) mode at 450 kHz in a 10 mm diameter, 1 mm wall thickness aluminum alloy 1100 pipe filled with diesel oil and immersed in water. (a) Cross sectional view ($r - z$). (b) Three-dimensional view ($r - \theta$).

Animate unchecked, only a static grid is plotted.

If you want to export plots and movies in an easy way, you can use the **Export settings** (A12). Use the check boxes **PDF** and **PNG** to specify the format of the exported plot. **PDF** is a vectorized graphic that I used for the plots made by DC for this manual. **PNG** in contrast is a rendered graphic for which you can set the **PNG resolution** in dots per inch (dpi). The better performance of a vectorized graphic becomes obvious upon zooming in. Compare Figs. 9(a) and 9(b), which are a **PDF**s, to one of the DC screenshots, which are **PNG**. The font in all plots made by DC is the standard LaTeX-font "Computer Modern". Type the **File name** and the **Directory** to specify where the plot/movie is exported to. If you check **Export plots** and then press any of the three **Plot** buttons (their labels change to **Export**), the respective plot will be exported. If you check **Animate** in (A11) and then press the **Export** button, the animation is displayed in a new window. Once all frames are captured, the window closes automatically and the movie be exported. The movie consists of five repetitions of the calculated cycle.

If you want to perform your own processing on the dispersion curves, you can export them. Enable the **Dispersion curves** check box and select from the top drop-down menu to the right which dimension the frequency axis should have, namely **kHz**, **MHz**, or **MHz·mm**. Then press the ***.mat**, ***.xlsx**, and ***.txt** buttons to export the data in the specific file format in the specified **Directory**. For each mode, phase velocity ($\mathrm{m\,ms^{-1}}$), the energy velocity components $c_{\mathrm{e}1}$ and $c_{\mathrm{e}2}$ ($\mathrm{m\,ms^{-1}}$), energy velocity magnitude $|\vec{c}_{\mathrm{e}}|$ ($\mathrm{m\,ms^{-1}}$), energy velocity skew angle (°), propagation time (µs), coincidence angle (°), wavelength (mm), wavenumber ($\mathrm{rad\,mm^{-1}}$), and attenuation ($\mathrm{Np\,m^{-1}}$) are exported. For a discussion on the energy velocity in anisotropic specimens, read Appendix A. In the second drop-down menu, you can select **Horizontal arrangement** or **Vertical arrangement** of the mode data in the files. You can also export the through-thickness profiles selected in the **Through-thickness profiles** (A10) panel. Enable the **Through-thickness** check box, and upon pressing the ***.mat**, ***.xlsx**, and ***.txt** buttons, the selected through-thickness **Quantity** of the selected **Mode** at the chosen **Frequency** is exported to the desired file format. If you have checked the **Phase** check box in the **Through-thickness profiles** (A10) panel, the phase information will be exported, too.

Adjust the **Plot layout settings** (A13) according to your preferences. For the plots in this manual, I switched the **Title** off. Use the **Legend location** drop-down menu to place the legend in the **Through-thickness profiles** (A10) inside or outside the axes. You can also adjust the **Box line width** and the **Curve line width**. Only for the dispersion curves, you can adjust the **Dispersion curve colors**, for symmetric modes in the field **S**, for anti-symmetric modes in the field **A**, and for non-symmetric modes in **B**. If you have selected **Rod** or **Pipe** as a **Geometry**, you can set the color for torsional and longitudinal modes in the **T,L** field and that of the flexural modes with $n = 1$ in the **F** field. If **Geometry** is **Circumferential**, the mode color can be set in the **C** field. You can compose any color by setting the proportions of red, green, and blue. Set a value between zero and one for each proportion, e.g., [1 0 .5]. Change the **Font size** of the **Title**, **Axes labels**, **Axes ticks**, **Mode labels**, and of the **Legend**. Press

Default, to restore the default settings.

4.3 Anisotropic

The **Anisotropic** tab (see Fig. 10) looks similar to the **Isotropic** tab. Only panels that are new or different will be explained in this section. In the **Anisotropic** tab, you can compute dispersion diagrams and mode shapes for guided waves in multilayered anisotropic plates such as the fiber reinforced composite sketched in Figs. 11(a) and 11(b), as well as in multilayered isotropic rods and pipes, as depicted in Fig. 11(c). In plates, **Hybrid** laminates containing layers of different materials and material symmetry classes are possible, while in rods and pipes only different isotropic layers can be set up, currently. Available material classes are **Orthotropic**, **Transversely isotropic**, **Cubic**, and **Isotropic**.

Let's put up some conventions for waveguides, and let us take a composite layup as an example for a **Plate**. Composites are stackings of m layers with layer thicknesses d_m, consisting of **Orthotropic** or **Transversely isotropic** fiber-epoxy combinations. For each layer, we assign a local (crystallographic) Cartesian coordinate system $x'_{i,m} = (x'_1, x'_2, x'_3)_m$ residing on the top of the mth layer, and we define the layers to lie parallel to the x'_1-x'_2-plane. The fibers are oriented along $x'_{1,m}$, while $x'_{3,m}$ is normal to the layer. To

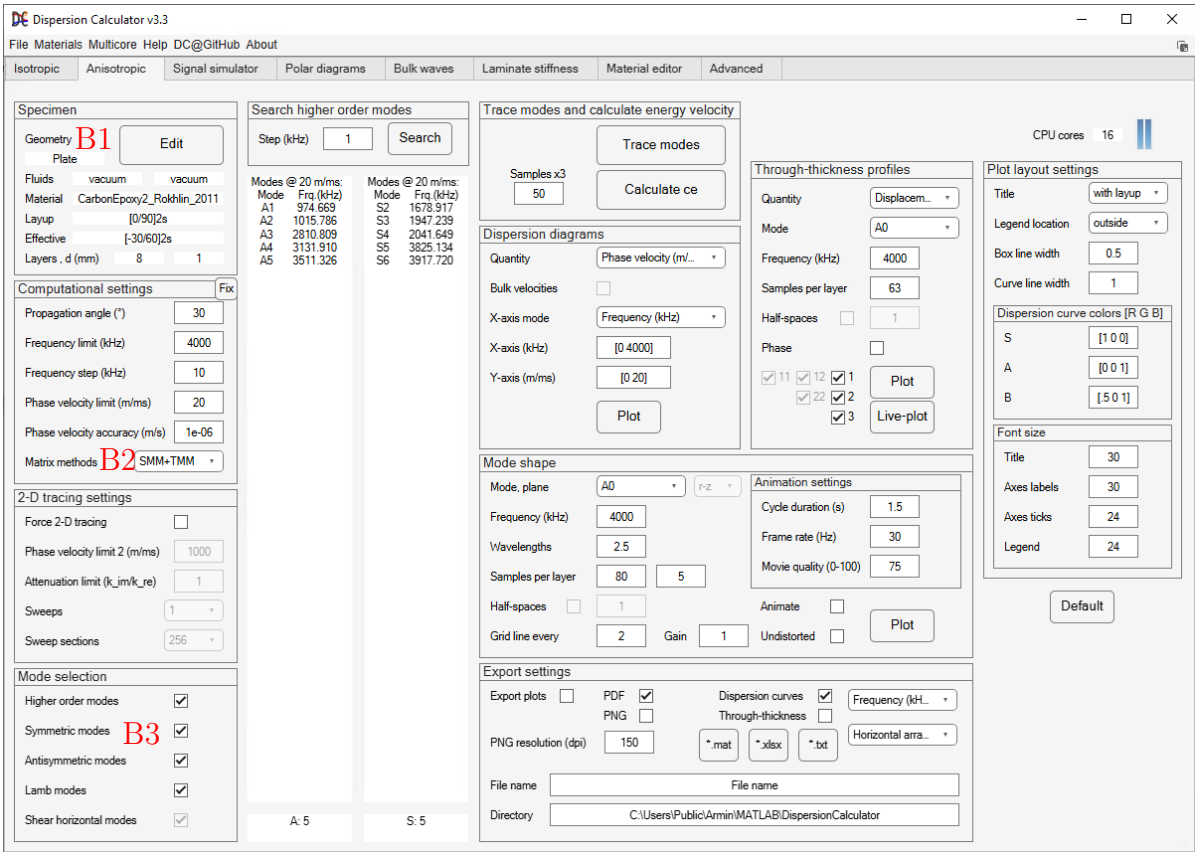


FIG. 10. The tab for the calculation of anisotropic specimens.

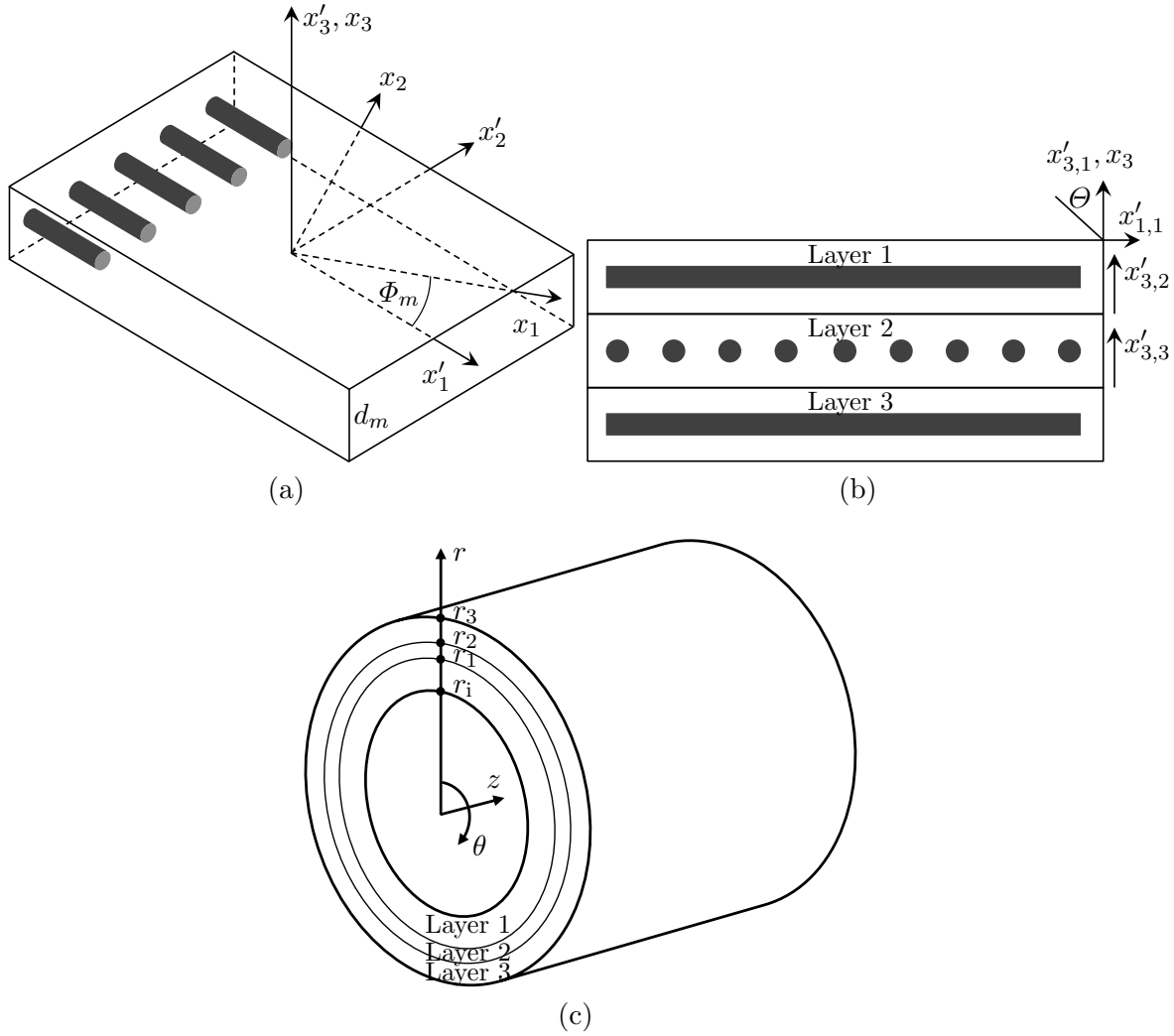


FIG. 11. (a) Single composite layer with local (crystallographic) coordinate system x_i' and global coordinate system x_i, $i = 1, 2, 3$. (b) Layered composite plate with [0/90/0] orientation with respect to the x_1'-axis. (c) Multilayered isotropic pipe with cylindrical coordinate system (r, θ, z).

describe this system with arbitrary layer orientations in a convenient way, we introduce the nonprimed global coordinate system $x_i = (x_1, x_2, x_3)$. We choose it such that the laminate lies in the x_1-x_2-plane, x_1 is the direction of guided wave propagation, and x_3 is the thickness direction of the laminate. With respect to the global coordinate system, the local coordinate systems are yielded by a counterclockwise rotation of an angle Φ_m between x_1 and $x_{1,m}'$ about the x_3-axis. Hence, x_3 and $x_{3,m}'$ coincide.

In cylindrical waveguides (**Geometry** is **Rod**, **Pipe**, or **Circumferential**), a cylindrical coordinate system (r, θ, z) sits at the center ($r = 0$), as depicted in Fig. 11(c). When modeling axial waves (**Rod**, **Pipe**), the propagation direction is z, and for **Circumferential** wave propagation, it is θ. As a cylindrical waveguide can only consist of isotropic layers, there is currently no directional dependency in the layer materials such

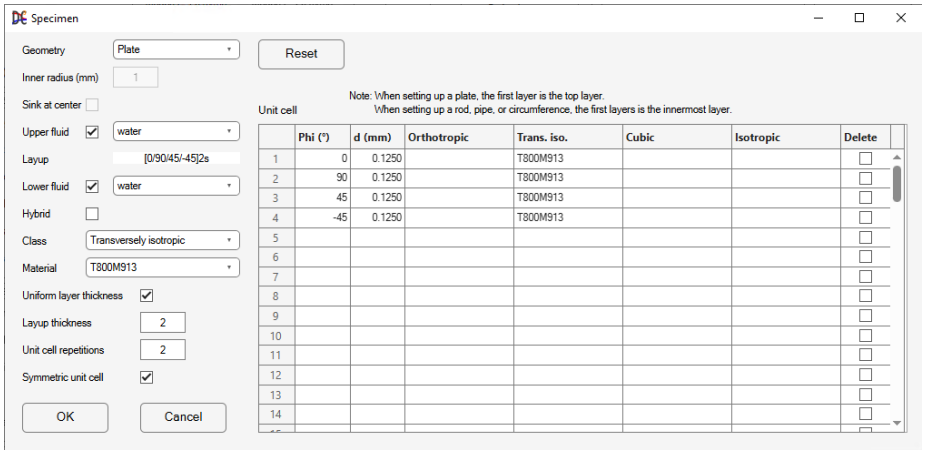
as the winding angle in a composite tube or pressure vessel.

To set up your waveguide, press **Edit** in the **Specimen** (B1) panel shown in Fig. 10. This opens the specimen setup window displayed for three different situations in Fig. 12. First choose from the **Geometry** drop-down menu the kind of waveguide problem you want to solve. The four available geometries can be classified into two categories. **Plate** is the only Cartesian type of waveguide while **Rod**, **Pipe**, and **Circumferential** are cylindrical problems. Note that switching between those two categories in the **Geometry** drop-down menu resets the definition to a standard single layer problem. In case of a cylindrical waveguide, it is an isotropic layer. The other options in the specimen setup window depend on the selected **Geometry**. Only if you choose **Pipe** or **Circumferential** the **Inner radius** field is active (the inner radius of a **Rod** is zero). **Sink at center** is only active when you choose a **Pipe**, and **Inner fluid** is activated. Notice also that the **Upper fluid** and **Lower fluid** drop-down menus, as they are called in case of a **Plate** correspond to the **Inner fluid** and **Outer fluid** menus in cylindrical waveguides; it is not the order one might expect intuitively. This reflects the fact that a plate is built up from top to bottom in the **Unit cell** table while a cylindrical waveguide is built from inside to outside (compare Fig. 11).

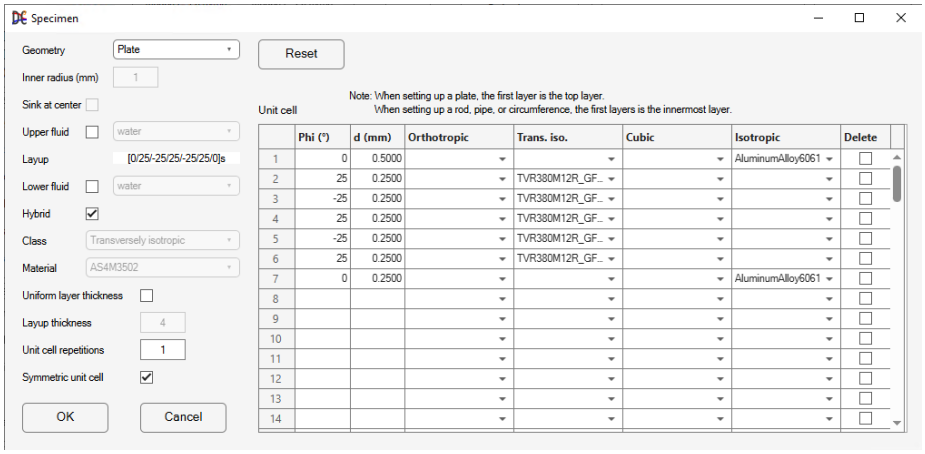
Important note:
When setting up a plate, the first row in the Unit cell table represents the top layer. When setting up a Rod, Pipe or Circumferential, the first row corresponds to the innermost layer.

To make it more clear to the user on which side he sets the fluids, there is the **Layup** field between the **Upper fluid/Inner fluid** and **Lower fluid/Outer fluid** menus. So in a **Pipe**, from inside to outside, there is the **Inner fluid**, then comes the pipe wall (**Layup**), and then comes the **Outer fluid**. When choosing a **Rod**, there can be no **Inner fluid**, of course. In case of **Circumferential**, there can currently be no fluid-loading or fluid-filling at all.

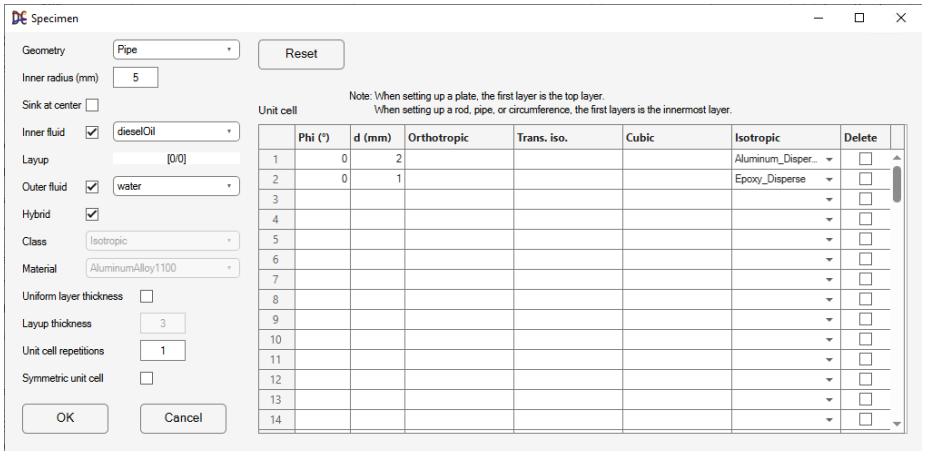
After having selected a **Geometry**, decide whether you want to consider fluid-loading. If you want to add a fluid to the upper half-space/inside of the pipe, check the **Upper fluid/Inner fluid** check box to activate the drop-down menu on the right, and then select a fluid from it. The same goes for the lower/outer half-space via the **Lower fluid/Outer fluid** check box and the respective drop-down menu. Leaving a check box unchecked means that the corresponding space is filled with a vacuum. You can fill one space with a fluid and leave the other one stress free, i.e., filled with a vacuum. Next, decide whether you want to set up a laminate containing layers from only one material, as the case in typical composites, or if you want to create a hybrid laminate containing layers of different materials, like for instance glass laminate aluminum reinforced epoxy (GLARE) [23]. In the first case, leave **Hybrid** unchecked, as shown in Fig. 12(a). Select a material class from the **Class** drop-down menu. According to your choice, the corresponding materials will be listed in the **Material** drop-down menu. Select a material. In the second case, check **Hybrid**, as shown in Fig. 12(b). The **Class** and **Material** drop-down menus become inactive, and in case of a **Plate**, the four columns



(a)



(b)



(c)

FIG. 12. The specimen setup. (a) Multilayered anisotropic composite plate, (b) hybrid laminate (GLARE), and (c) multilayered isotropic pipe.

in the **Unit cell** table named **Orthotropic**, **Trans. iso.**, **Cubic**, and **Isotropic** are now active (they are inactive in case **Hybrid** is unchecked). In case of a **Rod**, **Pipe**, or **Circumferential**, only the **Isotropic** column becomes active, see Fig. 12(c). For each layer, represented by one row in the **Unit cell** table, you can select any material from the active drop-down menus. Any selection will overwrite a preceding selection in the same row. Now, decide if the layup should have **Uniform layer thickness**. If yes, DC deduces the layer thicknesses from the **Layup thickness**, which you can enter below. You will see the layer thicknesses in the **d (mm)** column update automatically when you change the number of layers or the **Layup thickness**. Notice that the **Layup thickness** is the overall layup thickness, not just of the **Unit cell**. The values in the **d (mm)** column can not be edited if you have checked **Uniform layer thickness**. If you uncheck **Uniform layer thickness**, the **d (mm)** column will be active, and you must enter each layer's thickness in millimeters. In this mode, the **Layup thickness** field cannot be edited, but it will show the layup thickness calculated from your entries. Complete your definition of the **Unit cell** by filling the **Phi (°)** column with the layers' orientations. Typically, Phi (Φ) is the fiber direction in composite materials. Therefore, in an isotropic layer, you can leave the angle at 0°. In order to delete a layer, click the corresponding check box in the **Delete** column. The **Unit cell** table offers a large amount of 99 entries. This ensures that even the largest unit cells can be entered. By contrast, unit cells in common laminates are much smaller, often having only four or eight layers. Use **Unit cell repetitions** to specify how many repetitions of the unit cell are contained in the layup. Check **Symmetric unit cell** if the layup is symmetric with respect to the middle plane of the plate/wall. The **Layup** field updates automatically whenever the user edits the **Unit cell** table, **Unit cell repetitions**, or **Symmetric unit cell**. With the **Reset** button, you can reset the specimen definition to a **Plate** containing a single orthotropic layer. Press **OK** to accept the specimen definition or **Cancel** to discard it. The setup window closes and the specimen definition is displayed below the **Edit** button.

A few words should be said about setting up a unit cell in a proper way. With reference to the usual composite layup notation, we write the layup as $[unitCell]_{n(\mathrm{s})}$, where $n = 1, 2, ...$ is the number of unit cell repetitions and "s" stands for symmetric. Basically, the unit cell should be kept as simple as possible in order to maximize efficiency and robustness in the dispersion curve calculation. Let us consider a few examples:

- [0]. Users are tempted to set up a unidirectional laminate by multiple layers, e.g., [0/0/0/0] because this is how the laminate was manufactured. Layups containing layers with all the same orientation and material are not accepted. Replace them by one layer [0] having the layup thickness of the four layers. This is equivalent because DC assumes rigid bonding between the layers. Furthermore, it will be more efficient since the computational expense scales with the number of layers.

- [0/90/45/0/0/0/0]. The same thing, but DC does not prevent you from doing this. Avoid adjacent layers with the same layer orientation and material, but group them into one single layer of greater thickness. Uncheck **Uniform layer thickness** and enter the layers' thicknesses into the **d (mm)** column.

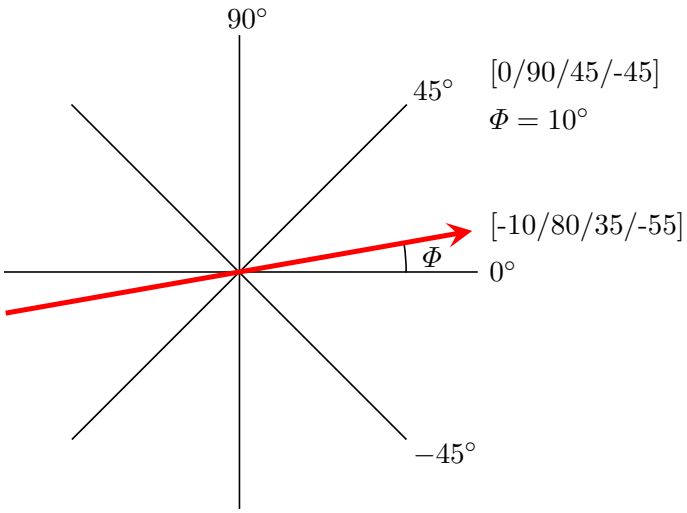


FIG. 13. Wave propagation at a **Propagation angle** of $\Phi = 10°$ in a [0/90/45/-45] **Plate** layup. A positive Φ results in a counterclockwise rotation of the propagation direction (red arrow) vs. the fiber orientations, yielding the **Effective** layup [0/90/45/-45]-10 = [-10/80/35/-55].

- $[0/90]_2$. Do not set up the unit cell as [0/90/0/90], but set it to [0/90] and set the **Unit cell repetitions** to two, similarly as shown in Fig. 12(a). This will be more efficient.

- $[0/90]_s$. Unlike the case in **Rod**, **Pipe**, and **Circumferential**, DC does not let you define symmetric unit cells like [0/90/90/0] or [0/90/0] in a **Plate**. Instead, set [0/90] and check **Symmetric unit cell**, as shown in Figs. 12(a) and 12(b). Only by doing so, DC automatically recognizes that symmetric and anti-symmetric modes can be distinguished (no such distinction exists in cylindrical waveguides). In case of [0/90/0], uncheck **Uniform layer thickness** and divide the thickness of the 90° layer in half in the **d (mm)** column. Such a situation is present in Fig. 12(b). The center layer of this GLARE laminate (taken from Ref. [23]) is a 0.5 mm thick aluminum layer. In row no. 7, it is cut in half.

- [Al/0/0/0/0/0/Al/0/0/0/0/0/Al]. This is another GLARE laminate taken from Ref. [23]. Set this up as $[Al/0/Al]_s$. The corresponding layer thicknesses in this unit cell are 0.5 mm, 1.25 mm, and 0.25 mm, adding up to a 4 mm thick laminate.

Now, go to the **Computational settings** (B2). When considering a **Plate**, you can define the **Propagation angle** Φ for the guided waves with respect to the layer orientations (in case of a **Rod** and a **Pipe**, it is always in the axial direction, z, and in a **Circumferential**, it is in the circumferential direction, θ). Consider for example the [0/90/45/-45] layup sketched in Fig. 13. A positive **Propagation angle** Φ results in a counterclockwise rotation of the propagation direction vs. the fiber orientations, yielding the **Effective** layup, respectively, the angles Φ_m [0/90/45/-45]-10 = [-10/80/35/-55]. The **Phase velocity limit** is set to 20 m s^{-1} by default, and it does not change automatically (except for isotropic single layers). Starting in DC v3.1, you can use the **Matrix**

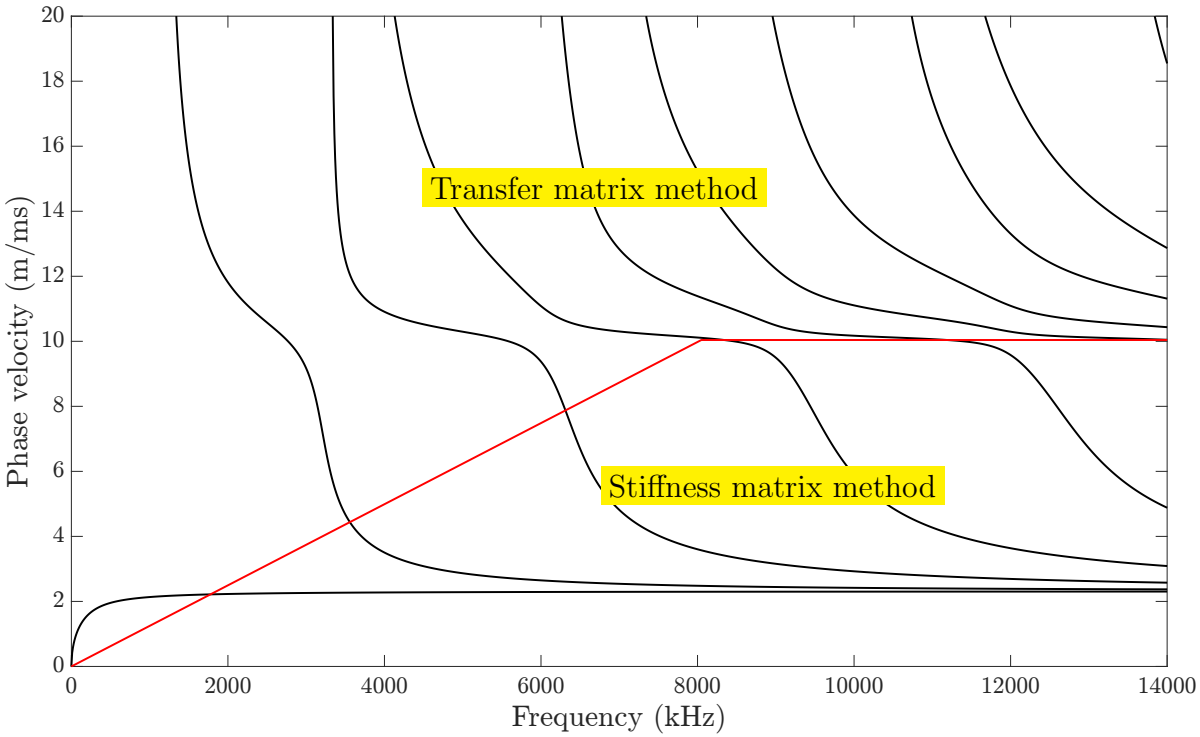


FIG. 14. Generic dispersion diagram. The transfer matrix method is used for dispersion curve tracing above the red line. The stiffness matrix method takes over below the red line.

methods drop-down menu to choose if the transfer matrix method (TMM) shall be used instead of the stiffness matrix method (SMM) whenever there are stable conditions for TMM. Stable conditions for TMM are above the red line in the generic dispersion diagram in Fig. 14. TMM becomes unstable when the bulk waves in the layers turn evanescent. Therefore, before starting dispersion curve tracing, DC performs a phase velocity sweep at the maximum frequency covered by the dispersion diagram (14 000 kHz in Fig. 14). DC starts at a high phase velocity, and as the phase velocity decreases, the bulk waves become evanescent and TMM becomes increasingly inaccurate and unstable eventually. DC determines the phase velocity where the inaccuracy sets in and adds a very significant safety on top. This determines the slope of the inclined part of the red line. The slope accounts for the fact that the inaccuracy range extends (from zero) to lower phase velocities at lower frequencies. However, the inaccuracy range can never exceed that phase velocity above which all bulk waves are homogeneous (not evanescent). This phase velocity is indicated by the horizontal part of the red line. There are four benefits of using TMM instead of SMM above the red line. Firstly, TMM computes faster, in particular when there are many layers. Secondly, the dispersion equation amplitudes obtained with TMM do not have maxima, which when using SMM can impede the tracing algorithm from finding the wanted mimima (see a dispersion equation amplitude mimimum in Fig. 33). Thirdly, TMM produces much more pronounced mimima with quasi-SH modes so that it is more robust than SMM in tracing such dispersion curves. And fourth, TMM is stable at very low frequencies and thicknesses where SMM

TABLE 2. Possible configurations in a fluid-loaded anisotropic plate.

symmetric	decoupled	modes
✓	✓	S, A, S^{SH}, A^{SH}
✓	✗	S, A
✗	✓	B, B^{SH} (S^{SH}, A^{SH})[a]
✗	✗	B

[a] If the half-spaces are covered by different fluids, but the layup is symmetric, S^{SH} and A^{SH} can be distinguished.

becomes unstable. Therefore, the default setting of the **Matrix methods** drop-down menu is **SMM+TMM**. However, it should be noted that using SMM alone (above and below the red line) brings better results sometimes. Choose **SMM** in this case. Try both options in difficult cases.

When computing a **Plate**, some of the check boxes can be inactive in the **Mode selection** (B3) panel, depending on the **Specimen** and on the **Propagation angle** in the following way:

Decoupled case

The decoupling into pure Lamb waves (only sagittal motion (x_1'-x_3'-plane)) and shear horizontal waves (motion only in x_2') depends on material symmetries and on the propagation direction of the guided waves. In case of decoupling, both mode families can be traced separately. Otherwise, sagittal and shear horizontal motion are coupled, and no separation into Lamb and guided shear horizontal modes is possible.

- **Orthotropic and transversely isotropic**. Decoupling occurs for wave propagation parallel or normal to the fibers ($\Phi = 0°, 90°$). This is possible in a single layer or in a cross-ply laminate $[0/90]_{n(s)}$, $n = 1, 2, ...$
- **Cubic**. Decoupling occurs for guided wave propagation along $\Phi = 0°, 45°, 90°$.
- **Isotropic**. Decoupling occurs for all propagation directions.

In case we have a hybrid layup, the whole system allows decoupling only in case all contributing layers support decoupling in the given situation.

Symmetric system

In symmetric systems, DC traces symmetric and anti-symmetric (leaky) Lamb and shear horizontal waves separately. If the system is non-symmetric, so are the modes and no separation is possible. The only exception from this are shear horizontal modes. If the layup is symmetric, but the upper and lower half-spaces are covered by different

fluids, the shear horizontal modes are still symmetric and anti-symmetric because shear horizontal modes do not couple to non-viscous fluids.

- A system is symmetric if the top half of the layup mirrors the bottom half in terms of layer (fiber) orientations, layer thicknesses, and layer materials. This is always the case in a single layer. In case of fluid-loading, a symmetric system requires in addition that the upper and lower half-spaces are filled with the same fluid.

The four possible configurations are listed in Table 2. If Lamb and shear horizontal waves are coupled, they are called Lamb waves anyway in reminiscence to Sir Horace Lamb who described Lamb waves mathematically for the first time in 1917 [46]. Lamb dealt only with isotropic media wherein the coupling does not occur. In contrast to pure modes, coupled modes have three nonzero displacement components. If the system is non-symmetric, the modes do not have a strictly symmetric or anti-symmetric character, except for shear horizontal modes, as said above. In DC, non-symmetric modes are unconventionally denoted with the letter "B". To the best of my knowledge, there does not exist an official notation for non-symmetric modes.

If you have a symmetric system, you can choose to calculate the **Symmetric modes** and/or the **Antisymmetric modes**, otherwise the corresponding check boxes will be inactive. If the above described exception for the shear horizontal waves occurs, symmetric and antisymetric shear horizontal modes will be calculated anyway. In the decoupled case, you can calculate **Shear horizontal modes** separately, otherwise the corresponding check box will be inactive.

4.4 Signal simulator

The **Signal simulator** (see Fig. 15) simulates the temporal and frequency responses of single and multiple guided wave modes after propagating a certain distance in a specimen, based on the dispersion curves obtained in the **Isotropic** and **Anisotropic** tabs (currently limited to plates). Therefore, the tool is valuable to any individual performing NDI or SHM using guided waves. The dispersion diagrams calculated before will help him to excite the desired mode in a specimen. However, it is often difficult to excite only one particular mode, especially when the frequency-thickness product becomes large. In situations where dispersion curves get close to each other in the dispersion diagram, it is very likely that multiple modes will be excited. Then, the inspector has to deal with multiple wave packets appearing in the measured signal of voltage versus time. The **Signal simulator** enables him identifying the modes in the signal. The tool is also useful to interpret signals obtained from acoustical emission experiments. Or suppose you want to perform long-range testing on a pipe or on an aircraft wing structure. The dispersive nature of guided waves leads to a spreading and to a reduction in amplitude of the wave packet with increasing propagation distance because the different frequency components propagate at different phase velocities. Therefore, you would want to excite a weakly dispersive Lamb wave, retaining the shape of its wave packet over long distances. Or simulate a **Chirp** signal to compensate for the dispersive effects. Another effect that becomes evident in the signals is damping due

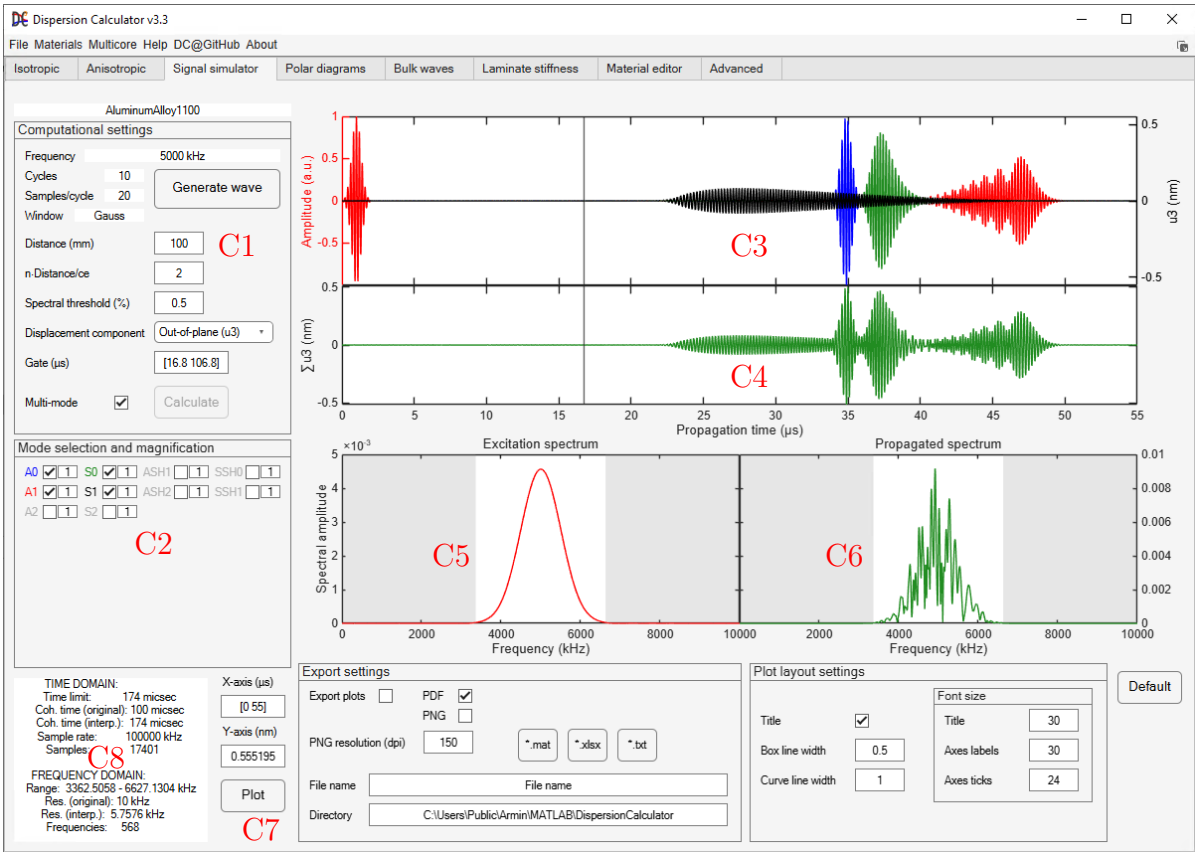


FIG. 15. The Signal simulator.

to fluid-loading and/or viscoelasticity. You would prefer a mode that suffers minimal damping loss in the required frequency range. With the **Signal simulator**, you can simulate the wave packets to find an appropriate mode for your task.

Notice that no particular method of excitation and reception is considered here. The assumption is that guided waves are generated *somehow* at $x = 0$, with the excitation signal plotted on the left-hand side in the signal window (C3), starting at $t = 0$. The propagated signal calculation is then based only on the dispersion diagram, which you calculated before, i.e., plane harmonic waves are superposed, having the phase velocities (and attenuation) given by the dispersion curve of the respective mode in the frequency range seen in the **Excitation spectrum** (C5). Power-normalization is performed on each contributing plane wave in the same way as done with the through-thickness profiles. The plotted amplitude can be chosen as the **Out-of-plane** displacement (u_3) for Lamb waves or the **In-plane** displacement (u_1) for Lamb waves or u_2 for shear horizontal waves at the top of the plate, which are assumed to correlate most directly to the detected signal of voltage versus time. The signal is simulated for ideal and simplified conditions, so experimentally measured signals will look more or less different due to many influences present under real conditions. These are, for instance, the specific shape of the ultrasonic beam or a certain angle of incidence, whereas in the simulation, the "perfect" (coincidence) angle (for each contributing phase velocity in the spectrum) is

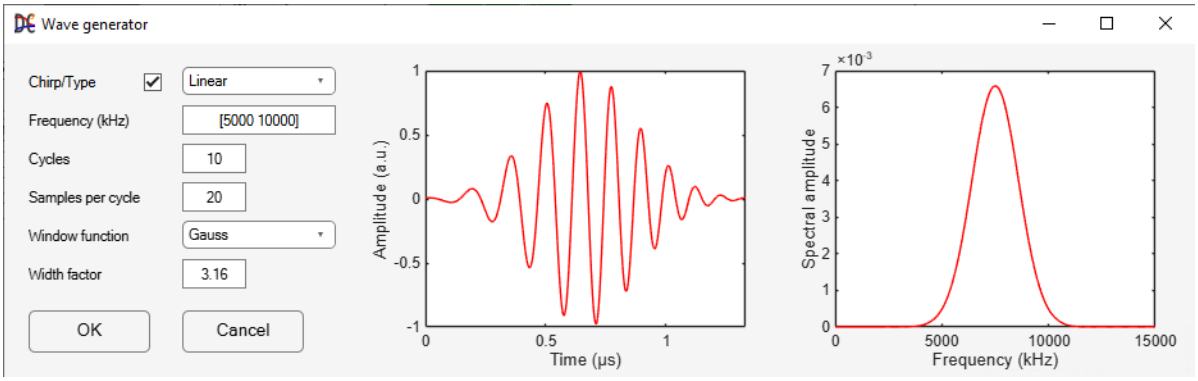


FIG. 16. The wave generator window.

assumed. Furthermore, waveguide imperfections are ignored such as surface roughness, layer debonding or other flaws. Calculating fully realistic signals requires a sophisticated model of the experimental setup, i.e., a detailed model of the sensors as well as of the waveguide to model the generated ultrasonic beam, the propagating wave field, and the received signal. This requires an FEM software such as COMSOL Multiphysics. You can, however, choose to plot the **In-plane** displacement or the **Out-of-plane** displacement, depending on what component your sensor is more sensitive to. Usually, in non-contact air-coupled ultrasonics, one would choose the **Out-of-plane** displacement, whereas in SHM, where you have sensors attached to the component, the **In-plane** displacement is more favored. In practice, the received signal will map a mixture of both components, and there are users indeed who compute both components with the **Signal simulator**, export them, and calculate a simulated signal from weighted fractions of both components. The key point is that the **Signal simulator** calculates the displacement caused by the guided waves at the top surface, but the signal displayed on your measurement screen depends on how your sensor reacts to this displacement, and this can obviously not be included in the **Signal simulator**.

Once you have calculated a dispersion diagram for a **Plate** in the **Isotropic** or **Anisotropic** tabs, the data will be available in the **Signal simulator**. The field above the **Computational settings** (C1) gives the name of the material whose dispersion curves are used. Therefore, notice that whenever you calculate a new dispersion diagram, the prior data will be replaced by the new ones. To set up the simulation, go to the **Computational settings** (C1) panel. Press **Generate wave** to open the **Wave generator** window displayed in Fig. 16. First decide if you want to simulate a carrier wave with constant frequency or a frequency **Chirp**. In the latter case activate the **Chirp** check box. Due to the dispersive nature of guided waves, the frequency components in a wave packet travel at different speeds, leading to a spreading of the wave packet. In a **Chirp** signal, the frequency changes with time. With a properly designed **Chirp** signal, undesirable dispersive effects in guided waves can be compensated for such that the wave packet maintains its shape while propagating large distances. From the **Type** drop-down menu you can select how the frequency changes with time in the chirp signal. This can be **Linear**, **Quadratic**, or **Exponential**. Now enter the carrier wave

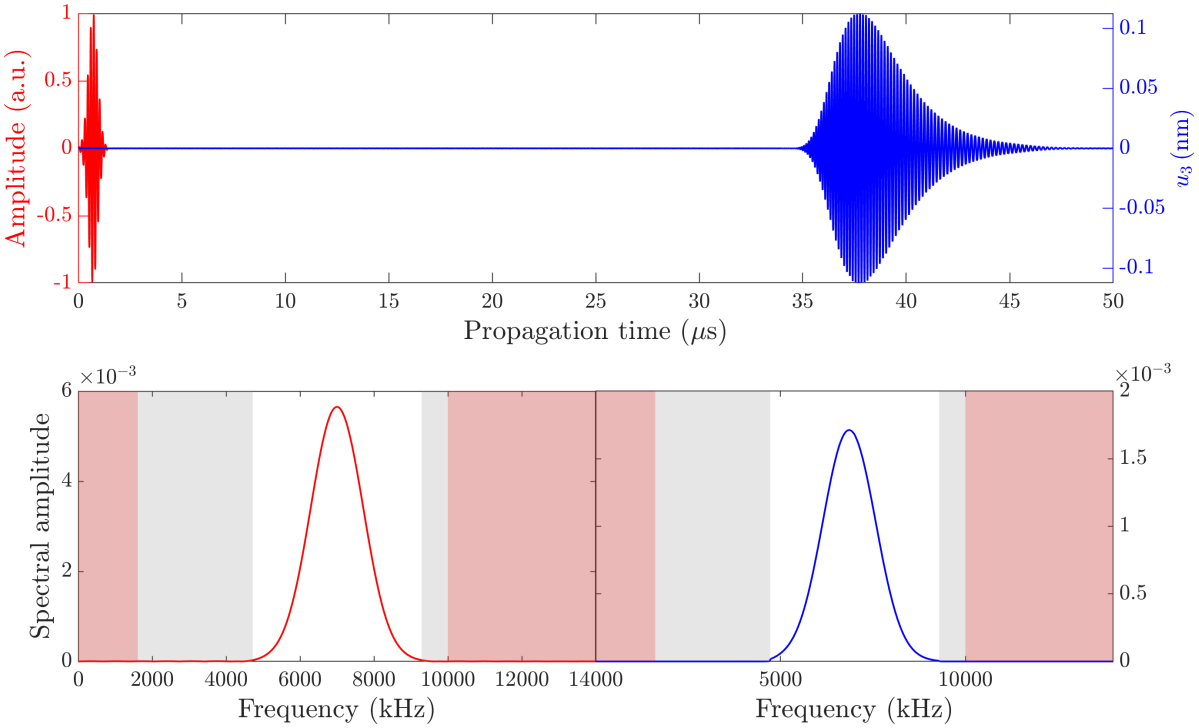


FIG. 17. Top: Simulated out-of-plane signal (ten cycles, Gaussian window) of the A_1 Lamb wave at 7 MHz after propagating a distance of 100 mm in a 1 mm thick free aluminum alloy 1100 plate. Bottom: Excitation spectrum (left) and propagated spectrum (right).

Frequency. When you have enabled **Chirp**, you must define the instantaneous frequencies at the start and at the end of the signal. The frequency can increase (up-chirp) or decrease (down-chirp) with time. If **Chirp** is unchecked, a single value is requested as **Frequency**. Then enter the number of **Cycles** in the wave packet. Enter the **Samples per cycle** in the excitation signal. This determines the sample rate, i.e., the resolution of the temporal signal. In a **Chirp** signal, the number corresponds to the shortest cycle. Then choose a window from the **Window function** drop-down menu. There are listed five functions, namely **Gauss**, **Tukey**, **Hann**, **Hamming**, and **Triangular**[1]. By multiplying the carrier wave with the selected window function, the shape of the wave packet is obtained. Finally set the **Width factor** of the selected **Window function**. In case of a **Gauss** window, the width is inversely proportional to **Width factor**. The default value is 3.16. In case of a **Tukey** window, this value sets the plateau width. A value of 1 returns a **Hann** window, and 0 gives a rectangular window. The default value is 0.5. The two plots on the right show the wave packet in the time and frequency domains; they update whenever the user changes any option. Press **OK** to accept the settings and close the **Wave generator** window.

After having designed the wave excitation signal, set the **Distance** which the modes are supposed to propagate in the specimen. The option **n·Distance/ce** defines the

[1] https://en.wikipedia.org/wiki/Window_function

temporal length t_{end} of the simulation in multiples n according to

$$t_{\mathrm{end}} = nx/c_{\mathrm{e1}}, \tag{1}$$

where x is the propagation distance and c_{e1} is the energy velocity component in the propagation direction of the chosen mode. n is set to two by default in order to ensure that the temporal axis covers the whole wave packet of a strongly dispersive mode. For short propagation distances, higher multiples of n can be necessary. If you have checked **Multi-mode**, the lowest energy velocity among all contributing modes is used for the calculation of t_{end}. The advantage of defining t_{end} in the described way is that an appropriate temporal range is obtained automatically so that the user does not have to try and calculate by himself. Notice that you can edit the temporal axis later in the **X-axis** (C7) field. The **Spectral threshold** defines which frequency range shall be used for the construction of the propagated wave packet(s). Only frequencies with spectral amplitudes exceeding this threshold in percent of the maximum spectral amplitude shown in the **Excitation spectrum** (C5) will contribute to the simulation. This spectral amplitude maximum in the excitation signal is located exactly at the center **Frequency** defined above (if it is not a **Chirp** signal). By decreasing the threshold, a wider frequency range will be taken into account. This contributing frequency range is indicated by the white area in the **Excitation spectrum** (C5) plot. When the threshold is changed, you will notice that the white area's width changes accordingly. The same happens also in the **Propagated spectrum** (C6) plot. If you set a **Frequency** close to a cut-off frequency or to the top frequency in the dispersion diagram, a red area will appear, showing where no data are available (see Fig. 17). Therefore, when you calculate a dispersion diagram, set the top frequency high enough to not miss data for simulating a signal at high frequencies. From the **Displacement component** drop-down menu, you can choose to calculate the **Out-of-plane (u3)** displacement or the **In-plane (u1/2)** displacement. Notice that the **Out-of-plane** displacement of shear horizontal waves is zero, so choose **in-plane (u1/2)** in this case. With the **Gate**, you can define the signal range for which the **Propagated spectrum** (C6) shall be calculated. This range is indicated by the two vertical black lines in the signal plots (C3) and (C4). In case you have multiple wave packets in the signal, as shown in Fig. 15, you can put the gate around a particular mode to get its spectral amplitude distribution (C6). If **Multi-mode** is unchecked, you can simulate only one propagated mode at a time. The **Mode selection** (C2) panel (it is renamed to **Mode selection and magnification** when you check **Multi-mode**) is updated automatically to show the modes available at the **Frequency** defined above. Check a check box to simulate the corresponding mode. In the top window (C3), the excitation signal will be plotted in red. Its amplitude is scaled to the LEFT amplitude axis, ranging from -1 to 1. The propagated signal (temporal response) is plotted in blue, and its amplitude is scaled to the RIGHT amplitude axis. The displacement amplitudes are power-normalized and given in nanometers in the same way as are the displacement components drawn in the through-thickness profiles of the **Isotropic** and **Anisotropic** tabs, as explained in Sec. 4.2, **Through-thickness profiles** (A10). The middle window (C4) appears only if **Multi-mode** is enabled. As mentioned above, you can change the default temporal axis by editing **X-axis** (C7). The displacement axis

in the signal window (C3) can be changed by editing **Y-axis** (C7). We have already discussed the **Excitation spectrum** (C5), displayed in the bottom left-hand window. The **Propagated spectrum** (C6) (frequency response) is displayed in the bottom right-hand window. For a better view and in order to read values using the data tip tool, you can **Plot** (C7) the simulations in a separate window. Figure 17 shows the simulated signal of the A_1 Lamb wave at 7000 kHz after propagating a distance of 100 mm in a 1 mm thick free aluminum alloy 1100 plate. The excitation signal consists of ten cycles multiplied with a Gaussian window. The wave packet arrives after a propagation time of 34.5 µs. Due to dispersion, it has spread significantly. Its spectrum, shown in the bottom right plot, is shifted to lower frequencies, peaking at a frequency of 6860 kHz, a shift of 140 kHz with respect to the excitation spectrum, shown in the bottom left plot. The output window (C8) gives some more information:

Time domain:

- **Time limit**: It is the calculated length of the temporal axis t_{end}.
- **Coherence time**: The coherence time t_{c} is the time after which the temporal response repeats itself, i.e., we will see a twin of the wave packet at each instant of time $n \cdot t_{\mathrm{c}}$, $n = 1, 2, ...$, away from the actual wave packet. The coherence time is calculated by
$$t_{\mathrm{c}} = 1/\Delta f, \tag{2}$$
where Δf is the **Frequency step**, i.e., the frequency resolution of the calculated dispersion curves. Therefore, to avoid unwanted wave packet twins, we require that $t_{\mathrm{c}} > t_{\mathrm{end}}$. You can satisfy this by setting a small enough value for **Frequency step**, but you don't have to. If you have $t_{\mathrm{c}} < t_{\mathrm{end}}$, DC will interpolate the dispersion curves with a value of Δf small enough to meet $t_{\mathrm{c}} > t_{\mathrm{end}}$. In this case, the output window will read **Coh. time (original)**, giving the coherence time resulting from the native frequency resolution, followed by **Coh. time (interp.)**, giving the coherence time resulting from the interpolated frequency resolution.
- **Sample rate**: It is calculated by multiplying the **Frequency** with the **Samples per cycles**.
- **Samples**: It gives the total number of sample points contained in the temporal signal.

Frequency domain:

- **Spectral range**: This is the frequency range contributing to the construction of the propagated wave packets. It is indicated by the white areas in the spectral amplitude plots in the windows (C5) and (C6).
- **Resolution**: It is the frequency resolution of the dispersion curves, as set by **Frequency step**. In case of $t_{\mathrm{c}} < t_{\mathrm{end}}$, the output window will read **Res. (original)**, giving the native frequency resolution, followed by **Res. (interp.)**, giving the interpolated frequency resolution.

- **Frequencies**: This is the total number of frequency sample points contained in the **Spectral range**.

In the example shown in Fig. 17, the spectral range is from 4720 kHz to 9290 kHz. The red areas indicate where no dispersion curve data are available. The cut-off frequency of A_1 at a phase velocity of 20 m s^{-1} is 1615 kHz so that no data are available below this value. The top frequency in the dispersion diagram is 10 000 kHz, just high enough to not cut the white area contributing to the simulation.

If you enable **Multi-mode**, as shown in Fig. 15, you must first **Calculate** all modes present at the selected **Frequency**. You can terminate the calculation by pressing the **Stop** button. After the calculation is finished, all modes will be plotted. You can show/hide any particular mode to see only those you want. The font colors in the **Mode selection and magnification** (C2) panel match the line colors of the corresponding modes plotted in the top window (C3). To the right of each check box, the amplitude magnification can be entered. The (displacement) amplitudes are calculated for modes carrying a power flow of unity. However, in reality, each of the modes will carry a different power flow because they are excited at different efficiency. To account for that, the user can magnify the amplitude of each mode individually. Notice again that the amplitudes of the propagated wave packets are scaled to the right amplitude axis, whereas the excitation signal (red) scales to the left one. The middle window (C4) shows the sum amplitude of all simulated modes, thereby representing the signal which the user measures in his experiment. By comparing it with the top window (C3), he can ascribe particular modes to the contributions in window (C4). The lower right-hand window now shows the **Propagated spectrum** (C6) of the sum amplitude shown in window (C4).

4.5 Polar diagrams

Often it is useful to have the guided wave dispersion versus the propagation direction in a laminate at a certain frequency rather than versus the frequency for a certain propagation direction. In the **Polar diagrams** tab (see Fig. 18), you can calculate polar diagrams that give a very intuitive picture of the anisotropic propagation characteristics of the fundamental modes in a laminate. These are set up as usual except that only unit cells that contain at least one zero degree layer are accepted. Notice that fluid-loading and viscoelasticity cannot be considered here. Then use the **Computational settings** (D1) panel. The dispersion curves are traced in the frequency space, as usual, up to the **Frequency limit** with a certain **Frequency step**. Notice that after calculating the polar dispersion diagram(s), data are available not only at the **Frequency limit** but at each **Frequency step** from zero to **Frequency limit**. With the **Propagation angle limit** drop-down menu, you can choose to calculate the dispersion curves for propagation angles ranging from 0° to 180° or only to 90°. In the first case, to obtain the full polar diagram, the data will be copied to the third an fourth quadrant. In the second case, the data of the first quadrant will be copied to the three remaining ones. Notice that the second option is insufficient for unit cells such as [0/45]. This

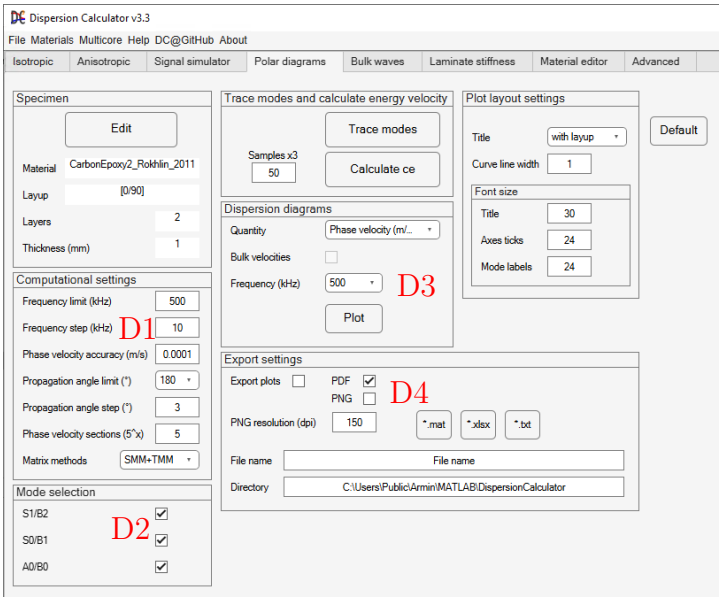


FIG. 18. The tab for the calculation of polar dispersion diagrams of the fundamental modes.

will become obvious by kinks in the polar diagrams at the quadrant borders. Use **Propagation angle step** to define the propagation angle increments for which the dispersion curves will be calculated. In the **Mode selection** (D2) panel, you can choose

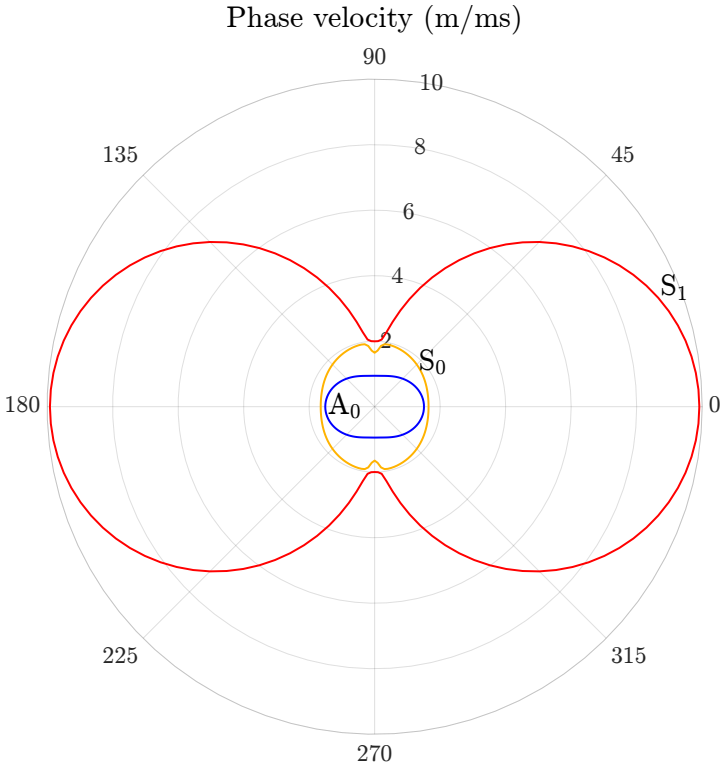


FIG. 19. Polar dispersion diagram at 500 kHz for 1 mm thick unidirectional T800M913.

which ones of the fundamental modes A_0/B_0, S_0/B_1, and S_1/B_2 shall be calculated. In the **Dispersion diagrams** (D3) panel, you can choose which **Quantity** to plot, namely the **Phase velocity**, **Energy velocity**, **Propagation time**, **Coincidence angle**, **Wavelength**, and **Wavenumber** versus propagation direction. From the **Frequency** drop-down menu, select a frequency for which to plot the polar diagram. Upon pressing **Plot**, a new window will open with a polar diagram like the one displayed in Fig. 19. In the **Export settings** (D4) panel, you can export the plots. Press the ***.mat**, ***.xlsx**, and ***.txt** buttons to export the polar dispersion curves at the selected **Frequency** to the desired file format.

4.6 Bulk waves

In the **Bulk wave** tab (see Fig. 20), you can model the behavior of elastic waves (bulk waves) in bulk material and their scattering on interfaces. In isotropic materials, the phase velocity of bulk waves is independent from the propagation direction, and the polarization is along the wave propagation direction for longitudinal waves (L) and perpendicular for shear vertical (SV) and shear horizontal waves (SH). Therefore, if we consider bulk waves emanating from a point source in an isotropic bulk material, the phase velocity surfaces for both wave types are spherical with the radii being the

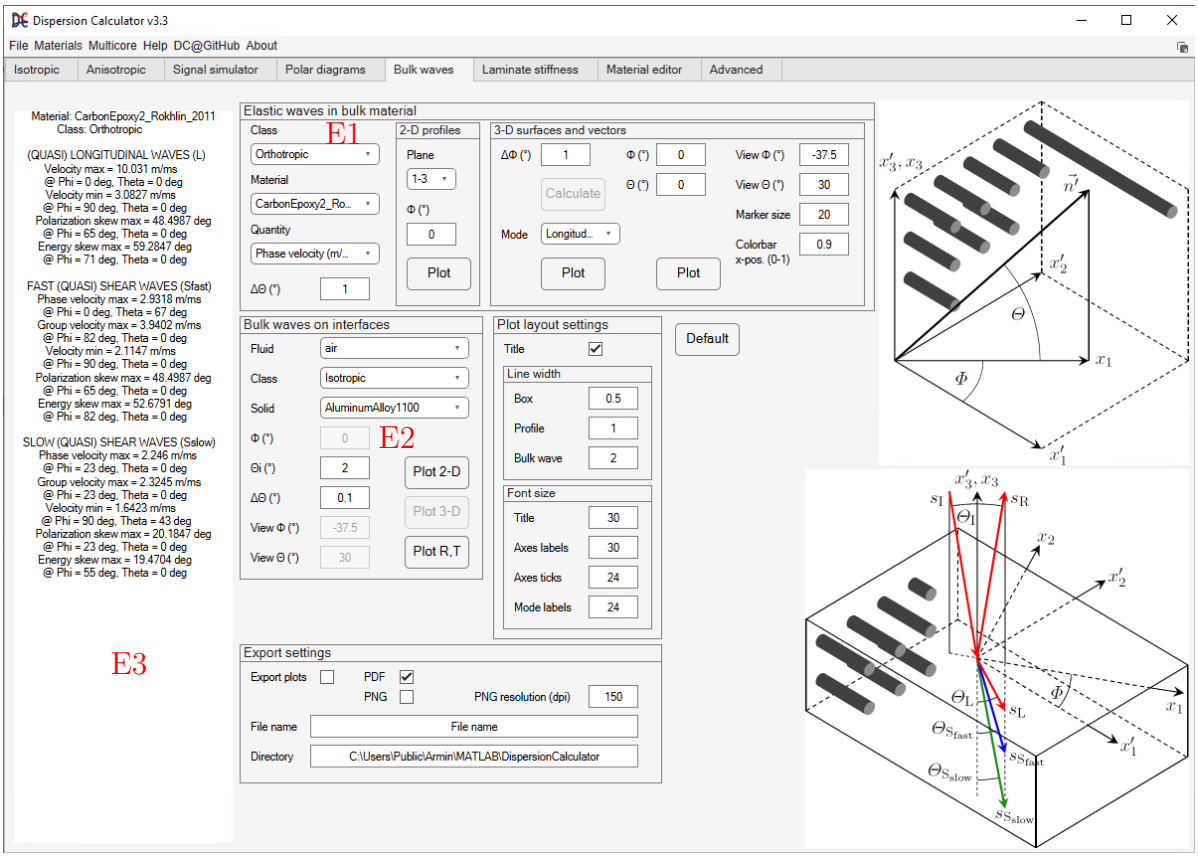


FIG. 20. The Bulk waves tab.

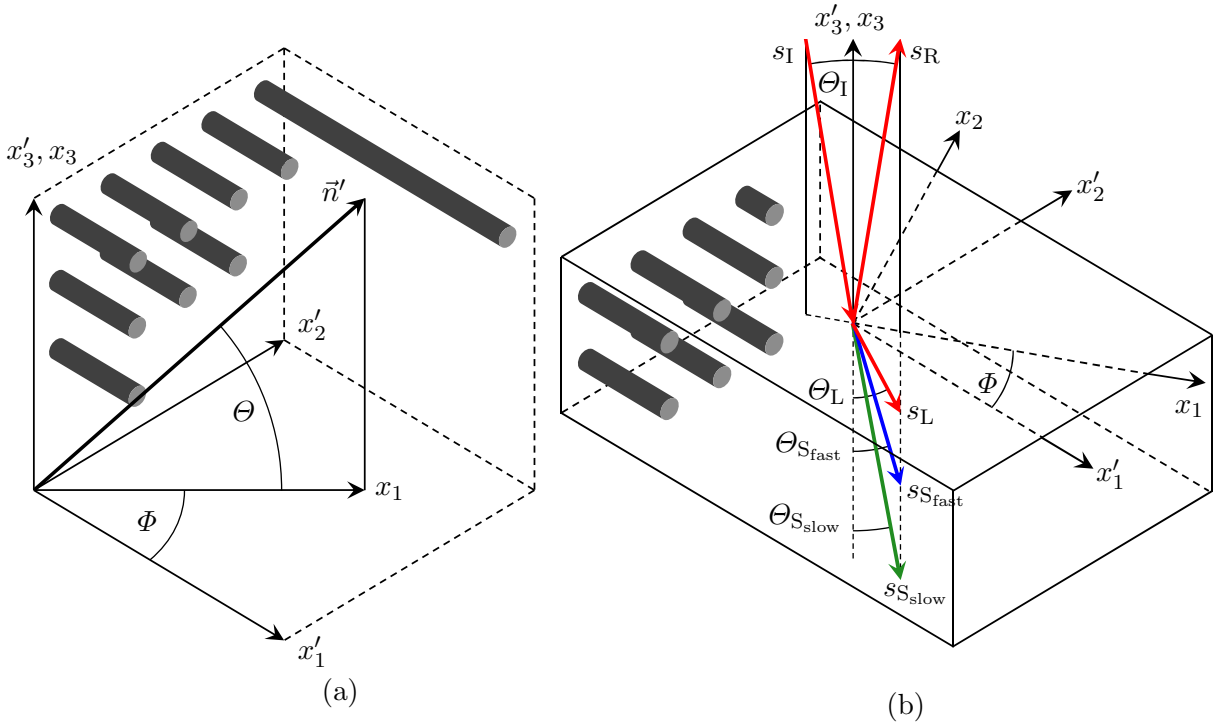


FIG. 21. Crystallographic and global coordinate systems x_i', x_i and fiber orientation in transversely isotropic material. (a) Elastic wave propagation in the bulk material as treated in the **Elastic waves in bulk material** (E1) panel. (b) A plane wave impinging from a fluid generates three bulk waves in transversely isotropic material propagating in the x_1-x_3-plane of the global coordinate system. This topic is covered in the **Bulk waves on interfaces** (E2) panel.

longitudinal velocity for L waves and the transverse velocity for SH and SV waves. These velocities are given in the **Material editor** (Sec. 4.8) tab. The group velocity surfaces coincide with the phase velocity surfaces. L waves are polarized along the radial direction, and SH and SV waves normal to it. In anisotropic materials, however, each of the three bulk wave types has its own phase velocity, which depends on the propagation direction $\vec{n}'$ in Fig. 21(a), and the group velocity surface does not coincide with the phase velocity surface. In general, the group velocity vector deviates from the phase velocity vector, where the latter is parallel to the propagation direction unit vector $\vec{n}'$. In absence of attenuation, the group velocity is the same as the energy flow velocity, and its vector represents the actual direction of the acoustic beam. See Appendix A for information on the similar problem of the group velocity of guided waves in anisotropic media. This situation leads also to differences in the magnitude of the phase and group velocities. The bulk waves' polarization is either along or perpendicular to the propagation direction, respectively, only in case of propagation along or perpendicular to the fibers (in these cases, the modes are called 'pure'), except for one shear type bulk wave in transversely isotropic materials, which has pure character for any propagation direction. The shear waves do not have the same phase velocity anymore. Because of that, the bulk waves in anisotropic materials are called quasilongitudinal waves (L), fast quasishear waves ($\mathrm{S_{fast}}$),

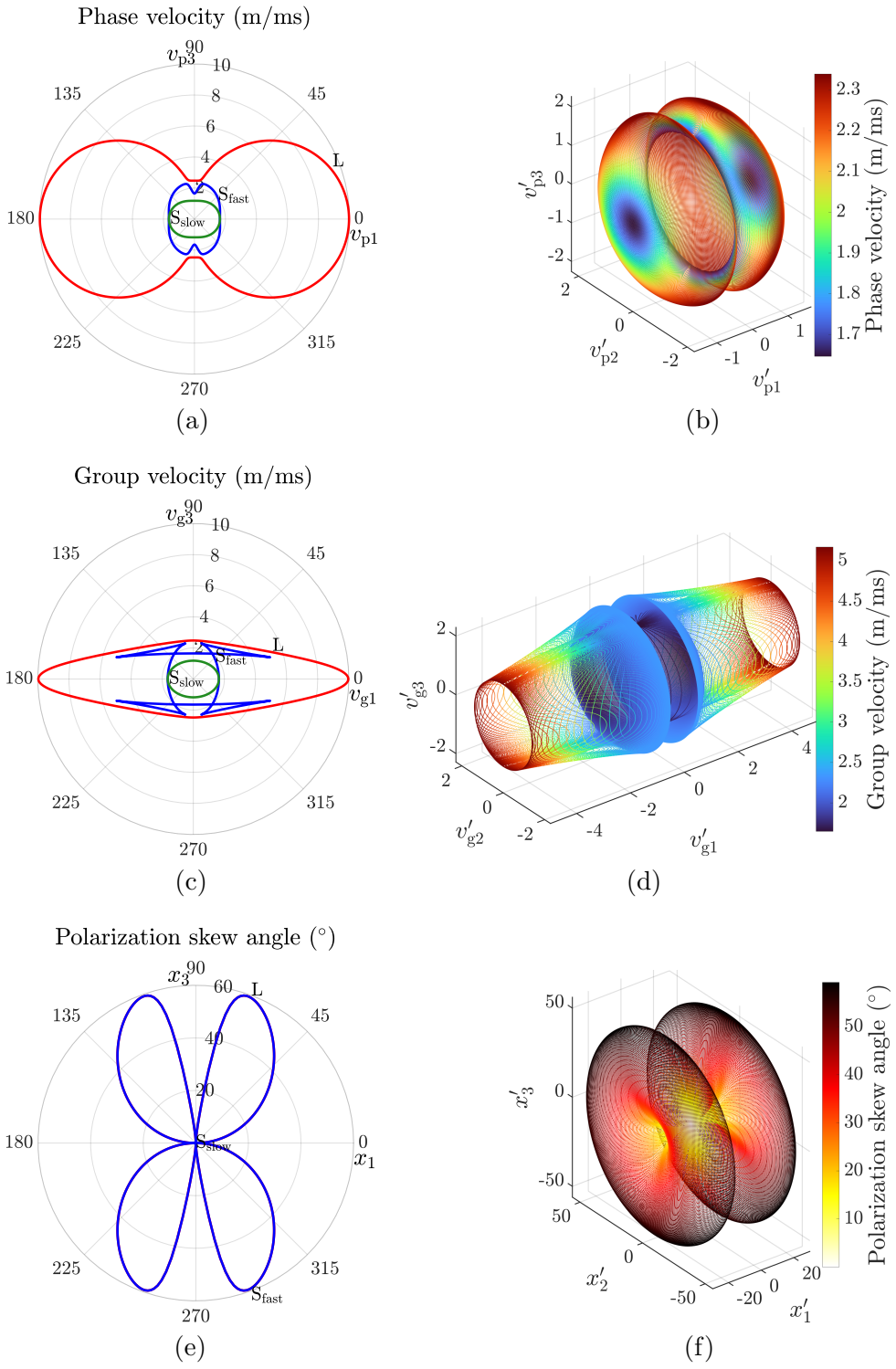


FIG. 22. Bulk waves in T800M913. (a) Phase velocity, (c) group velocity, and (e) polarization skew angle profiles at $\Phi = 0°$, and (b) S_{fast} phase velocity, (d) S_{fast} group velocity, and (f) L and S_{fast} polarization skew angle surfaces.

and slow quasishear waves (but slow shear waves in transversely isotropic material) (S_{slow}). Figure 21(a) displays the conventions used for the elastic wave propagation in a fiber-epoxy system. Here, x_1' is along the fiber direction, and x_2' and x_3' are normal to it. Θ defines the wave propagation angle with respect to the x_1'-x_2'-plane (elevation), and Φ is the angle between the propagation direction projected into the x_1'-x_2'-plane and x_1' (azimuth).

In the **Elastic waves in bulk material** (E1) panel, you can plot the phase velocity, group velocity, the corresponding slownesses s, the energy skew angle γ between the bulk waves' phase velocity vectors and the group velocity (or energy) vectors, and the polarization skew angle δ of bulk waves in anisotropic materials in 2-D and 3-D. The slowness is the inverse phase velocity, and the polarization skew angle is the deviation angle from transverse polarization ($\beta = 90°$) in case of S_{fast} and S_{slow}, and longitudinal polarization ($\beta = 0°$) in case of L. In transversely isotropic material, δ for S_{slow} is always zero since it is a pure mode for any propagation direction in this material class. From the **Class** drop-down menu, select which material class you want to consider. The **Material** drop-down menu will be updated accordingly. Select a material from the **Material** drop-down menu. Choose the **Quantity** to be plotted and use **ΔΘ** to determine the Θ resolution. In the **2-D profiles** sub-panel, you can plot the quantity for a given angle **Φ**, as shown in Figs. 22(a), 22(c), and 22(e) for T800M913 at $\Phi = 0\,°$. By using the **Plane** drop-down menu, you can select to plot the 1-3-plane or the 1-2-plane. This is useful for orthotropic materials where the modules E_2 and E_3 are not equal. If you select the 1-2-plane, the axes x_2' and x_3' in Fig. 21(a) are swapped. In the **3-D surfaces and vectors** sub-panel, you can plot the complete surface. Now, you have to set **ΔΦ** to determine also the Φ resolution. Upon pressing **Calculate**, the point cloud made up by **ΔΘ** and **ΔΦ** will be calculated and the key data be displayed in the output window (E3). Use the **Mode** drop-down menu to select which one of the three bulk wave mode surfaces you want to plot. Figures 22(b) and 22(d) show the phase and group velocity surfaces of S_{fast} in T800M913, respectively, and Fig. 22(f) shows the (coinciding) polarization skew angle from pure mode polarization of L and S_{fast}. It can readily be seen that the skew angle becomes zero for wave propagation parallel and normal to the fibers. These are the propagation directions for which pure L and S_{fast} modes occur. In the center column of the **3-D surfaces and vectors** sub-panel, you can plot the slowness and polarization vectors of bulk waves propagating along a specific direction. Define the propagation direction using **Φ** and **Θ**, and press **Plot** to get an image like the one shown in Fig. 23. In addition to the **Plot layout settings**, which apply to all plots on the current tab, you find some additional layout settings in the **3-D surfaces and vectors** sub-panel, which apply only to the 3-D plots. You can adjust the view by using **View Φ** and **View Θ**. Of course, you can also use the **Rotate 3-D** tool inside the plot window. You can also change the **Marker size** and the x-position of the colorbar.

In the **Bulk waves on interfaces** (E2) panel, you can model the bulk waves that are generated in an isotropic or anisotropic material when a plane wave impinges from a fluid. This situation is depicted in Figs. 21(b) and 24(a). Here, the upper half-space is covered by water and the lower half-space by transversely isotropic T800M913. The

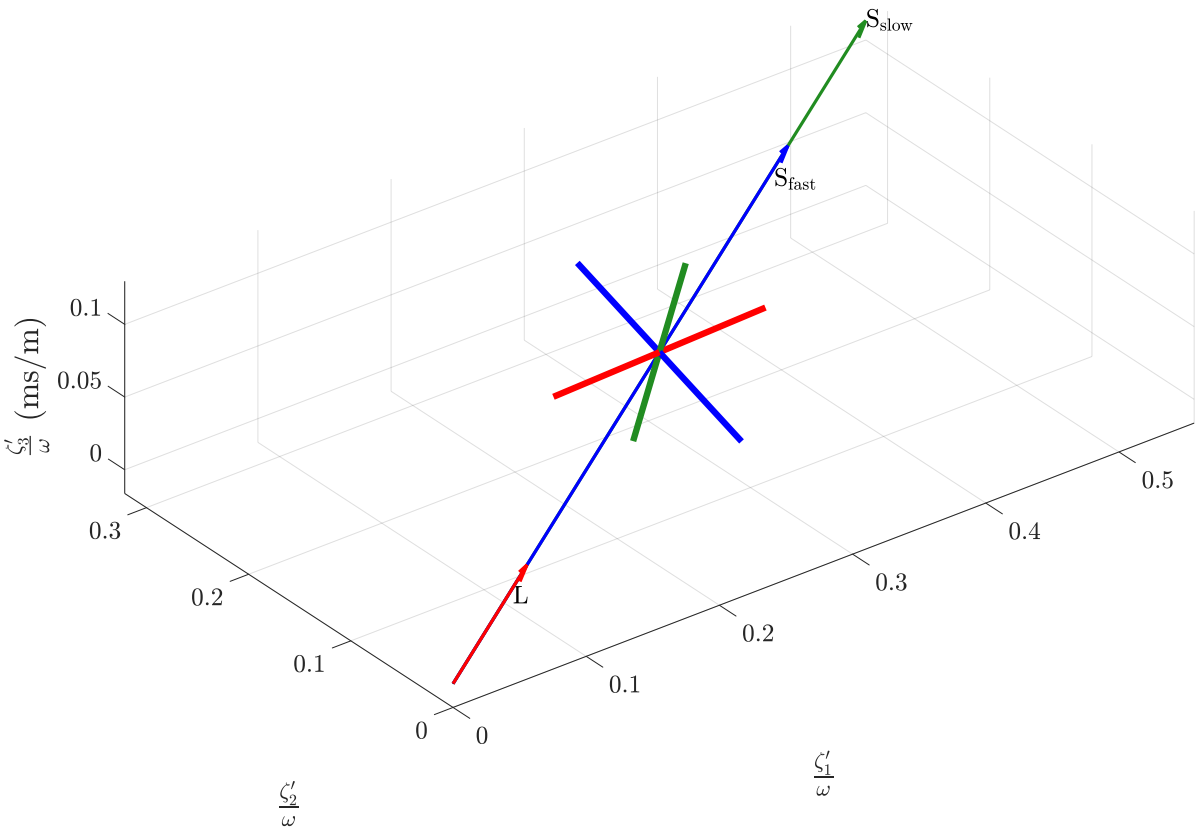


FIG. 23. Bulk wave slownesses and polarizations in T800M913 for propagation along $\Phi = 30°$, $\Theta = 10°$. The polarization vectors are always mutually orthogonal.

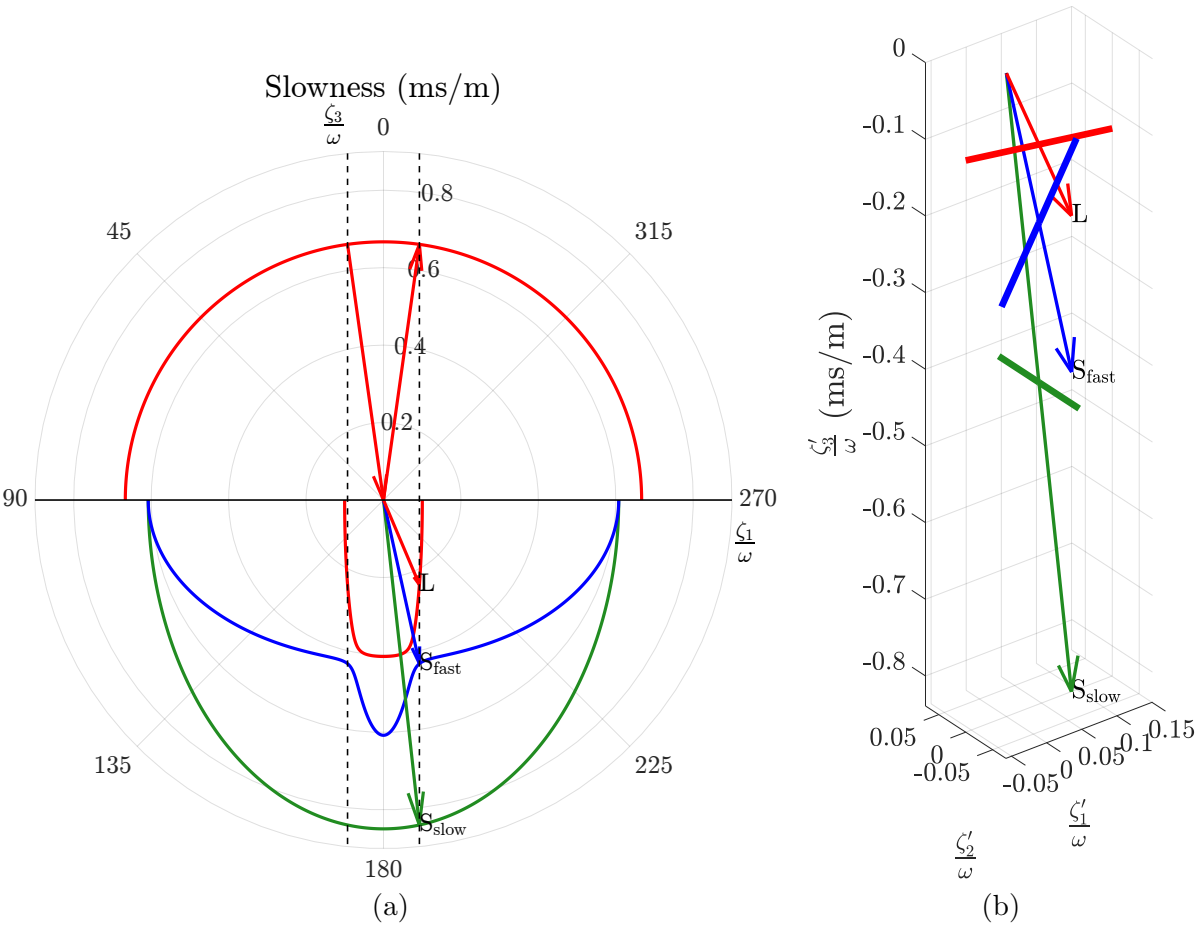


FIG. 24. Bulk waves generated in T800M913 through plane wave incidence from water at $\Phi = 0\,^\circ$, $\Theta_\mathrm{I} = 8\,^\circ$. The upper half-space is covered by water, the lower half-space by T800M913. (a) Construction of the bulk wave slownesses based on Snell's law. (b) Slownesses and polarizations of the generated bulk waves.

incident as well as the reflected and transmitted waves propagate in the $\Phi = 0\,^\circ$ plane, i.e., $x_1 = x_1'$. The curves represent the slownesses in water and T800M913. Water as well as any other nonviscous fluid supports only the longitudinal wave type (because the shear modulus is zero), represented by the red vectors in the upper half-space. The plane wave is incident under an angle of $\Theta_\mathrm{I} = 8\,^\circ$. Its wavenumber ζ_I is given by

$$\zeta_\mathrm{I} = \frac{\omega}{v_\mathrm{I}} = \omega s_\mathrm{I}, \tag{3}$$

where ω is the angular frequency $2\pi f$, with the frequency f, and v_I is the phase velocity in water. At the boundary to the solid, a certain proportion of energy is reflected, and the rest is transmitted into T800M913 where three bulk waves with the wavenumbers $\zeta_\mathrm{L}, \zeta_{\mathrm{S}_\mathrm{fast}}, \zeta_{\mathrm{S}_\mathrm{slow}}$ are generated. For these waves, Eq. (3) applies with their respective phase velocities and slownesses $v_\mathrm{L}, v_{\mathrm{S}_\mathrm{fast}}, v_{\mathrm{S}_\mathrm{slow}}, s_\mathrm{L}, s_{\mathrm{S}_\mathrm{fast}}, s_{\mathrm{S}_\mathrm{slow}}$. According to Snell's law, all scattered waves (reflected and transmitted ones) must have the same projected wavenumber

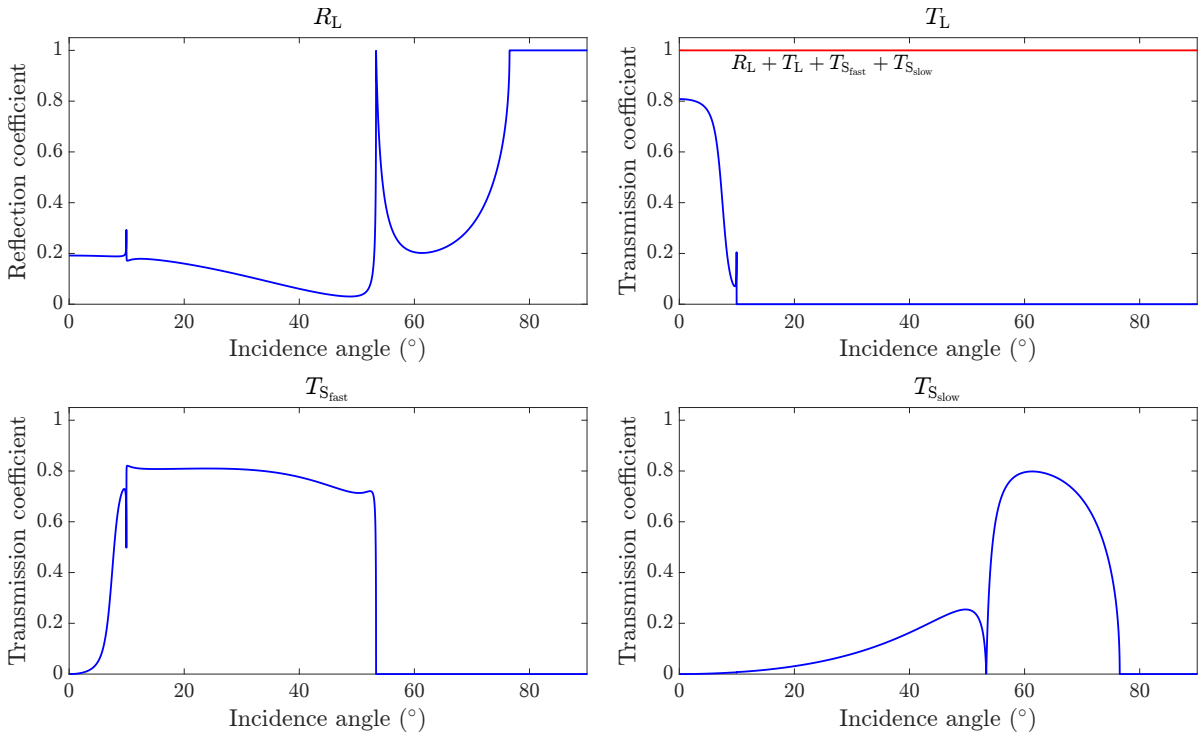


FIG. 25. Reflected and transmitted energy scattering coefficients versus incidence angle for the bulk waves scattered on the interface between water and T800M913 at $\Phi = 30\,^{\circ}$

along the boundary $\zeta_{\mathrm{I}x_1}$ as the incident wave, i.e.,

$$\zeta_{\mathrm{I}} \sin\Theta_{\mathrm{I}} = \zeta_{(k)} \sin\Theta_{(k)} = \zeta_{\mathrm{I}1}, \qquad k = 1, 2, ...4, \tag{4}$$

where $\zeta_{(k)}$ stands for the wavenumber of any scattered wave, and $\Theta_{(k)}$ is the corresponding reflection or refraction angle, respectively. With the use of Eq. (3), we can write Eq. (4) in terms of the slownesses of the scattered waves $s_{(k)}$ as

$$s_{\mathrm{I}} \sin\Theta_{\mathrm{I}} = s_{(k)} \sin\Theta_{(k)} = s_{\mathrm{I}1}. \tag{5}$$

This important constraint is indicated by the two vertical guide lines in Fig. 24(a). The utility of this picture is that the wave slowness vectors $\vec{\zeta}_{(k)}/\omega$ can be constructed simply by drawing lines from the origin to the points, where the right guide line intersects the slowness profiles. Then, by assigning a frequency, the actual wavenumber can be obtained. From Fig. 24(a), it becomes also clear that, as the incidence angle increases, critical angles will be reached beyond which a slowness vector for a given bulk wave cannot be constructed anymore. Physically, these waves are evanescent, i.e., their amplitude decays exponentially with depth, and they are confined to a certain area below the boundary. Basically, these are surface waves (Rayleigh waves). By contrast, below the critical angles, we have propagating bulk waves. Now, let us look at Fig. 24(b) to find out more about the polarization of the bulk waves in T800M913. Here, only the lower half-space from Fig. 24(a) is displayed. The bold lines emanating from the middle of the wave vectors indicate their polarization.

The final question is how much energy each of the scattered waves carries. Clearly, the conservation of energy has to be satisfied, so the sum of the energies of the reflected and transmitted waves must be the same as that of the incident wave. Figure 25 displays the reflected and transmitted energy scattering coefficients versus incidence angle for the bulk waves scattered on the interface between water and T800M913 at $\Phi = 30\,°$. Four subplots represent the reflected longitudinal wave (upper left), the transmitted longitudinal wave (upper right), the transmitted fast quasishear wave (lower left), and the transmitted slow shear wave (lower right). The red line in the upper right subplot indicates the summation of all four energy scattering coefficients, confirming the result of unity. The kinks and peaks in the blue curves mark exactly the critical angles. Consider for instance the upper right subplot showing the transmitted longitudinal wave. There is a notable peak at 9.9°. Beyond this angle, L will not be generated in the solid anymore. The other critical angles are 53.4° and 76.6°. Beyond the latter, total reflection occurs, i.e., the reflected energy coefficient reaches unity.

To do your modeling, select a **Fluid** from the drop-down menu. Then select the **Class**. Depending on your choice, you can now select an isotropic, a cubic, a transversely isotropic, or an orthotropic **Solid**. All other parameters in this panel work similarly to those in the **Elastic waves in bulk material** (E1) panel. Whenever you make an entry, information about the generated bulk waves in the solid, including critical angles and energy scattering coefficients, will be displayed in the output window (E3). Press **Plot 2-D**, **Plot 3-D**, and **Plot R,T** to generate (and export) the kind of plots shown in Figs. 24 and 25.

4.7 Laminate stiffness

The **Laminate stiffness** tab depicted in Fig. 26 is an additional tool that allows to calculate the laminate stiffness matrix, which is also referred to as the *homogenized stiffness tensor*. The laminate stiffness matrix is the stiffness matrix of a layered, anisotropic laminate. It is calculated using the *classical laminate theory* (CLT). First, the layer stiffness matrices are transformed into the global coordinate system to account for their individual fiber orientations, and then the transformed stiffness matrices are multiplied by their respective layer thickness-to-unit cell thickness ratio and summed up to obtain the unit cell stiffness matrix. Since every unit cell has the same stiffness matrix, and because of the multiplication with the thickness ratios, the laminate stiffness matrix does not depend on the repetitions of the unit cell, i.e., it is equal to the unit cell stiffness matrix.

The **Specimen** (F1) setup is reduced accordingly. The **Laminate stiffness components** (F2) are displayed for the chosen **Azimuthal angle**. This matrix has the form of monoclinic material symmetry. In the polar diagram (F3), every laminate stiffness component can be plotted with respect to the wave propagation angle in the defined laminate. Use the check boxes on the left accordingly. You can export the laminate stiffness matrix displayed in (F2) to the desired file format.

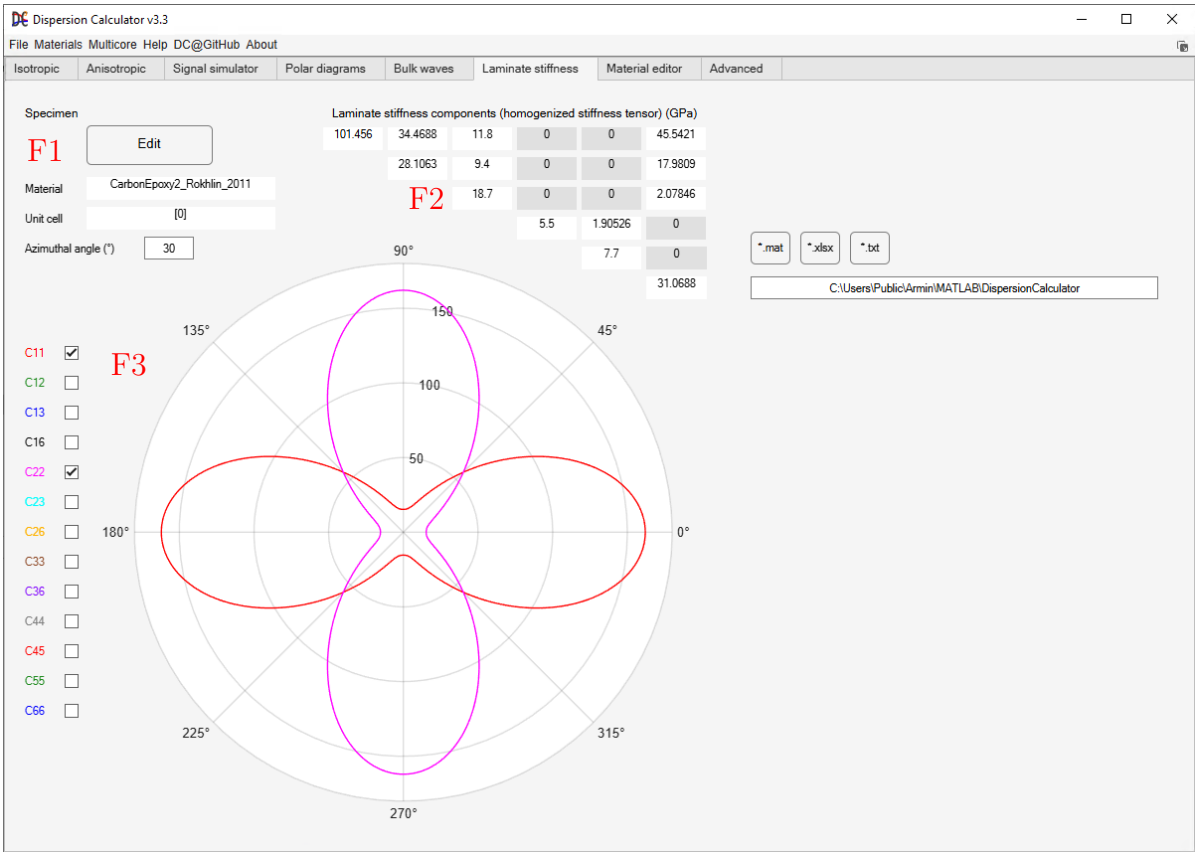


FIG. 26. The Laminate stiffness tab.

4.8 Material editor

Starting in DC v2.1, viscoelastic damping can be considered. To account for this, the **Engineering constants** and **Stiffness components (GPa)** are complex. With only non-zero **Real part**s, one describes a perfectly elastic material, as it was prior to DC v2.1. To consider viscoelastic damping, you need to enter non-zero **Imaginary part**s. Many complex default materials are available in DC. These are listed in Appendix C. To distinguish them from the perfectly elastic materials, they carry the suffix "_viscoelastic". For the guided wave modeling in viscoelastic materials, DC uses the hysteretic damping model, corresponding to a linear increase of attenuation with frequency, i.e., a constant loss per wavelength.

In the **Material editor** tab, you can view the parameters of the available materials and add or remove materials. In the **Isotropic materials** (G1) panel shown in Fig. 27, you can choose an isotropic **Material** from the drop-down menu. If you want to add a new material, enter its **Mass density**, and then you can choose one of the following three options:

1. **Engineering constants**. Enter Young's modulus **E (GPa)** and Poisson's ratio **ν**. In case you want to consider viscoelastic damping, enter non-zero **Imaginary part**s in addition to the **Real part**s.

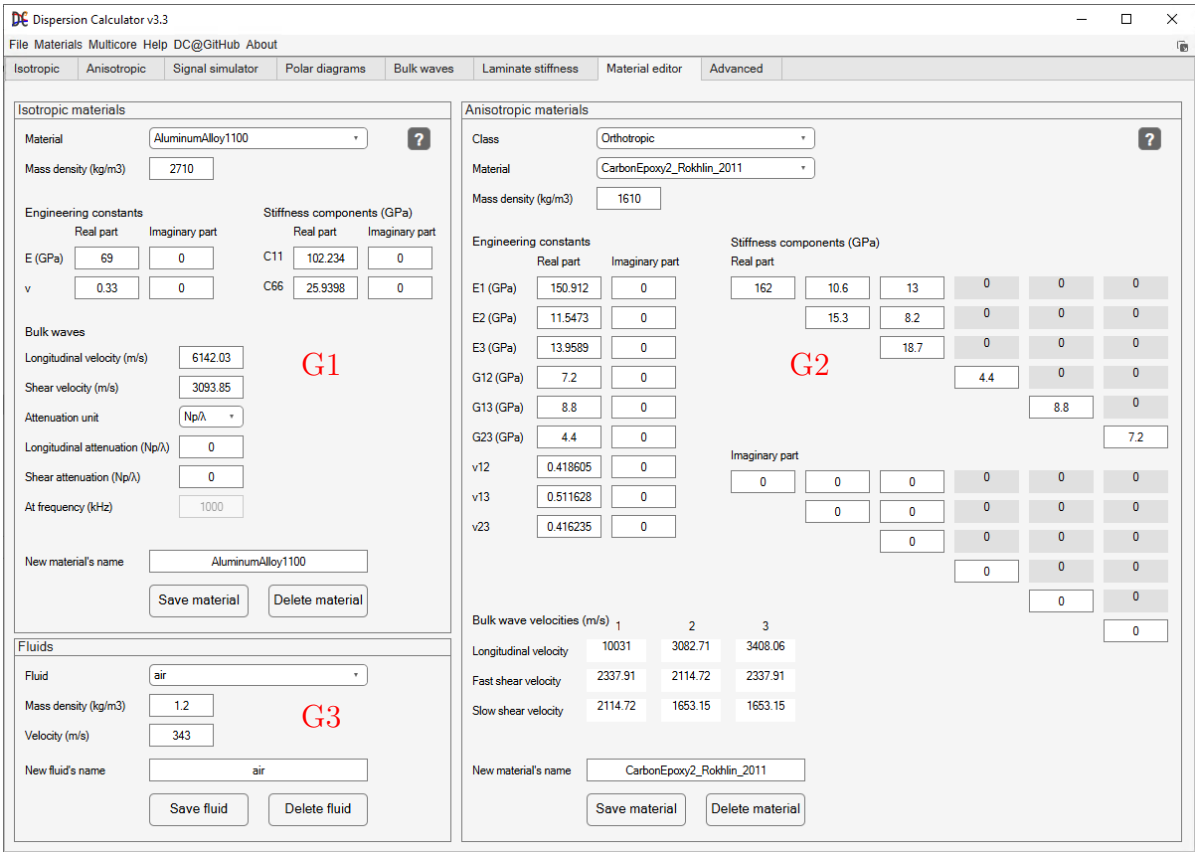


FIG. 27. The Material editor.

2. **Stiffness components (GPa)**. Enter **C11** and **C66**. Similarly as for the above, enter non-zero **Imaginary part**s for viscoelastic damping.

3. **Bulk waves**. Enter the **Longitudinal velocity (m/s)** and the **Shear velocity (m/s)**. To consider viscoelastic damping, you also need to fill **Longitudinal attenuation** and **Shear attenuation**. From the **Attenuation unit** drop-down menu, you can choose one of four units in which you enter the attenuation values. These are **Np/λ**, **dB/λ**, **Np/m**, and **dB/m**. If you choose **Np/m** or **dB/m**, the **At frequency (kHz)** field becomes active. Enter the frequency at which the **Longitudinal attenuation** and the **Shear attenuation** were measured.

Notice that if you change a material parameter, all parameters which depend on it are updated automatically. After you have edited your material, assign a **New material's name** and press **Save material**. If you want to remove a material from the list, select it from the **Material** drop-down menu and press **Delete material**.

The **Anisotropic materials** (G2) panel works in the same way except that you can only enter the **Engineering constants** or the **Stiffness components (GPa)**, but not the **Bulk wave velocities (m/s)**. These are calculated from the **Stiffness components (GPa)** and the **Mass density** by using the equations listed in Appendix B.

Notice, that the direction 1 is along the fibers, while 2 and 3 are normal to it. 2 is in-plane and 3 is out-of-plane. You can use the **Class** drop-down menu to choose between **Orthotropic**, **Transversely isotropic**, and **Cubic** materials. If you select **Transversely isotropic**, four of the nine engineering constants and four of the nine stiffness components will be inactive so that you just need to enter the remaining five independent engineering constants or stiffness components. If you select **Cubic**, six engineering constants and six stiffness components will be inactive so that you just need to enter the remaining three independent engineering constants or stiffness components.

In the **Bulk wave velocities** section, you get the **Longitudinal velocity**, **Fast shear velocity**, and the **Slow shear velocity** of the three bulk wave types for propagation along the three principal axes 1, 2, and 3. Notice that in transversely isotropic materials, the bulk wave velocities are equal for propagation along the axes 2 and 3 (normal to the fibers), i.e., for propagation in the plane of (transverse) isotropy. In cubic materials, all longitudinal velocities are equal and all transverse velocities are equal. In Appendix B, it is shown how to calculate those velocities.

In the **Fluids** (G3) panel, you can add fluids or solid coupling media like the Olympus wedge. The latter is meant for the calculation of the coincidence angle dispersion diagrams available in the **Isotropic**, **Anisotropic**, and **Polar diagrams** tabs. Fluids are currently treated as non-viscous.

4.9 Advanced

The **Advanced** tab offers options, which do not need to be changed on a regular basis, but which help to overcome specific dispersion curve tracing problems. There are three panels. The **Phase velocity sweeps** (H1) and the **Frequency sweeps...** (H2) panels take effect on the standard 1-D tracing algorithm, while the **2-D tracing settings** (H3) panel affects the 2-D tracing algorithm. **Phase velocity sweeps** (H1) controls the range where the 1-D tracing algorithm takes equal steps in frequency. **Frequency sweeps...** (H2) only controls the narrow range close to the cut-off frequency, where the algorithm takes equal steps in phase velocity. In (H1) and (H2) are two columns, where

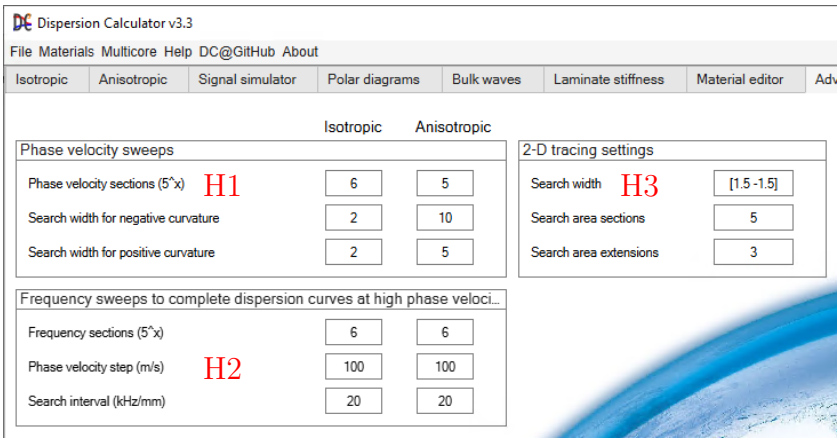


FIG. 28. The Advanced tab.

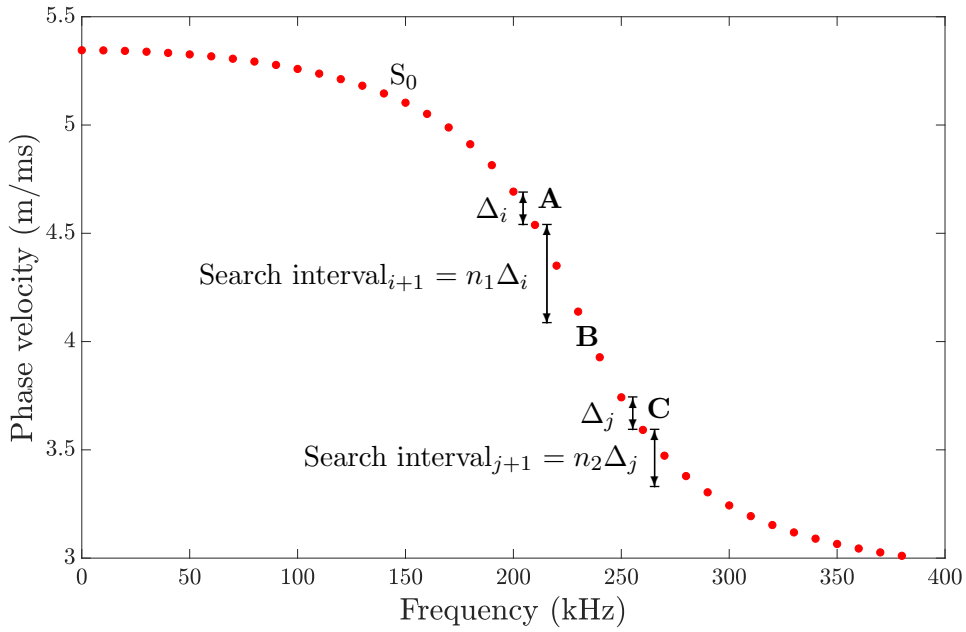


FIG. 29. The search interval as it is applied by the 1-D tracing phase velocity sweeps.

the left one is for dispersion curve tracing in the **Isotropic** tab, and the right one is for the **Anisotropic** tab.

Let us discuss the **Phase velocity sweeps** (H1) panel. The number of **Phase velocity sections** determines the ability to find a modal solution. At each frequency step, starting at low frequency, a first guess is made in which phase velocity search interval the modal solution for a particular mode should lie. This is necessary to avoid an extremely slow curve tracing. It is then checked, whether there is a modal solution in the search interval. A modal solution is defined by a simultaneous sign change and minimum in the characteristic function amplitude. If yes, then a bisection algorithm starts to converge upon the root until the desired **Phase velocity accuracy** is reached. If no root is found, an x-loop starts to divide the search interval into 5^x sections, until a root in a section is found, and then the bisection of the critical section starts unless the desired **Phase velocity accuracy** has already been reached. If the search interval covers any previously traced lower mode(s), those solution(s) will be ignored in the tracing of the current mode. The number of **Phase velocity sections** x as the power of five sets the maximum number of sections into which the search interval is divided. In order to understand the **Lamb wave search width for negative curvature**, please have a look at Fig. 29. It shows the tracing of the S_0 Lamb wave in a 10 mm thick free aluminum alloy 1100 plate. Consider situation **A**. Suppose the sample at 210 kHz has already been obtained, and we are now searching for the 220 kHz sample. It is expected at a lower phase velocity. It is therefore taken the difference of the phase velocities between the 200 kHz and the 210 kHz sample Δ_i. The curvature is determined to be negative here. The 220 kHz sample is expected to lie within the search interval$_{i+1}$, which reaches from the 210 kHz sample's phase velocity down by the multiple $n_1 \cdot \Delta_i$, where n_1 is the **Lamb**

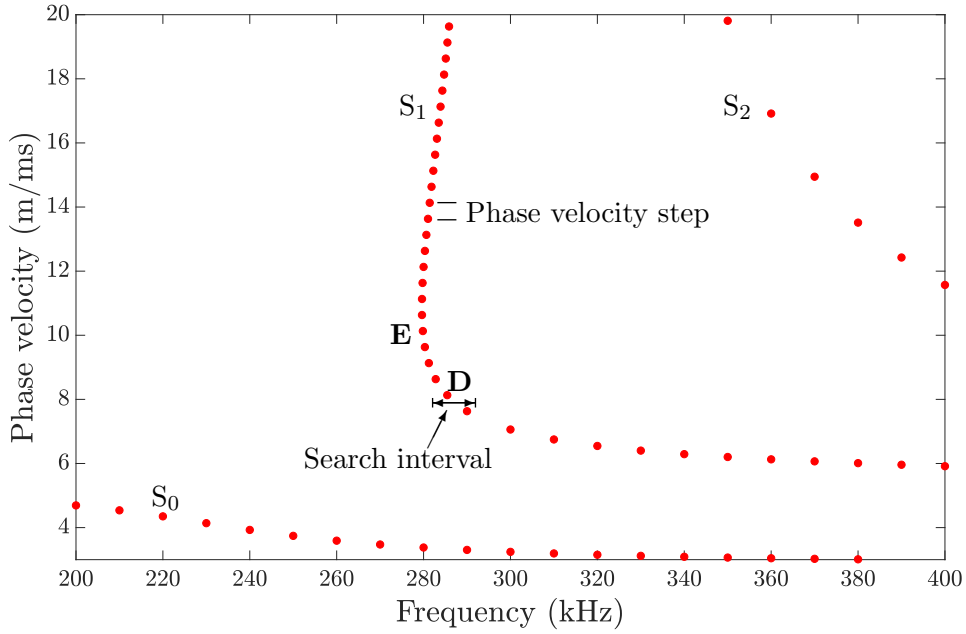


FIG. 30. The search interval and the phase velocity step as used by the 1-D tracing frequency sweeps.

wave search width for negative curvature. The 220 kHz is indeed found there. As the tracing progresses, we pass the turning point **B** and reach situation **C**. Since the curvature is now positive, DC uses the **Lamb wave search width for positive curvature** n_2, which is smaller than the value for negative curvatures n_1. The **Shear horizontal wave search width** is used for shear horizontal waves in anisotropic media only. In isotropic materials, the shear horizontal modes are calculated by an analytic equation so that dispersion curve tracing is unnecessary.

Since the higher order modes are more or less parallel to the phase velocity axis at high phase velocities, the usual phase velocity sweeps at fixed frequencies do not work there very well, and the dispersion curves are often incomplete. Therefore, **Frequency sweeps to complete dispersion curves at high phase velocity** (H2) are performed up to the top phase velocity required in the dispersion diagram. The **Frequency sections** work similarly to the **Phase velocity sections**. Higher values in **Phase velocity sections** and **Frequency sections** increase the chance to find modal solutions, and therefore accomplish the tracing successfully at the cost of processing time. However, setting low numbers does not necessarily cause problems in the curve tracing because there is a routine involved that replaces missing samples. This routine fits the already known samples and extrapolates them to the current frequency step. However, if three consecutive samples have been missed, the tracing of the current curve stops and continues with the next mode. Please consider Fig. 30. After the tracing of the fundamental modes, the higher order modes are traced. S_1 comes first. The cut-off frequency at 20 m s^{-1} is 286.129 kHz. Since we have a **Frequency step** of 10 kHz, the phase velocity sweeps starts at 290 kHz, leaving the curve incomplete below 290 kHz. The frequency

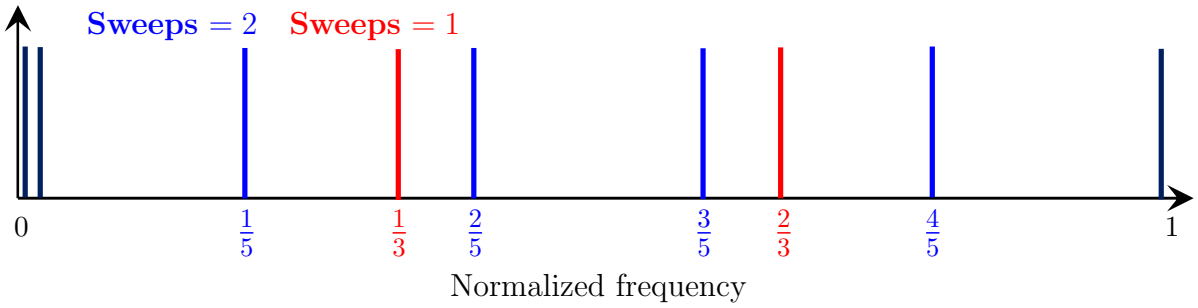


FIG. 31. Distribution of the extra sweeps along the frequency range. Black: two fixed sweeps are performed close to zero frequency and one close to **Frequency limit**. Red: when the **Sweeps** factor is set to 1 (= default), two evenly distributed sweeps are performed at 1/3 and 2/3 of **Frequency limit**. Blue: when setting the **Sweeps** factor to 2, four sweeps (= 2 · **Sweeps**) are performed. The maximum possible number of evenly distributed sweeps is $2 \cdot 8 = 16$.

sweeps continue the curve at point **D**, which is above the 290 kHz sample by an amount of **Phase velocity step**. This point is within a search interval spanning from a certain offset above 290 kHz to 290 kHz−**Search interval** in kHz. The aforementioned offset is needed to overcome point **E**, where the curve turns to higher frequencies again. A more detailed description of this dispersion curve tracing algorithm can be found in Ref. [43].

Prof. Lowe describes a different tracing routine in Refs. [32, 60, 61], which he applies in DISPERSE. An extrapolation method is used to make a guess for the next sample point. This works well in the wavenumber space since the curves are more straight there than in the frequency space.

Let us now focus on the 2-D dispersion curve tracing algorithm. This algorithm is a new development introduced in DC v3.2, replacing the first one that came with DC v2.0. The term "2-D" refers to the fact that the search for roots in a dispersion equation is performed in the complex wavenumber plane, rather than on the real wavenumber axis only, as the case in the 1-D tracing algorithm described above. DC computes the complex wavenumber ξ by

$$\xi = \frac{\omega}{c} = \frac{\omega}{c_{\text{real}} + \mathrm{i}c_{\text{imag}}}, \tag{6}$$

where ω is the angular frequency ($= 2\pi f$) and c is the complex velocity with the real part c_{real} and the imaginary part c_{imag}. From this, the phase velocity c_{p} is given by

$$c_{\text{p}} = \frac{\omega}{\xi_{\text{real}}}, \tag{7}$$

and attenuation in Nepers per meter by

$$\alpha = \xi_{\text{imag}}. \tag{8}$$

The algorithm starts by performing sweeps at certain frequencies to find modal solutions, which serve as starting points for dispersion curve tracing. There are three sets of sweeps:

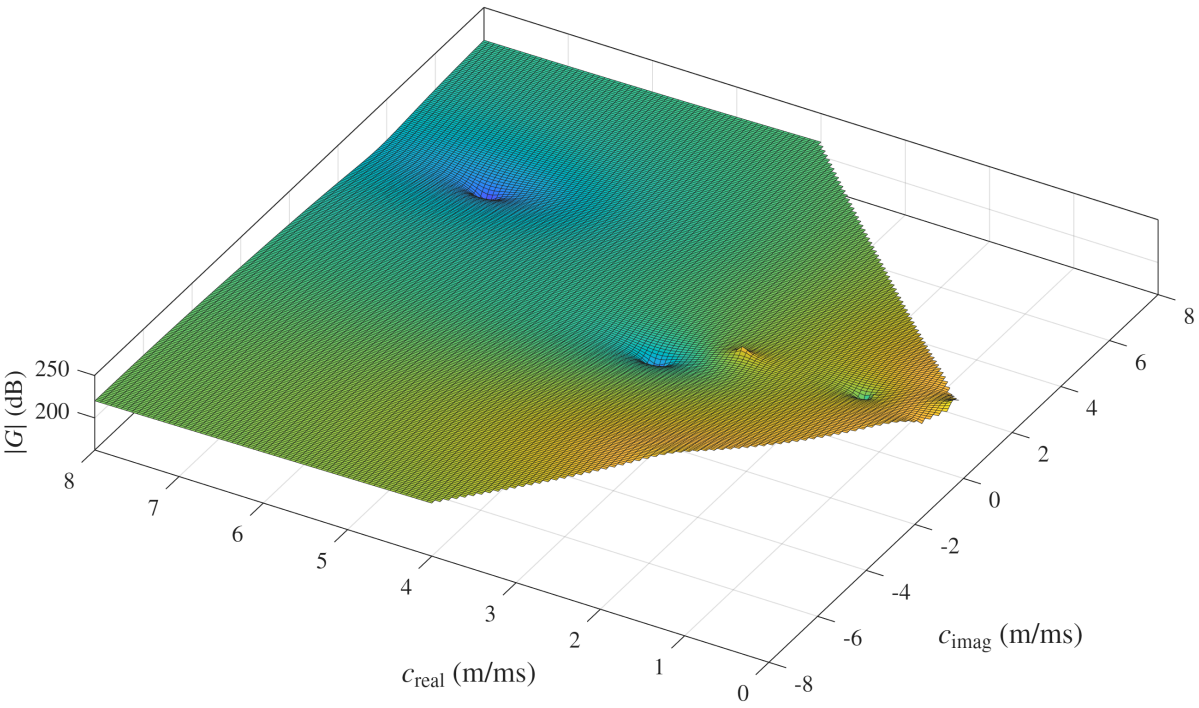


FIG. 32. Absolute value of the complex symmetric Rayleigh-Lamb equation calculated for a 1 mm thick viscoelastic epoxy plate (DC material: "Epoxy_DISPERSE_Viscoelastic") in vacuum at 1080 kHz during an initial sweep. The equation is evaluated on a 128×256 grid in the complex velocity plane. There are three minima from which DC starts dispersion curve tracing.

- Sweeps at the cut-off frequencies obtained by the frequency sweep (**Search**).
- Extra sweeps, namely two close to zero frequency (only one for SH and T modes), one close to the **Frequency limit**, and a number of (two multiplied by **Sweeps**) evenly distributed sweeps, see Fig. 31.
- In case of fluid-loading (but not for SH and T modes), the above extra sweeps are carried out first for finding the modes growing in the fluid (leaky modes) and are then repeated for modes decaying in the fluid (Scholte modes).

Figure 32 shows the dispersion equation amplitude absolute value obtained in the first type of initial sweep at a cut-off frequency of 1080 kHz in a 1 mm thick viscoelastic epoxy plate. As the complex velocity is independent of the frequency, DC chooses the search range in the complex velocity plane, and then calculates the wavenumbers from it according to Eq. 6. This is so in the initial sweeps and in the following dispersion curve tracing as well. In the complex velocity plane, c_{real} and c_{imag} are related to the phase velocity and attenuation, respectively. For $c_{\text{imag}} = 0$, we have $c_{\text{p}} = c_{\text{real}}$. The search range in this type of sweep is limited by $c_{\text{real}}^{\text{max}} =$ **Phase velocity limit**, and $c_{\text{imag}}^{\text{max}} = \pm$**Phase velocity limit**. Searching for negative c_{real}, respectively c_{p}, is pointless as the solutions are equivalent to those for positive c_{p}, just propagating in the opposite

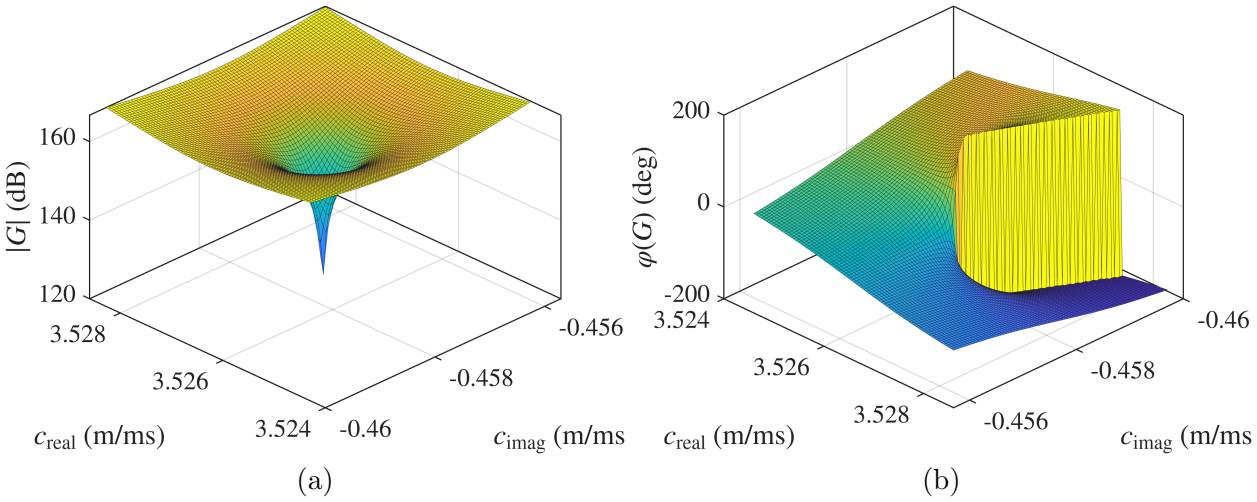


FIG. 33. (a) Absolute value and (b) phase angle of the complex symmetric Rayleigh-Lamb equation calculated for a 1 mm thick viscoelastic epoxy plate in vacuum at 1080 kHz. The minimum in (a) represents a leaky Lamb wave solution located at $c_{\mathrm{p}} = 3.585\,96\,\mathrm{m\,s^{-1}}$, $\alpha = 245.573\,\mathrm{Np\,m^{-1}}$. The strong phase change between sample points neighboring the minimum signifies that it is associated with a zero and that it is therefore a valid modal solution.

direction. However, DC does search for solutions in the positive and negative ranges of c_{imag}. Following the rule of division of complex numbers, the *negative* range of c_{imag} corresponds to the *positive* range of ξ_{imag}, respectively to positive damping, and vice versa. In Fig. 32, there are three minima indicating modal solutions. Two of them are in the negative c_{imag} range. These are forward propagating modes where the phase and energy velocities have the same sign. The third minimum has a positive c_{imag}, representing a backward propagating mode with opposite signs in the phase and energy velocities. Backward propagating modes have negative attenuation since the propagation direction is defined with respect to the phase velocity, not the energy velocity. To ignore more strongly damped solutions with complex velocities for which $\xi_{\mathrm{imag}}/\xi_{\mathrm{real}} > 2$ in the sweep, the **Attenuation limit** has been set to 2. However, it is important to note that once DC starts dispersion curve tracing, it does *not* stop tracing when **Attenuation limit** is exceeded. In Fig. 32, the dispersion equation amplitude has been calculated for a coarse quadratic grid of 128 samples in c_{real} and $2 \cdot 128$ samples in c_{imag}. In DC, the minimum number of **Sweep sections** can be set to 256, resulting in a grid of 256×512 samples. Increasing this number improves the chance to find weakly pronounced minima in the dispersion equation amplitude.

If such a sweep does not find any modal solution at the cut-off frequency, the sweep is repeated at the next few frequency steps until a frequency/thickness range of $50\,\mathrm{kHz\,mm^{-1}}$ has been covered. This procedure continues for all cut-off frequencies. After that, the extra sweeps are performed to find modes that could possibly have been missed so far. The difference in these sweeps is that $c_{\mathrm{real}}^{\mathrm{max}}$ and $c_{\mathrm{imag}}^{\mathrm{max}}$ are limited to a smaller speed; in

the **Isotropic** tab, the longitudinal wave speed is used. As the same number of **Sweep sections** is applied, this leads to a more dense grid. When fluid-loading is present, the extra sweeps are repeated with $c_{\text{real}}^{\text{max}}$ and $c_{\text{imag}}^{\text{max}}$ limited by the speed of sound in the fluids because Scholte waves cannot exceed this speed. In contrast to the sweeps at the cut-off frequencies, extra sweeps are performed exclusively at selected frequencies, i.e., there is no continuation at the next frequency steps if no new solution has been found. Importantly, DC ignores previously traced dispersion curves in the sweeps as well as during dispersion curve tracing.

After each sweep, DC performs dispersion curve tracing, starting from the found solutions: first by increasing the frequency until **Frequency limit** is reached, and then by decreasing the frequency until (nearly) zero frequency. The tracing algorithm stops intentionally only when either **Frequency limit**, zero frequency, or **Phase velocity limit 2** are reached. While changing the frequency step by step, the algorithm tries to follow the solution as it moves in the complex velocity plane. At each frequency step, the tracing algorithm uses linear extrapolation based on the two preceding solutions to guess where the solution will be at the current frequency. Around the supposed complex velocity, DC places a search range spanned in the real and imaginary parts by the shifts between the two preceding solutions multiplied by the **Search width**, defined in the **2-D tracing settings** (H3). The so defined search area is sampled in a standard 5×5 equally spaced grid, and the specific dispersion equation is evaluated at these 25 samples. Figure 33(a) shows the absolute value of the dispersion equation amplitude G, evaluated at 81×81 sample points in this example. The minimum indicates a modal solution candidate. Next, DC uses two-dimensional bisection to converge on the minimum: the algorithm takes the outer boundaries of the four squares adjacent to the minimum and evaluates another grid of 25 samples within these boundaries. This process continues until the desired **Phase velocity accuracy** has been reached in both c_{real} and c_{imag}. DC prefers the bisection method over more efficient root-finding algorithms such as Muller's method because it is the most robust method, and robustness is considered more important than speed. As a final check, the phase angles of the complex amplitude at the four samples directly neighboring the minimum are calculated. The criterion for a zero is that there must be a phase change of at least 90° between one of the two opposing pairs. Passing this test means that a valid modal solution has been found. This is indeed the case in Fig. 33(b).

There are two possible reasons why the algorithm cannot find a solution in a defined search area in the complex velocity plane. Either the solution lies outside the search area, or the 5×5 samples grid is too coarse to detect a minimum. The algorithm has effective and sophisticated measures to deal with such situations. In short, in case a solution has not been found, the search will be repeated with a finer grid, i.e., with a higher density of samples, but with an increased size in c_{real} and c_{imag}, too. There exist different strategies of how the number of samples is increased, and whether the grid will be square or rectangular. These strategies are applied to solve various complicated situations that can occur. **Search area sections** gives the number of how many times a yet unsuccessful search is performed with an ever finer grid covering an ever larger complex plane area. **Search area extensions** gives the number of how many times the above

search loop is repeated with an extra increase in the size of the search grid. Suppose you have five **Search area sections** and three **Search area extensions**. In case a modal solution is not found in the first search with the 5×5 grid, the search will be repeated a maximum of four more times, every time with a finer grid until a solution gets detected. After that, if still unsuccessful, the size of the grid will be extra enlarged by a factor of two in both dimensions so that it will be four times larger in area. At the same time, the number of samples will be increased also, but only a little bit so that the density will be less than in the previous grids. The five searches will be repeated a second time, with the base grid enlargement factor of four and with further enlargements in each of the five iterations, followed by a third time again with a four times larger base grid plus further enlargements in every of the five iterations. If the solution is still not found, the previous solutions will be fitted and extrapolated. If no solutions have been found at three consecutive frequency steps, the dispersion curve tracing stops and continues with the next mode.

To improve robustness, the 2-D tracing algorithm has quite some flexibility in dealing with complicated situations. Only some prominent features can be described here briefly:

- When the slope of a dispersion curve exceeds a certain critical value, DC takes gradually smaller steps in frequency.
- DC ignores any minimum in the grid that coincides with any previously traced dispersion curve.
- DC discards a solution as an outlier in certain rare and well-defined cases if it lies too far off from the previous solution.
- In case multiple minima are found in the grid, DC will pick the one which lies closest to the last solution found on the currently traced dispersion curve.
- If DC does not find a minimum, it checks if there are boundary minima, i.e., valleys in the dispersion equation amplitude, which cut into one of the four boundaries of the grid and which get deeper toward the boundary. Such a valley can point toward a valid minimum. The grid is therefore extended toward the suspected minimum until it is found. Boundary minima on the four corners are ignored.
- In case multiple boundary minima are found in the grid, DC will extend the grid towards the one which lies closest to the last solution found on the currently traced dispersion curve.

5 Examples

5.1 T800M913 single layer plate @ 45°

We want to obtain the dispersion diagram for a unidirectional layer of the fiber-matrix system T800M913. The thickness shall be 2 mm and the wave propagation along an angle of 45° with respect to the fiber orientation. Since a single layer is symmetric, we can determine symmetric and anti-symmetric modes. Lamb and shear horizontal waves are coupled. Using the default settings, there are two problems:

1. Take a look at situation **A** in Fig. 34. If you are using the default **Step** of 0.5 kHz for the frequency sweep in the **Search higher order modes** panel, the cut-off frequency of A_{16} is not found and the mode not traced. Set a smaller **Step** of 0.05 kHz. A_{16} will now appear in the upper left output window below the **Search higher order modes** panel and can be traced.

2. Consider situation **B**. If you are using the default **Frequency step** of 5 kHz, the tracing of A_{12} fails to cross through the lower order modes at point **B** so that the part beyond 4.5 MHz is missing. To solve the problem, set a **Frequency step** of 2 kHz.

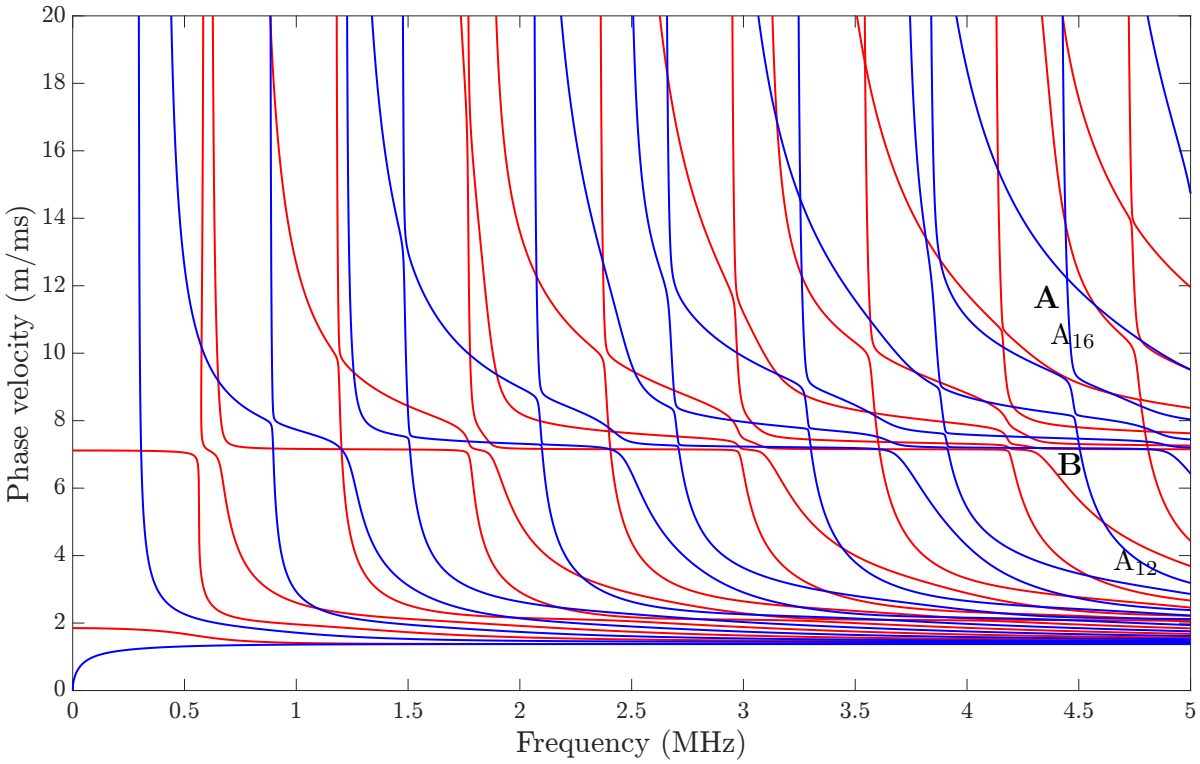


FIG. 34. Dispersion diagram for wave propagation along 45° with respect to the fiber direction in 2 mm thick unidirectional T800M913.

5.2 T800M913 single layer plate @ 45° (Force 2-D tracing)

We stay with the previous problem, but now we want to find some nonpropagating modes, too. As it is a case with no damping, we need to activate **Force 2-D tracing** to find these extra modes. When using 2-D tracing in nonattenuated cases, it is generally a good idea to use a smaller than default **Frequency step**: set it to 1 kHz here. Then set **Attenuation limit** to 10 and **Sweep sections** to 512 for a more thorough search. Notice that the 2-D tracing algorithm traces the dispersion curves up to a **Phase velocity limit 2** of 1000 m s^{-1} by default. Adjust the **Y-axis** in the **Dispersion diagrams** panel to see the full range.

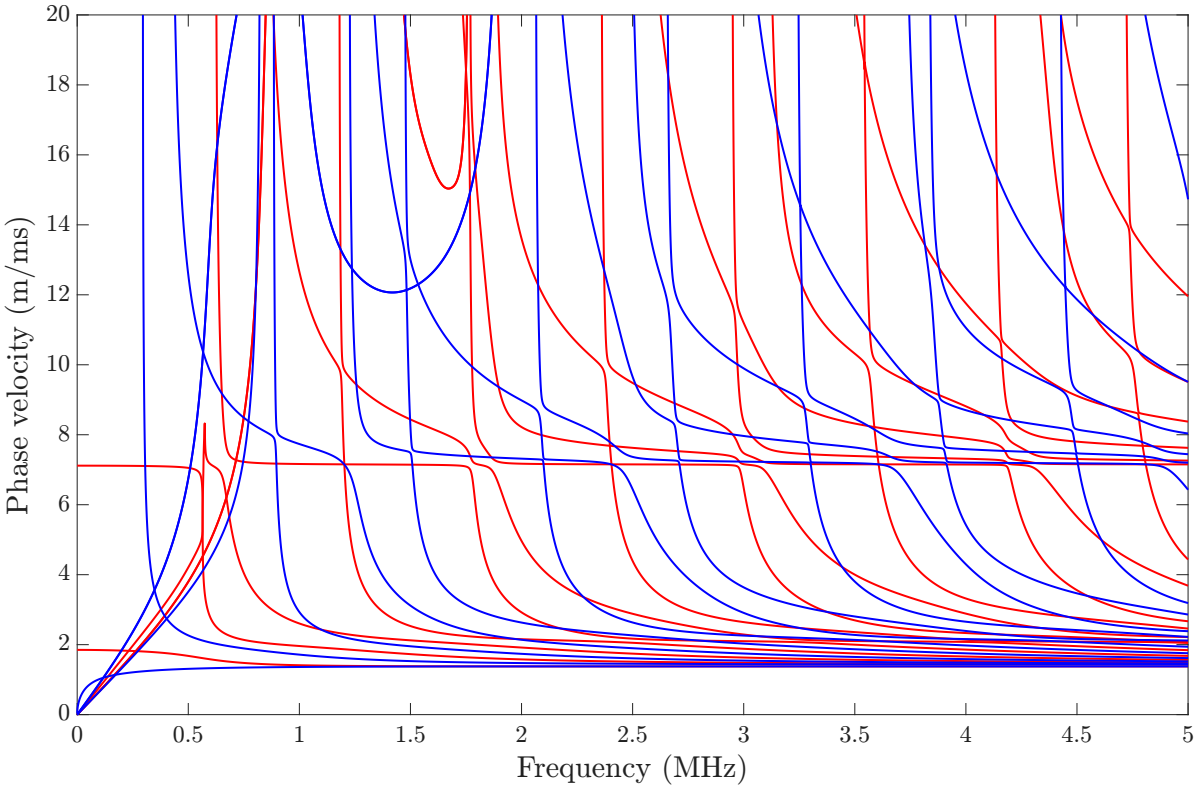


FIG. 35. Dispersion diagram for wave propagation along 45° with respect to the fiber direction in 2 mm thick unidirectional T800M913. By using the 2-D dispersion curve tracing algorithm, we can find some extra nonpropagating modes.

5.3 T800M913 $[0/90]_4$ plate @ 45°

Now, let us compute a 2 mm thick layup $[0/90]_4$ from the same material, and again we let the guided waves propagate along 45°. Since the layup is non-symmetric, symmetric and anti-symmetric waves cannot be distinguished. Lamb and shear horizontal waves are coupled. The dispersion diagram in Fig. 36 poses a significant challenge to the dispersion curve tracing algorithm because dispersion curves are very close to each other over wide frequency ranges, and there are difficult mode crossings. Using the default **Frequency step** of 5 kHz and **Lamb wave search width for negative curvature** of 10 and **Lamb wave search width for positive curvature** of 5 in the **Advanced** tab, dispersion curves are incomplete in many occasions. However, DC can master the challenge by setting a smaller **Frequency step** of 2 kHz and **Lamb wave search width for negative curvature** of 5 and **Lamb wave search width for positive curvature** of 5. The latter two options narrow down the phase velocity search range for each modal solution so that more neighboring modes are excluded and the modal solutions are found faster. The risk is that the tracing cannot follow a mode through sections where it bends abruptly (as the case in Sec. 5.1), but this is not the case here; the curves bend rather smoothly.

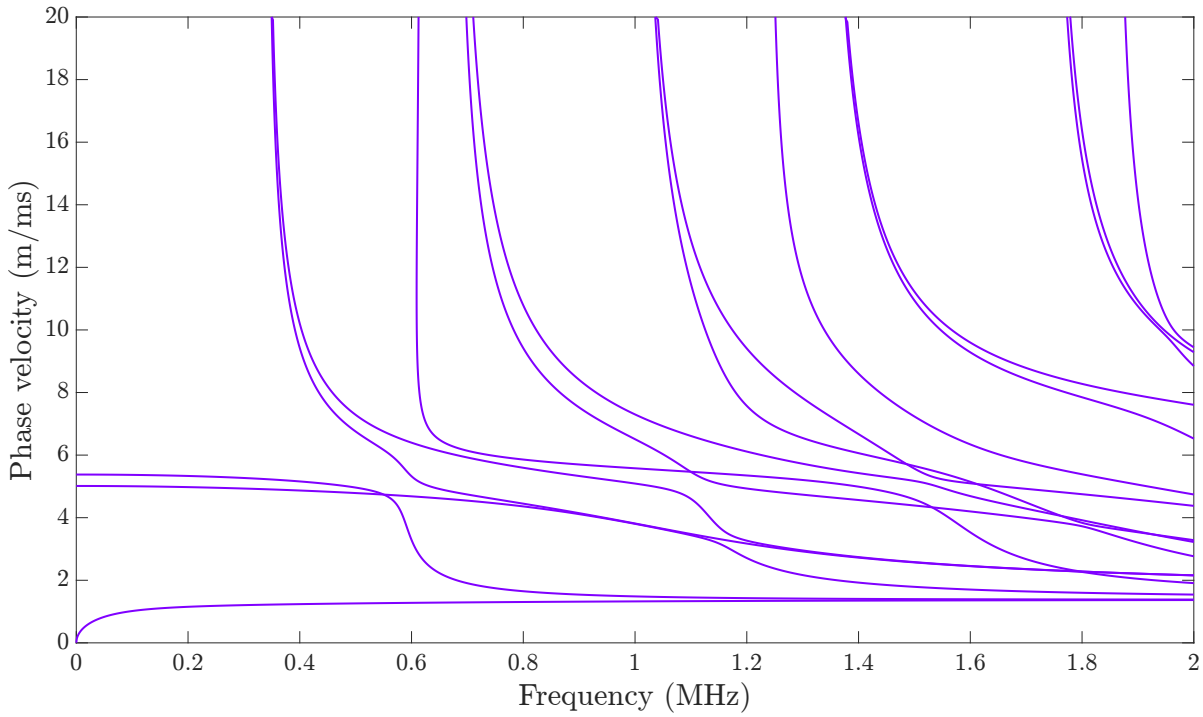


FIG. 36. Dispersion diagram for wave propagation along 45° in a 2 mm thick layup $[0/90]_4$ T800M913.

5.4 T800M913 $[0/90]_{100s}$ plate @ 0°

It was mentioned in the introduction that modern rocket booster pressure vessels can consist of up to 400 layers in certain areas. We want to calculate such kind of laminate. We assume a layup $[0/90]_{100s}$ with layer thicknesses of 0.125 mm yielding a 50 mm thick laminate. Despite the 400 layers, the dispersion diagram in Fig. 37 computes in less than three seconds on my work station (using multicore computing (**Multicore → Enable**)), thanks to optimizations done in DC v3.0. Lamb and shear horizontal waves decouple and can therefore be calculated separately. Figure 38 shows the displacement and stress field components of the S_0 Lamb wave at 50 kHz. The periodicity of the layup becomes obvious in the displacement u_1 and in the stresses σ_{11} and σ_{22}.

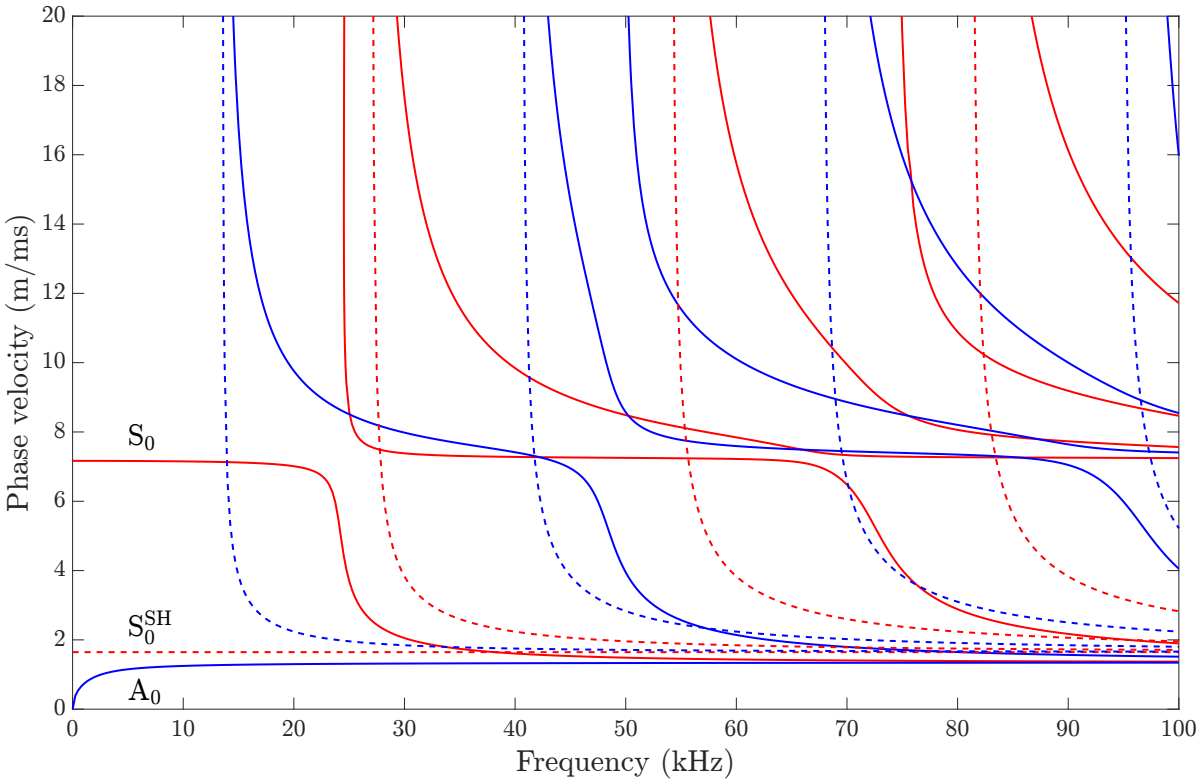
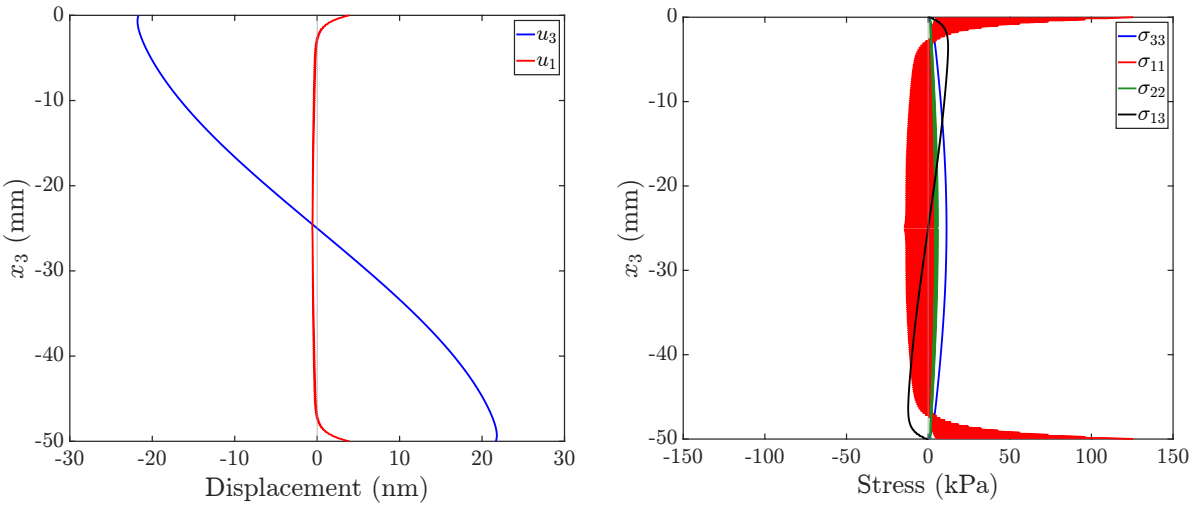


FIG. 37. Dispersion diagram for wave propagation along 0° in a 50 mm thick layup $[0/90]_{100s}$ T800M913.



(a) (b)

FIG. 38. (a) Displacement and (b) stress field components of the S_0 Lamb wave at 50 kHz in a 50 mm thick layup $[0/90]_{100s}$ T800M913.

5.5 CarbonEpoxy_Hernando_2015_Viscoelastic $[0/90/45/-45]_{2s}$ plate @ 45°

Let's see how the 2-D dispersion curve tracing algorithm performs in a viscoelastic $[0/90/45/-45]_{2s}$ CarbonEpoxy_Hernando_2015_Viscoelastic composite layup. The thickness shall be 2 mm and the wave propagation along an angle of 45° with respect to the fiber orientation. Set **Attenuation limit** to 10 to find more of the nonpropagating modes. There is no problem, the 2-D dispersion curve tracing algorithm just makes a good job.

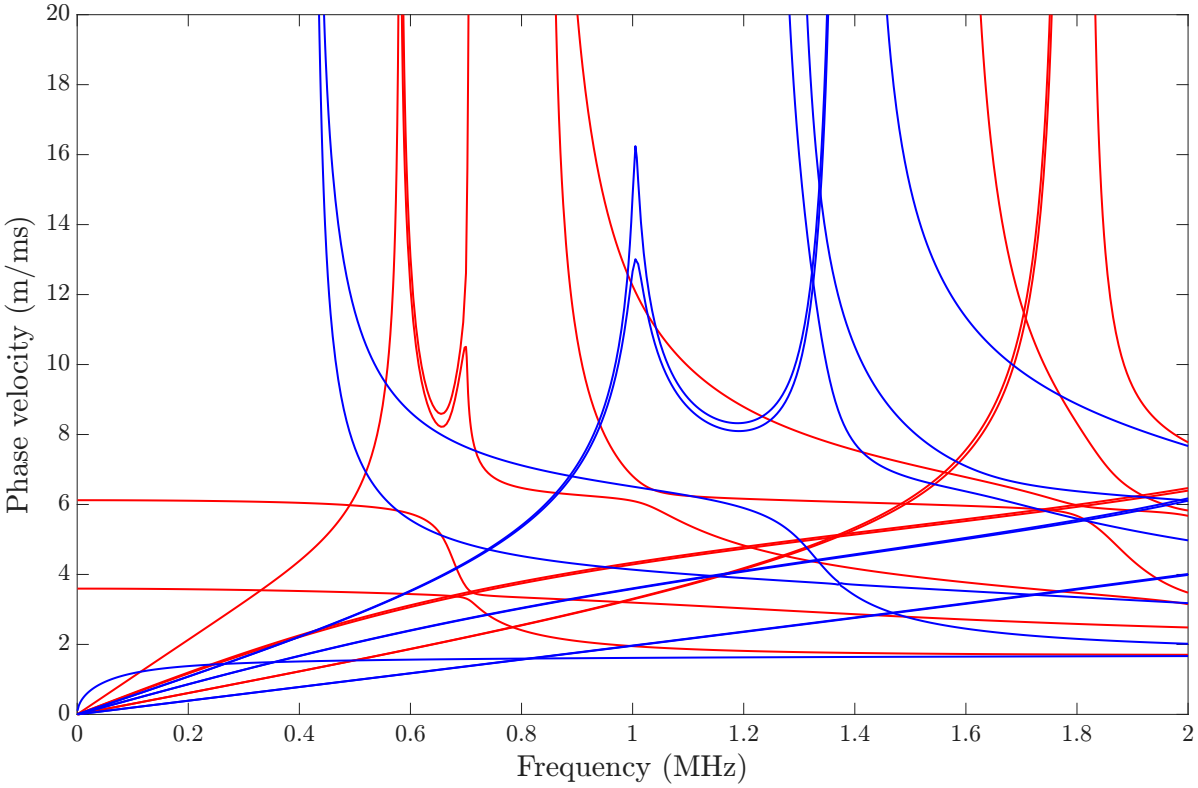


FIG. 39. Dispersion diagram for wave propagation along 45° in a 2 mm thick layup $[0/90/45/-45]_{2s}$ CarbonEpoxy_Hernando_2015_Viscoelastic.

5.6 Aluminum alloy 1100 pipe filled with water

We now consider an aluminum alloy 1100 pipe with an outer diameter of 10 mm and with an inner diameter of 8 mm. The pipe is filled with water. We calculate the dispersion diagram twice, and we only plot the longitudinal modes (red curves) and the flexural modes with the circumferential order $n = 1$ (blue curves). In the first run, we do not place a sink at the center (use a **Frequency step** of 1 kHz instead of 1.25 kHz to avoid a small problem in the tracing). This system has no attenuation. The dispersion diagram is shown in Fig. 41(a). In the second run, we place a sink at the center. This adds damping to the system. The result is in Fig. 41(b). Comparing both dispersion diagrams, we notice that, when the sink is placed, far less modes are present compared to the case without sink. Most of the modes in Fig. 41(a) are propagating almost entirely in the fluid-filling, with hardly any energy transport in the pipe wall. Different from these are a few modes which carry most of their energy in the pipe wall. These pipe modes remain when we put a sink at the center, which suppresses the bulk waves propagating from one side of the pipe wall through the fluid to the opposite side of the pipe wall. Usually, in pipe inspection, we are interested only in modes which propagate predominantly in the pipe wall. Let us compare for example F(1,12) and F(1,13) at 1.7 MHz from Fig. 41(a). The through-thickness displacements are plotted in Fig. 40. From Fig. 40(a), we see clearly that F(1,12) is a pipe mode, whereas F(1,13) in Fig. 40(b) is a fluid mode, which is rather unsuitable for inspecting a pipe. Hence, for this task, it is advantageous to suppress the fluid-modes by placing the sink.

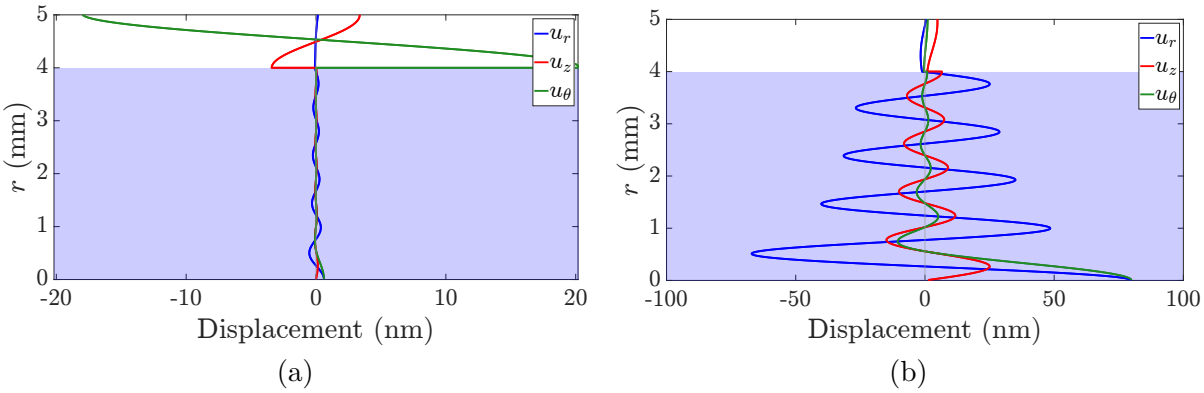
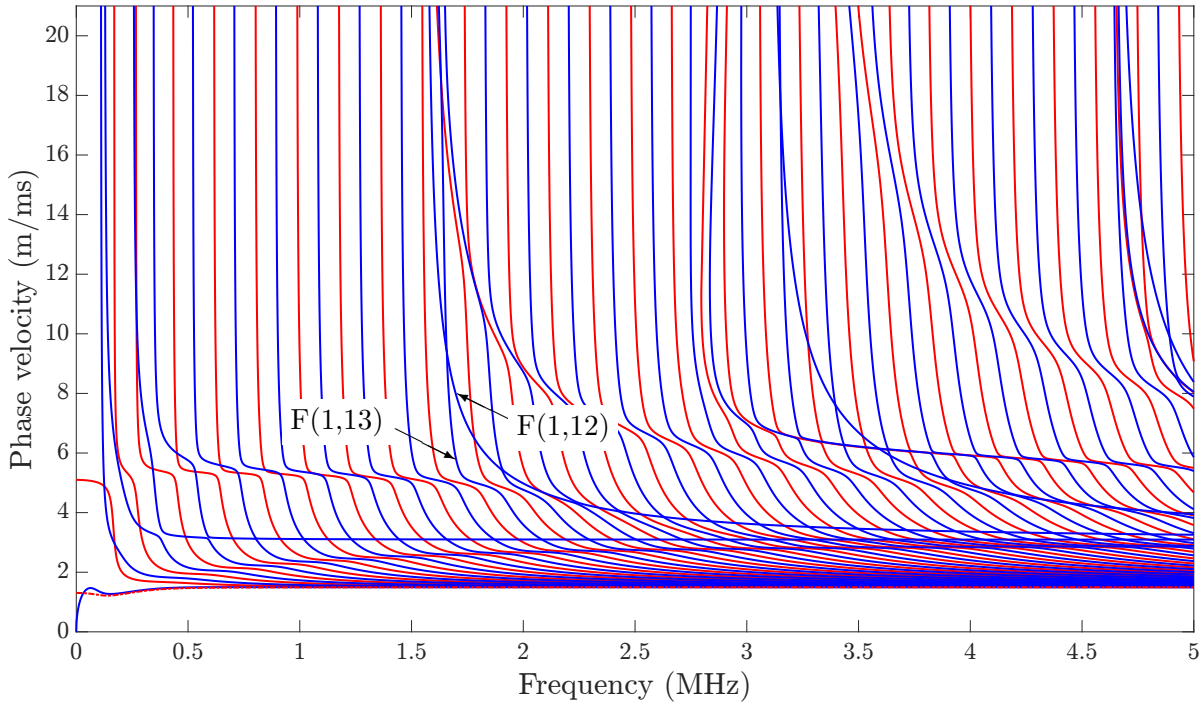
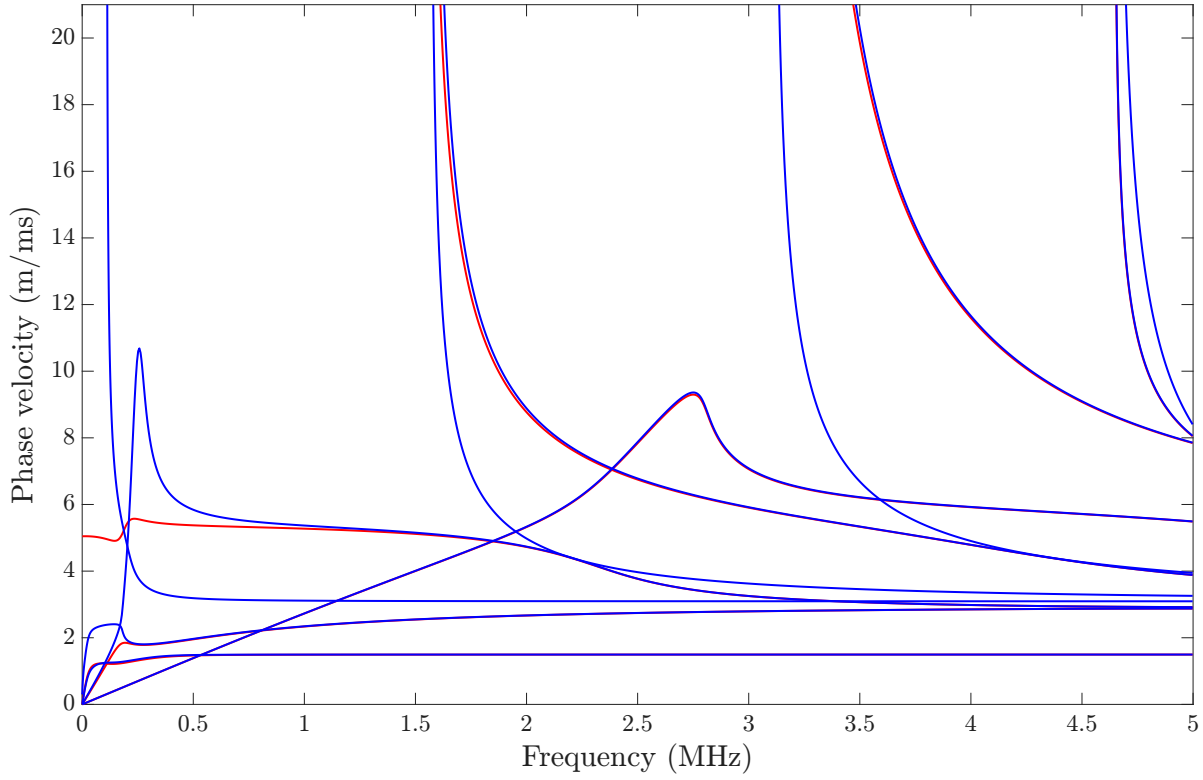


FIG. 40. Through-thickness displacements at 1.7 MHz for a 10 mm diameter, 1 mm wall thickness aluminum alloy 1100 pipe filled with water (no sink at the pipe center). (a) F(1,12). (b) F(1,13).



(a)



(b)

FIG. 41. Dispersion diagrams for a 10 mm diameter, 1 mm wall thickness aluminum alloy 1100 pipe filled with water. (a) No sink at the center. (b) Sink at the center.

5.7 Aluminum alloy 1100 circumference

Consider an aluminum alloy 1100 pipe with an outer diameter of 100 mm and with an inner diameter of 60 mm. We are interested in the Lamb-type and shear horizontal-type guided waves propagating in the circumferential direction. As the waveguide is, from the point of view of the guided waves, like a curved plate, the circumferential waves are non-symmetric. The phase velocity of circumferential waves is a function of the radius, and in DC, the phase velocity at the outer pipe surface (outer diameter, d_o) is displayed in the dispersion diagram, see Fig. 42. Notice that the fundamental shear horizontal mode C_0^{SH} is dispersive, unlike the case for S_0^{SH} in a flat plate. Also, what is S_0 in a flat plate turns out to be a higher order mode C_1 in a circumference.

Now consider an aluminum alloy 1100 pipe with an outer diameter of 100 mm and with an inner diameter of 98 mm. This is a thin-walled pipe. The dispersion diagram for circumferential waves is displayed in Fig. 43(a). Notice how the cut-off frequency of C_1 has shifted to very low frequency. As the ratio of the inner diameter to the outer diameter approaches unity, the waveguide behaves more and more like a flat plate. In the case considered here, the ratio is 0.98, and starting at this value, DC issues a recommendation to calculate a flat plate with a thickness of the wall thickness of the assumed pipe. Figure 43(b) shows the dispersion diagram for a 1 mm thick aluminum alloy 1100 plate for comparison. Already at a ratio of 0.98, both dispersion diagrams look very similar. As the ratio increases further, the equations for the circumferential become numerically unstable so that calculating a plate becomes absolutely necessary.

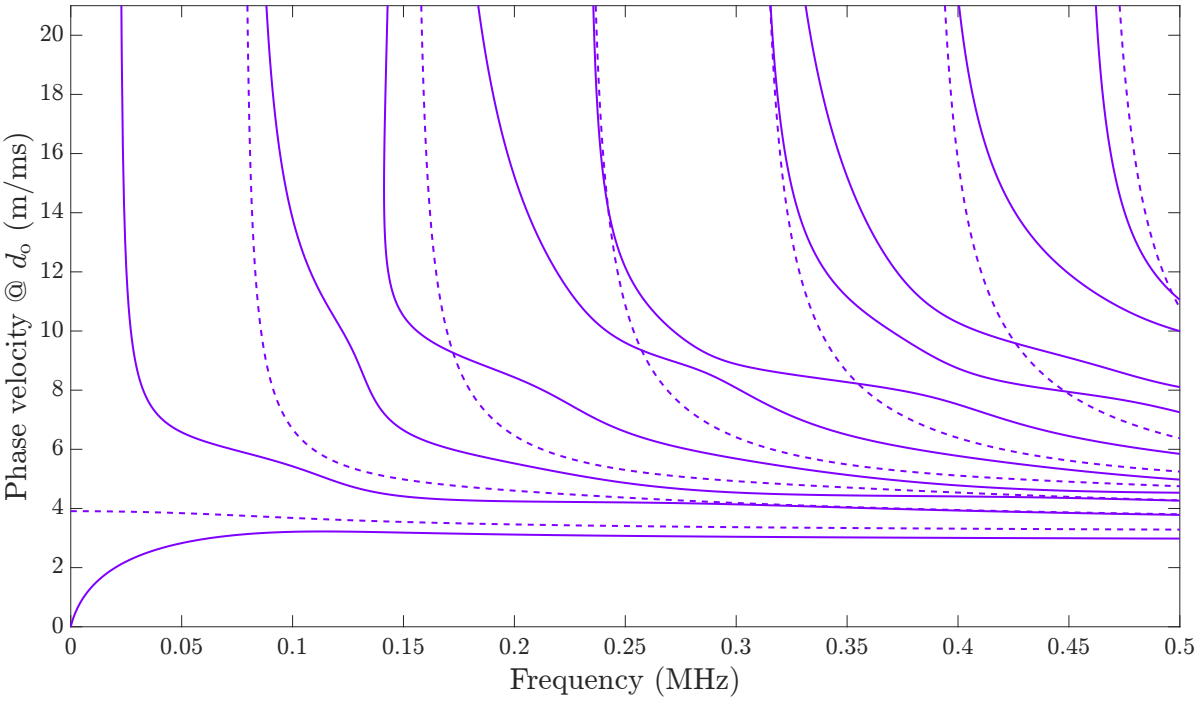
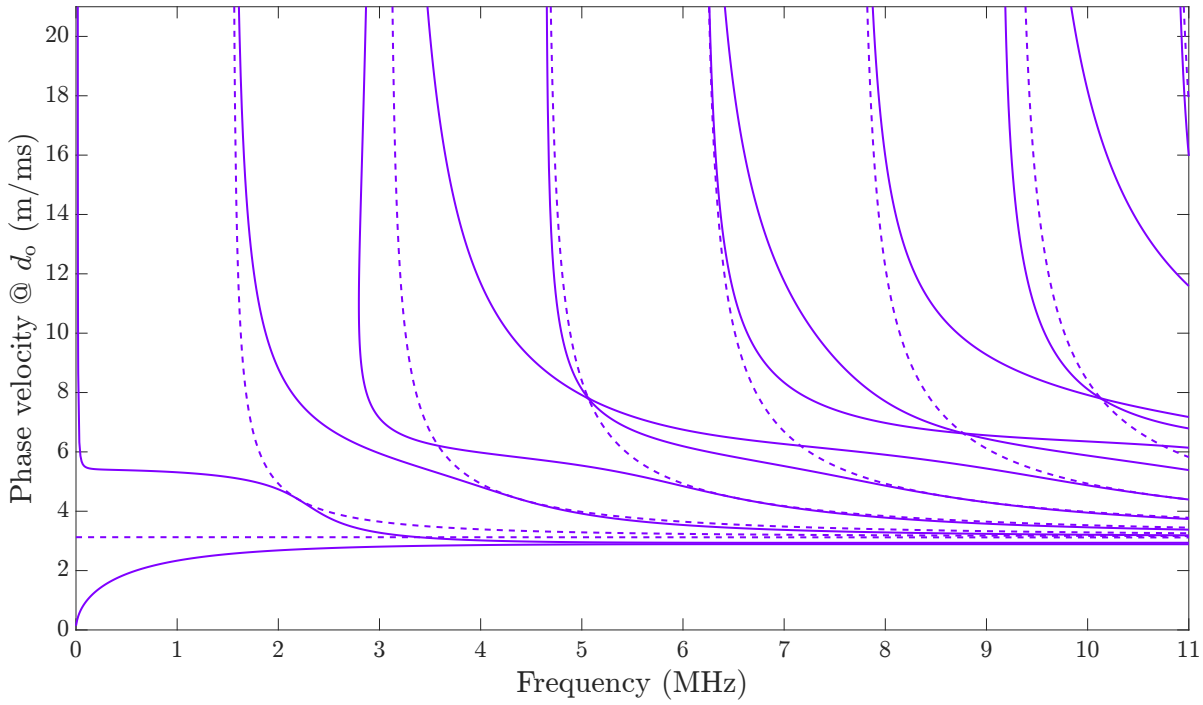
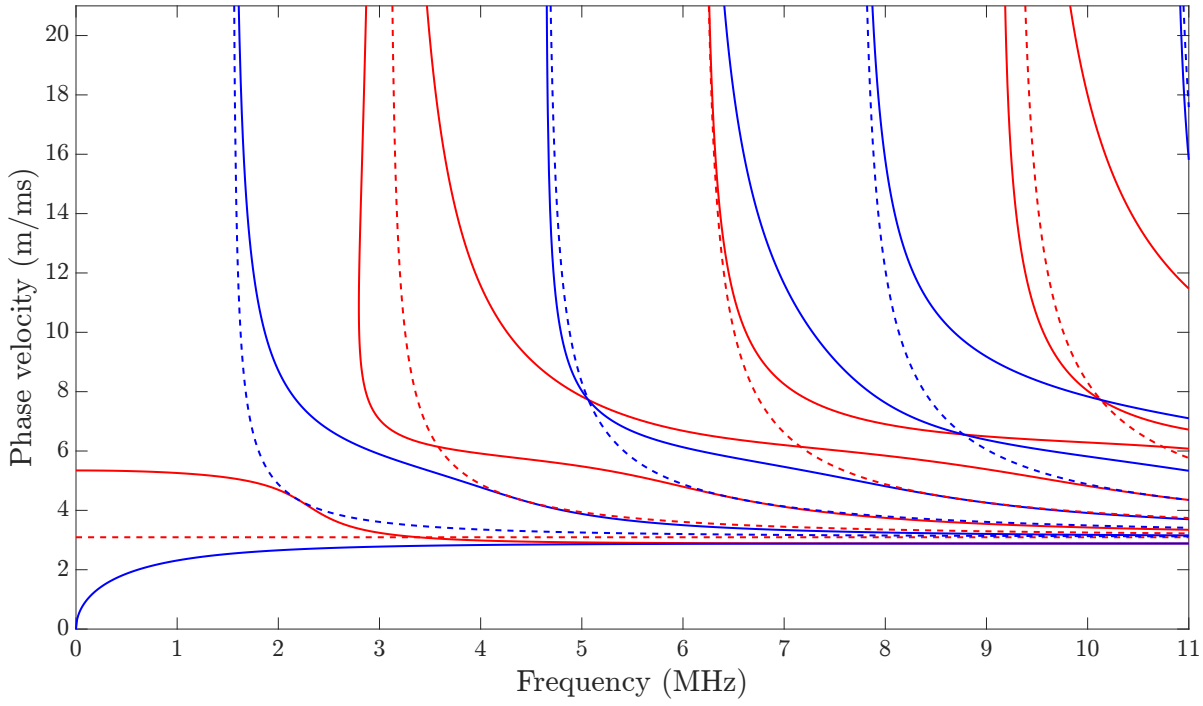


FIG. 42. Dispersion diagram for circumferential waves in a 100 mm diameter, 20 mm wall thickness aluminum alloy 1100 pipe.



(a)



(b)

FIG. 43. (a) Dispersion diagram for circumferential waves in a 100 mm diameter, 1 mm wall thickness aluminum alloy 1100 pipe. (b) Dispersion diagram for a 1 mm thick aluminum alloy 1100 plate.

A Group velocity and energy velocity of guided waves

The group velocity of a guided wave is the speed at which a wave packet propagates, i.e., the speed at which energy is transported. It is calculated by the well-known equation

$$c_{\mathrm{g}} = \frac{\partial \omega}{\partial \xi}, \tag{9}$$

where ω is the circular frequency ($= 2\pi f$), and ξ is the wavenumber of the guided wave. For simple dispersion equations like the Rayleigh-Lamb equations, which are given in the implicit forms $\Omega_{\mathrm{S}}(\omega, k) = 0$ and $\Omega_{\mathrm{A}}(\omega, k) = 0$, implicit differentiation can be used, i.e.,

$$c_{\mathrm{g}} = \frac{\partial \omega}{\partial \xi} = -\frac{\partial \Omega / \partial \xi}{\partial \Omega / \partial \omega}, \tag{10}$$

as shown by Bernard et al. [62]. Otherwise, Eq. (9) can be manipulated into

$$c_{\mathrm{g}} = c_{\mathrm{p}}^2 \left(c_{\mathrm{p}} - f \frac{\partial c_{\mathrm{p}}}{\partial f} \right)^{-1}. \tag{11}$$

To carry out Eq. (11) programmatically, the phase velocity dispersion curve is fitted so that its derivative can be taken from the fitted function. Although Eq. (10) is used in DC where possible, Eq. (11) is not applied as a general solution for two reasons. Firstly, the group velocity has physical significance only in non-dissipative systems, i.e., without attenuation caused by viscoelasticity or energy leakage to the surrounding medium (fluid-loading). With attenuation, the wavenumber turns complex so that Eq. (9) does not make sense anymore. The energy velocity is the more generic quantity. It is valid both with and without attenuation, where in the latter case, group and energy velocity are equivalent. Secondly, Eq. (9) gives only the group velocity component in the direction of wave propagation. Therefore, DC calculates the energy velocity whenever Eq. (10) cannot be used.

Let us introduce a coordinate system $x_i = (x_1, x_2, x_3)$, where a plate lies in the x_1-x_2-plane, x_1 is the propagation direction of the guided wave, and x_3 is normal to the plate. The total energy E_{total} is the energy per unit volume carried by a guided wave. It has two contributions, namely the strain energy E_{strain} and the kinetic energy E_{kin} so that $E_{\mathrm{total}} = E_{\mathrm{strain}} + E_{\mathrm{kin}}$, where

$$E_{\mathrm{strain}} = \frac{1}{2} \int_S \mathrm{Re}(\varepsilon_{ij} \sigma_{ij}^*) \mathrm{d}S, \qquad i, j = 1, 2, 3, \tag{12}$$

with the stress tensor σ_{ij} and the strain tensor ε_{ij}, and

$$E_{\mathrm{kin}} = \frac{1}{2} \rho \int_S v_i^2 \mathrm{d}S, \tag{13}$$

where integration of the respective energy densities is performed over the cross section S of the laminate, and where ρ is the material density and $v_i = \dot{u}_i$ is the particle velocity vector.

The power flow P_j is the energy flow per unit volume and unit time carried by a guided wave. It is obtained by integration of the power flow density (*Poynting vector*) p_j, indicating the magnitude and direction of the power flow

$$P_j = \int_S p_j \mathrm{d}S = -\frac{1}{2}\int_S \mathrm{Re}(\sigma_{ij} v_i^*)\mathrm{d}S = -\frac{1}{2}\int_S \mathrm{Re}\begin{bmatrix}\sigma_{11}v_1^* + \sigma_{21}v_2^* + \sigma_{31}v_3^* \\ \sigma_{12}v_1^* + \sigma_{22}v_2^* + \sigma_{32}v_3^* \\ \sigma_{13}v_1^* + \sigma_{23}v_2^* + \sigma_{33}v_3^*\end{bmatrix} \mathrm{d}S. \tag{14}$$

The star (*) indicates the complex conjugate. In absence of attenuation, there is only power flow along the propagation direction P_1, (and P_2 in coupled cases in anisotropic media) while P_3 (normal to the plate) is zero.

The energy velocity components $c_{\mathrm{e}1}$ and $c_{\mathrm{e}2}$ are calculated from the ratios of the power flow P_j to the total energy E_{tot} as

$$\vec{c}_{\mathrm{e}} = \frac{1}{E_{\mathrm{tot}}}\begin{bmatrix}P_1 \\ P_2 \\ 0\end{bmatrix} = \begin{bmatrix}c_{\mathrm{e}1} \\ c_{\mathrm{e}2} \\ 0\end{bmatrix}, \tag{15}$$

where the total energy is obtained with Eqs. (12) and (13) by

$$\begin{aligned} E_{\mathrm{tot}} = \frac{1}{2}\int_S \mathrm{Re}(\varepsilon_{11}\sigma_{11}^* + \varepsilon_{33}\sigma_{33}^* + \varepsilon_{23}\sigma_{23}^* + \varepsilon_{13}\sigma_{13}^* + \varepsilon_{12}\sigma_{12}^*)\mathrm{d}S, \\ + \frac{1}{2}\rho\int_S (v_1^2 + v_2^2 + v_3^2)\mathrm{d}S, \end{aligned} \tag{16}$$

(ε_{22} is zero in a plate), and the power flow components are calculated from Eq. (14) by

$$\begin{aligned} P_1 &= -\frac{1}{2}\int_S \mathrm{Re}(\sigma_{11}v_1^* + \sigma_{12}v_2^* + \sigma_{13}v_3^*)\mathrm{d}S, \\ P_2 &= -\frac{1}{2}\int_S \mathrm{Re}(\sigma_{12}v_1^* + \sigma_{22}v_2^* + \sigma_{23}v_3^*)\mathrm{d}S. \end{aligned} \tag{17}$$

Obviously, Eq. (15) describes a vector with a component $c_{\mathrm{e}1}$ along and a component $c_{\mathrm{e}2}$ normal to the wave propagation direction x_1. The magnitude of the energy velocity vector is given by

$$c_{\mathrm{e}} = |\vec{c}_{\mathrm{e}}| = \sqrt{c_{\mathrm{e}1}^2 + c_{\mathrm{e}2}^2}. \tag{18}$$

$\vec{c}_{\mathrm{e}}$ points in the direction of energy flow so that it indicates the actual direction of the guided wave beam (ray angle Φ_{r}). In general, Φ_{r} deviates from the wave propagation angle Φ by the skew angle γ, given by

$$\gamma = \Phi - \Phi_{\mathrm{r}}. \tag{19}$$

With the energy velocity vector components obtained from Eq. (15), we can calculate

$$\gamma = -\tan^{-1}\frac{c_{\mathrm{e}2}}{c_{\mathrm{e}1}}. \tag{20}$$

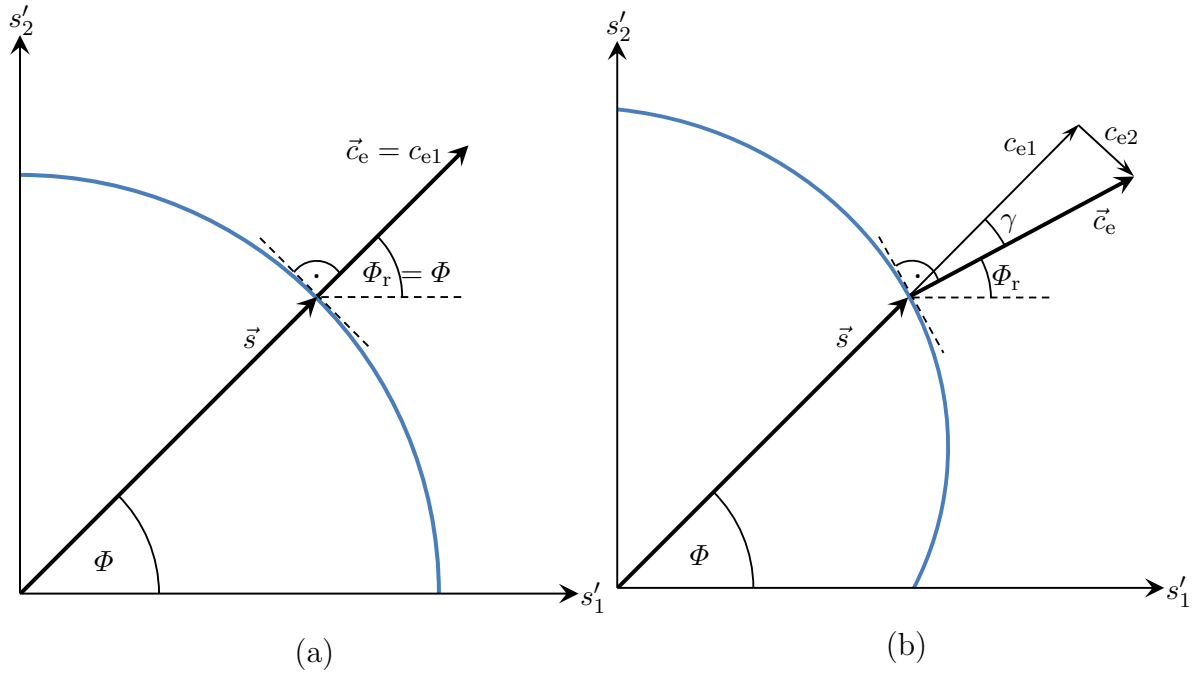


FIG. 44. The energy velocity vector. (a) In the decoupled case such as in an isotropic medium, the energy velocity direction coincides with wave propagation direction. (b) In the coupled case in anisotropic media, the energy velocity direction skews from the wave propagation direction.

The negative sign in Eq. (20) accounts for the fact that the guided wave beam, i.e., the energy flow, skews towards the fiber direction because energy can be transported more efficiently in that direction.

In decoupled cases such as in isotropic media or in anisotropic media for wave propagation along or normal to the fibers, Φ_r is equal to Φ. This situation is illustrated in Fig. 44(a), where the slowness is plotted versus the wave propagation angle in an isotropic medium. The slowness is the inverse phase velocity $s = c_p^{-1}$, and the energy velocity vector $\vec{c}_e$ is oriented perpendicular to the slowness profile. Since the slowness profile in an isotropic medium is a circle, the energy velocity vector is always aligned with the slowness vector and with the wave propagation direction, i.e., $\gamma = 0$ so that $c_{e2} = 0$ and $\vec{c}_e = c_{e1}$. By contrast, in coupled cases, i.e., for wave propagation other than along axes of symmetry in an anisotropic medium, the energy velocity direction is not anymore aligned with the wave propagation direction, but skews by an angle $\gamma \neq 0$, as illustrated in Fig. 44(b).

DC applies Eqs. (15) to (20) to every sample point in every phase velocity dispersion curve to obtain results as those presented in Fig. 46. Therein, the energy velocity components are illustrated for a coupled case, namely for wave propagation along an angle of $\Phi = 45°$ in a 1 mm thick layer T800M913. Figure 46(a) shows the energy velocity component along the wave propagation direction c_{e1}. The plot is useful for scenarios like air-coupled ultrasonic inspection where you send a guided wave in a certain (wave propagation) direction, and detect the signal after propagating a certain distance

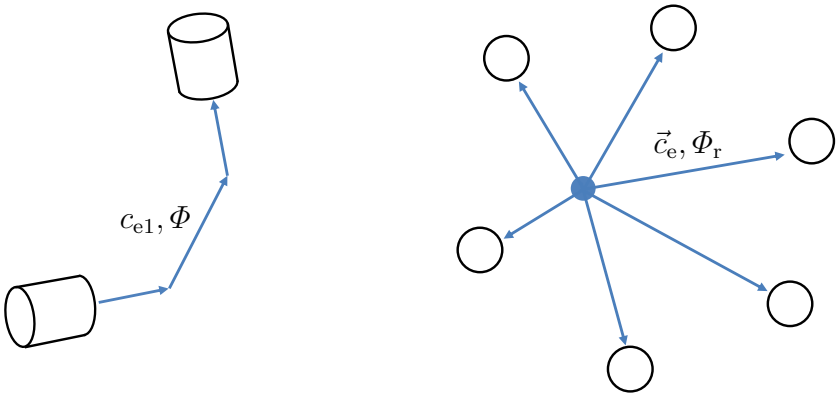


FIG. 45. The two main scenarios encountered in guided wave-based non-destructive inspection. Left: A guided wave is sent along a chosen propagation angle Φ, like in air-coupled ultrasonic inspection. Right: Guided waves are emanating outwards from a point source such as from a crack to be detected by a sensor array.

in that direction (see on the left in Fig. 45). Figure 46(b) shows the energy velocity component c_{e2} normal to the propagation direction. It can take positive and negative values, meaning that the energy velocity vector, i.e., the energy flux, can skew at some angle γ to the left and to the right side with respect to the wave propagation direction. This relates to the negative and positive skew angles seen in Fig. 46(d). The energy velocity magnitude c_e is displayed in Fig. 46(c). It is the speed at which energy is transported in the direction determined by the ray angle Φ_r, in this case $\Phi_r = 45° - \gamma$. Figure 46(e) displays a propagation time dispersion diagram for a propagation distance of 100 mm. These kind of diagrams are useful for acoustical emission experiments to identify modes in the measured signals of voltage versus time. The propagation time diagram is calculated based on c_{e1}, namely $t_{prop} = distance/c_{e1}$. Similarly, the signals simulated in the **Signal simulator** correlate with the energy velocity component c_{e1}. The onsets of the wave packets appear as predicted by the propagation time dispersion diagram. Similarly as stated for the energy velocity component c_{e1}, the propagation time dispersion diagram is valid for modes excited and detected in a chosen wave propagation direction angle Φ.

The difference between c_{e1} and $\vec{c}_e$ becomes even more striking when we look at polar dispersion diagrams. Figure 47(a) shows the energy velocity component c_{e1} of the fundamental modes at 200 kHz versus the wave propagation angle Φ in 1 mm thick T800M913. By contrast, Fig. 47(b) shows the energy velocity magnitude c_e versus the ray angle Φ_r. This plot, also called wavecrest plot, shows how the guided waves generated by a point source advance outwards with time. Therefore, these plots are useful for scenarios like acoustical emission where an array of sensors is placed on a specimen to detected ultrasonic signals emitted from a point source (see on the right in Fig. 45). It is striking how different the profiles of the extensional modes in Fig. 47(a) look compared to those in Fig. 47(b). They are equal only at $\Phi = 0°, 90°$ because these are the axes of symmetry where the skew angle and c_{e2} become zero so that $c_e = c_{e1}$. It shows how much the fiber direction $\Phi = 0°$ is preferred for the actual transport of energy. In the profile of

S_0, we notice that multiple energy velocity values are possible for a given ray angle. This is closely related to the skew angle plotted versus the wave propagation direction in Figure 47(c). Finally, Fig. 48 presents the corresponding polar propagation time for a propagation distance of 100 mm. Figure 48(a) is calculated from c_{e1}, so use it only for directed signals, whereas Fig. 48(b) is deduced from c_e and appropriate for point sources radiating outwards.

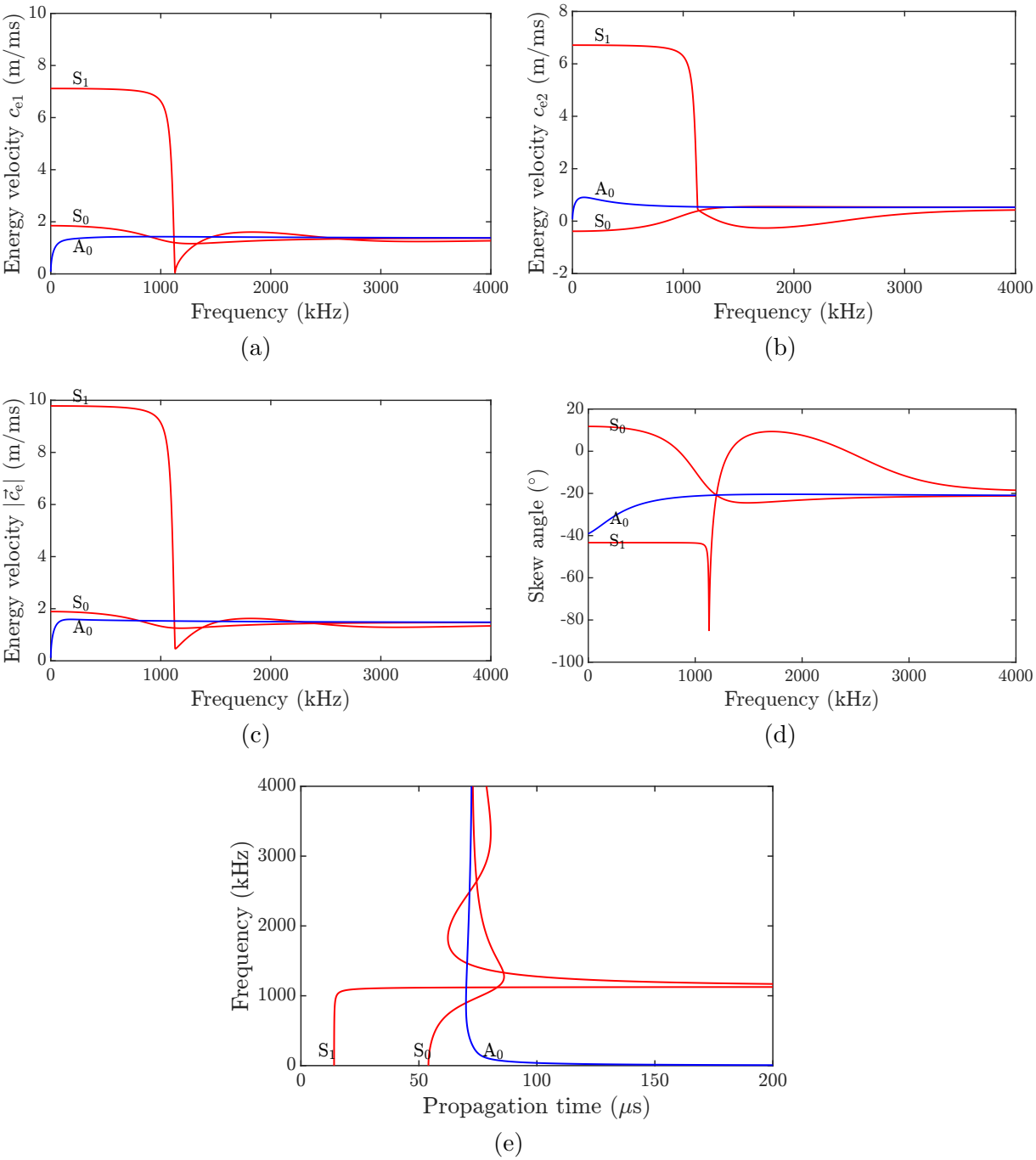


FIG. 46. Energy velocity and propagation time dispersion diagrams of the fundamental modes for propagation along $\Phi = 45°$ in a 1 mm thick layer T800M913. (a) Energy velocity component c_{e1}, (b) component c_{e2}, (c) magnitude $|\vec{c}_e|$, and (d) skew angle γ. (e) The propagation time for a distance of 100 mm is calculated from c_{e1}.

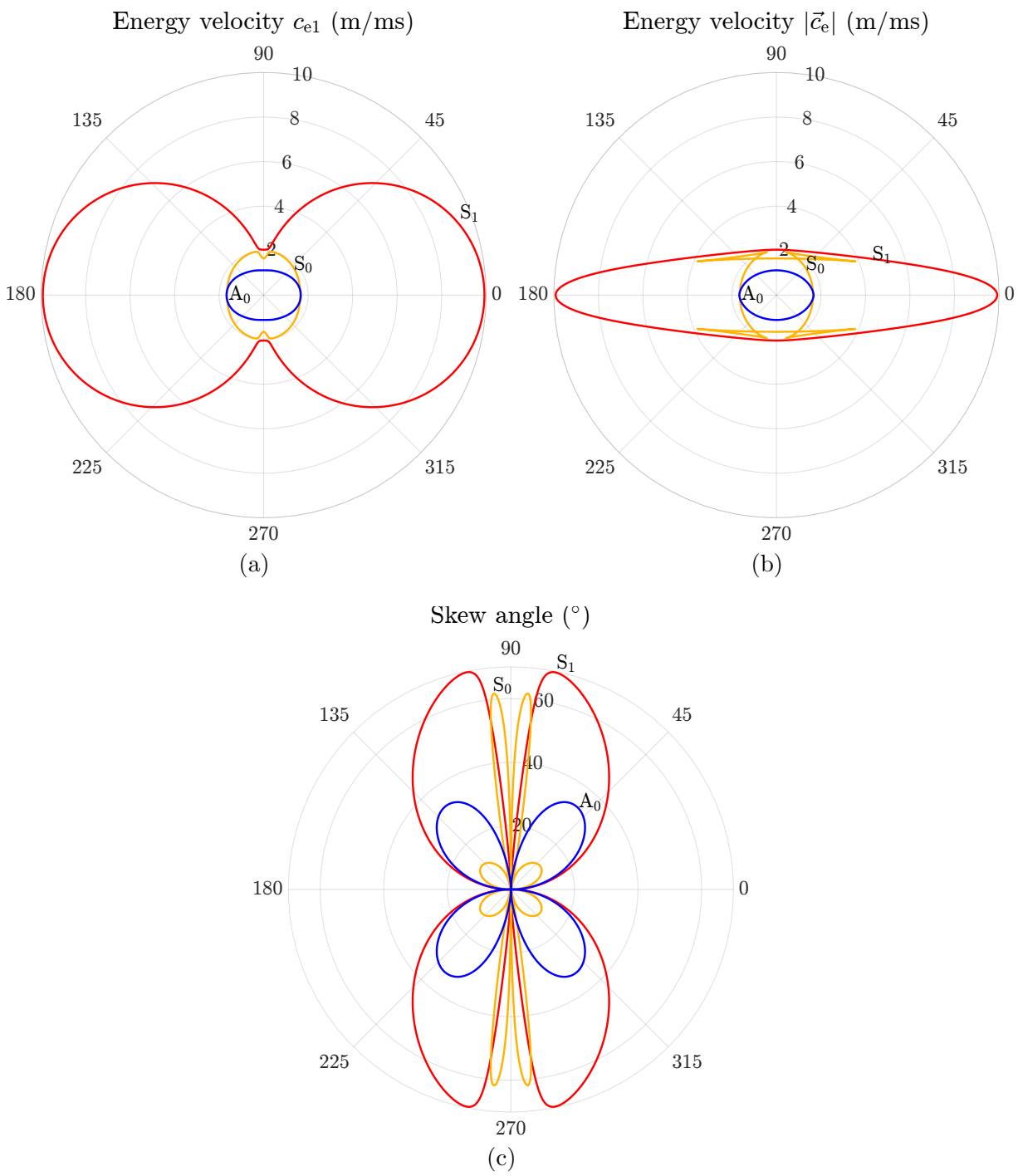


FIG. 47. Polar energy velocity dispersion diagrams of the fundamental modes at 200 kHz in a 1 mm thick layer T800M913. (a) Energy velocity component c_{e1}, (b) magnitude $|\vec{c}_e|$, and (c) skew angle γ.

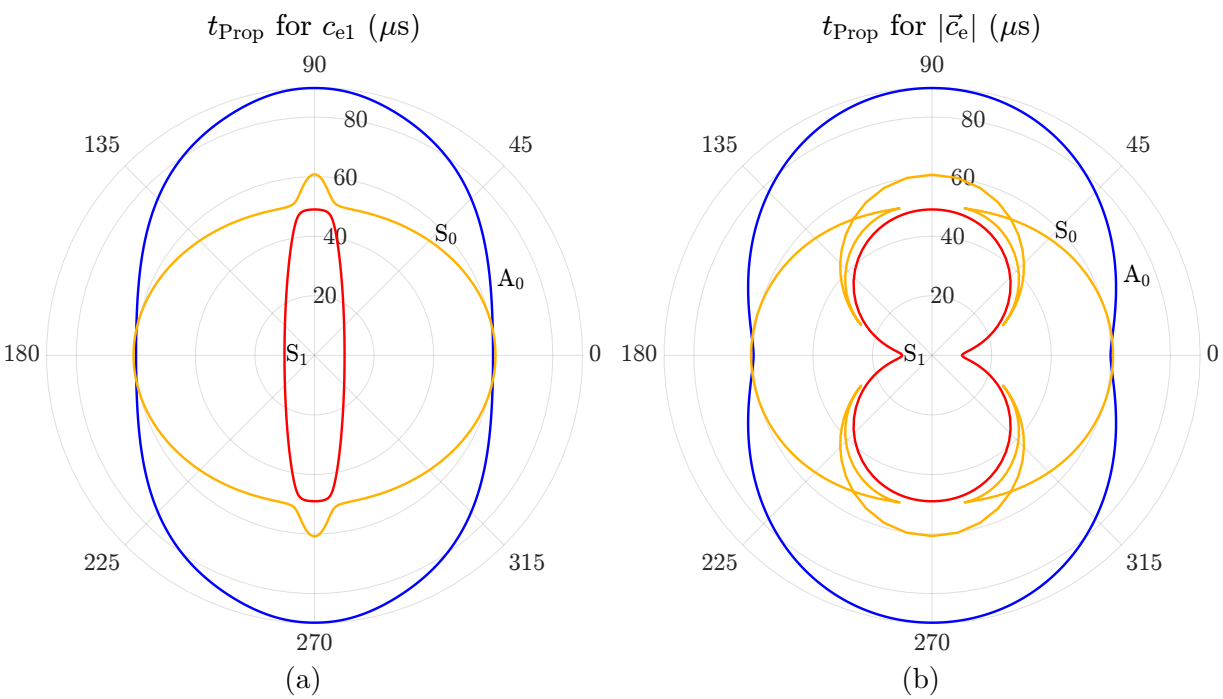


FIG. 48. Propagation time dispersion diagrams of the fundamental modes at 200 kHz in a 1 mm thick layer T800M913. (a) Propagation time based on c_{e1} and (b) based on $|\vec{c}_{\mathrm{e}}|$.

B Bulk wave velocities in elastic media

Of particular significance are the bulk waves' phase velocities for propagation along the principal axes in elastic media. The Table 3 lists the velocities as function of the stiffness components C_{ij} and of the density ρ. The direction 1 is along the fibers, while 2 and 3 are normal to it. In transversely isotropic media, the velocities are equal for propagation along 2 and 3 (and in fact for any direction within the 2-3-plane) since $C_{33} = C_{22}$ and $C_{55} = C_{66}$. In cubic materials, all longitudinal velocities are equal since $C_{11} = C_{22} = C_{33}$, and all shear waves are also equal because $C_{44} = C_{55} = C_{66}$. Both is true also in isotropic media with the specialty that $C_{44} = C_{55} = C_{66} = 0.5(C_{11} - C_{12})$. Notice that in case of complex stiffnesses C_{ij}, only the real parts are used in the below equations.

TABLE 3. Bulk wave velocities along principal axes.

Velocity	1	2	3
Longitudinal	$= \sqrt{\frac{C_{11}}{\rho}}$	$= \sqrt{\frac{C_{22}}{\rho}}$	$= \sqrt{\frac{C_{33}}{\rho}}$
Fast shear	$= \sqrt{\frac{C_{55}}{\rho}}$	$= \sqrt{\frac{C_{66}}{\rho}}$	$= \sqrt{\frac{C_{55}}{\rho}}$
Slow shear	$= \sqrt{\frac{C_{66}}{\rho}}$	$= \sqrt{\frac{C_{44}}{\rho}}$	$= \sqrt{\frac{C_{44}}{\rho}}$

C List of default materials

More material data are found on MatWeb. For a list of more than two hundred viscoelastic material data sets, consult the work by Ono [63–65].

C.1 Isotropic

If not stated otherwise, the materials are taken from Ref. [66].

	ρ (kg/m^3)	E (GPa)	ν	α_L (Np/λ)	α_T (Np/λ)
AluminumAlloy1100	2710	69	0.33	0	0
AluminumAlloy2011_Ono_2020_Viscoelastic [64]	2830	72.64	0.3355	0.0046	0.0075
AluminumAlloy2024	2770	72.4	0.33	0	0
AluminumAlloy2024_Ono_2020_Viscoelastic [64]	2770	71.56	0.3442	0.0012	0.0011
AluminumAlloy6061	2700	69	0.33	0	0
AluminumAlloy6061_Ono_2020_Viscoelastic [64]	2700	73.53	0.3298	0.0105	0.0092
AluminumOxide	3980	380	0.22	0	0
Aluminum_Disperse [2]	2700	70.76	0.3375	0	0
Aluminum_Disperse_Viscoelastic [2]	2700	70.76	0.3375	0.003	0.01
CastIronG3000	7300	101.5	0.26	0	0
CastIronG4000	7300	124	0.26	0	0
Concrete	2400	31	0.2	0	0
Copper110_Ono_2020_Viscoelastic [64]	8940	135.7	0.3174	0.0222	0.0244
CopperAlloyBrass260_Ono_2020_Viscoelastic [64]	8500	112.1	0.3382	0.0311	0.0191
CopperAlloyBrass360_Ono_2020_Viscoelastic [64]	8530	108.1	0.3333	0.0136	0.0092
CopperAlloyC17200BerylliumCopper	8250	128	0.3	0	0
CopperAlloyC26000CartridgeBrass	8530	110	0.35	0	0
CopperAlloyC36000FreeCuttingBrass	8500	97	0.34	0	0
CopperAlloyC71500CopperNickel30	8940	150	0.34	0	0
CopperAlloyC93200BearingBronze	8930	100	0.34	0	0
CopperAlloyCuAgZr_Ono_2020_Viscoelastic [64]	9920	149.7	0.3324	0.0132	0.025
CopperAlloyCuAl2O3_Ono_2020_Viscoelastic [64]	8900	137.6	0.2982	0.0122	0.0116
CopperOFHC_Ono_2020_Viscoelastic [64]	8960	171.8	0.1323	0.013	0.0389
DiamondNatural	3510	950	0.2	0	0
DiamondSynthetic	3360	863	0.2	0	0
Epoxy_Disperse [2]	1170	3.94	0.39	0	0
Epoxy_Disperse_Viscoelastic [2]	1170	3.94	0.39	0.04	0.1
Epoxy_Ono_2020_Viscoelastic [64]	1390	7.039	0.3319	0.0867	0.1636
GlassBorosilicate	2230	70	0.2	0	0
GlassCeramic	2600	120	0.25	0	0
GlassQuartz	2200	73	0.17	0	0
GlassQuartz_Ono_2020_Viscoelastic [64]	2560	69.06	0.2538	0.0126	0.0178

GlassSodaLime	2500	69	0.23	0	0
GlassSodaLime_Ono_2020_Viscoelastic [64]	2480	74.29	0.2227	0.0059	0.0065
Gold	19320	77	0.42	0	0
Ice [2]	900	9.55	0.33	0	0
Inconel625	8440	207	0.31	0	0
Inconel625_Ono_2020_Viscoelastic [64]	8440	209.2	0.3064	0.0104	0.0075
Inconel718_Ono_2020_Viscoelastic [64]	8190	199.5	0.3005	0.0144	0.0278
Inconel738_Ono_2020_Viscoelastic [64]	8110	199.4	0.2958	0.0122	0.0262
Iron_Ono_2020_Viscoelastic [64]	7870	211.4	0.304	0.09	0.0606
Lead	11340	13.5	0.44	0	0
MagnesiumAlloyAZ31B	1770	45	0.35	0	0
MagnesiumAlloyAZ61_Ono_2020_Viscoelastic [64]	1800	42.31	0.3147	0.0013	0.0014
Magnesium_Ono_2020_Viscoelastic [64]	1740	43.95	0.2974	0.0044	0.0056
Molybdenum	10220	320	0.32	0	0
Monel400	8800	180	0.32	0	0
Monel400_Ono_2020_Viscoelastic [64]	8800	179.3	0.3276	0.0214	0.0065
Nickel200	8890	204	0.31	0	0
Nickel200_Ono_2020_Viscoelastic [64]	8910	206.7	0.3156	0.0398	0.1029
NickelAlloyNiA3l_Ono_2020_Viscoelastic [64]	8100	214.7	0.2261	0.1178	0.1292
Nylon	1140	2.69	0.39	0	0
Nylon_Ono_2020_Viscoelastic [64]	1140	4.506	0.3735	0.0827	0.0991
Platinum	21450	171	0.39	0	0
Plexiglass	1190	2.74	0.31	0	0
Plexiglass_Ono_2020_Viscoelastic [64]	950	4.625	0.2421	0.0253	0.0408
Polycarbonate	1200	2.38	0.36	0	0
Polycarbonate_Ono_2020_Viscoelastic [64]	1200	3.178	0.383	0.1653	0.2008
PolyethyleneHighDensity_Ono_2020_Viscoelastic [64]	940	3.499	0.3613	0.0642	0.136
PolyethyleneLowDensity_Ono_2020_Viscoelastic [64]	910	3.825	0.3483	0.133	0.1785
Polypropylene_Ono_2020_Viscoelastic [64]	900	4.465	0.3062	0.1358	0.2002
Polystyrene	1050	2.78	0.33	0	0
PolyvinylChloride	1440	3.28	0.38	0	0
PolyvinylChloride_Ono_2020_Viscoelastic [64]	1400	4.665	0.3533	0.0641	0.1059
SiliconNitride	3300	304	0.3	0	0
Silver	10490	74	0.37	0	0
StainlessSteelAlloy304	8000	193	0.3	0	0
StainlessSteelAlloy304_Ono_2020_Viscoelastic [64]	8000	203.1	0.2875	0.014	0.0062
StainlessSteelAlloy405	7800	200	0.3	0	0
SteelAlloy1020	7850	207	0.3	0	0
SteelAlloy1020_Ono_2020_Viscoelastic [64]	7820	208	0.2908	0.0067	0.0058
SteelAlloy4340_Ono_2020_Viscoelastic [64]	7840	202.8	0.295	0.004	0.0024
Tantalum	1660	185	0.35	0	0
Teflon	2170	0.475	0.46	0	0
Tin	7170	44.3	0.33	0	0
Titanium	4510	103	0.34	0	0

TitaniumAlloyTiA6l4V	4430	114	0.34	0	0
TitaniumAlloyTiA6l4V_Ono_2020_Viscoelastic [64]	4500	114.7	0.3267	0.0271	0.014
TitaniumAlloyTiA6l6V2Sn_Ono_2020_Viscoelastic [64]	4540	111.5	0.3289	0.01	0.011
ToolSteelT8_Ono_2020_Viscoelastic [64]	8430	220.2	0.2757	0.0375	0.0135
Tungsten	19300	400	0.28	0	0
Tungsten_Ono_2020_Viscoelastic [64]	19250	405.2	0.2867	0.008	0.0039
Zinc	7140	104.5	0.25	0	0
Zirconium	6510	99.3	0.35	0	0

C.2 Cubic

$C_{33} = C_{22} = C_{11}$, $C_{13} = C_{23} = C_{12}$, $C_{44} = C_{55} = C_{66}$

ρ in kg/m^3, C_{ij} in GPa

	ρ	C_{11}	C_{12}	C_{66}
GaAs [59]	5310	118.8	53.8	59.4
InAs [59]	5670	83.29	45.26	39.59

C.3 Transversely isotropic

$C_{13} = C_{12}$, $C_{33} = C_{22}$, $C_{44} = 0.5(C_{22} - C_{23})$, $C_{55} = C_{66}$

ρ in kg/m^3, C_{ij} in GPa

	ρ	C_{11}	C_{12}	C_{22}	C_{23}	C_{66}
AS4M3502 [67]	1550	147.1	4.1	10.6	3.1	6
CarbonEpoxy_Rokhlin_2011 [17]	1610	162	11.8	17	8.2	8
SAERTEX7006919RIMR135 [43]	1454	122.692	4.31945	9.21241	4.20204	6.02
SigrafilCE125023039 [67]	1500	131	3.7	8.1	3.1	6.1
T300M914 [68]	1560	143.8	6.2	13.3	6.5	5.7
T700M21 [69]	1571	129.986	6.06192	11.1918	5.19179	4.135
T700PPS [67]	1600	152.7	4.7	11.7	4.4	4.5
T800M913 [67]	1550	154	3.7	9.5	5.2	4.2
T800M924 [70]	1500	164.708	5.45328	11.2997	4.73939	6
T800_Castaings	1510	184	7.09964	12.9996	6.99943	5.5
TVR380M12R_GFRP [23]	1800	49.7605	6.16767	17.1601	5.68314	5.233

C.4 Orthotropic

ρ in kg/m^3, C_{ij} in GPa

CarbonEpoxy2_Rokhlin_2011 [17]

ρ	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
1610	162	10.6	13	15.3	8.2	18.7	4.4	8.8	7.2

CarbonEpoxy_Castaings

ρ	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
1500	125	6.3	5.4	14	7.1	14	3.45	5.4	5.4

CarbonEpoxy_Castaings_Viscoelastic

ρ	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
1500	125 +2.5i	6.3 +0.126i	5.4 +0.108i	14 +0.28i	7.1 +0.1421i	14 +0.28i	3.45 +0.069i	5.4 +0.108i	5.4 +0.108i

CarbonEpoxy_Hernando_2015 [36]

ρ	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
1500	132	6.9	5.9	12.3	5.5	12.1	3.32	6.21	6.15

CarbonEpoxy_Hernando_2015_Viscoelastic [36]

ρ	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
1500	132 +0.4i	6.9 +0.001i	5.9 +0.016i	12.3 +0.037i	5.5 +0.021i	12.1 +0.043i	3.32 +0.009i	6.21 +0.015i	6.15 +0.02i

HIO_3 [71]

ρ	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
4640	30.1	16.1	11.1	58	8	42.9	16.9	20.6	15.8

Oak [2]

ρ	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
597	8.61313	2.17462	2.77298	1.73528	1.06095	2.40251	0.3	0.89	0.92

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